technica engineering
Member of KPIT Group

IVN2Eth Capture Module:

CM 1000 High

USER MANUAL

March 2024

Document version: 1.1 Firmware: v021.002.XXX Hardware: v5.4

CONTENT

1 Introduction 8
1.1 About this document 8
1.1.1 Scope 8
1.1.2 Validity 8
1.1.3 Glossary 9
2 Safety and warranty 11
2.1 General operating and safety strategy 11
2.2 General design rules for the power supply 11
2.3 Warranty and Safety Information 12
2.4 RoHS Certificate of Compliance 15
3 Product description 16
3.1 Functionality 17
3.2 Features 18
3.2.1 Logging 18
3.2.2 TECMP encapsulation 20
3.2.2.1 Timestamping 20
3.2.2.2 Interface identification 21
3.2.2.3 Counter 21
3.2.3 Cascading 22
3.2.4 Time synchronization 23
3.2.5 Startup buffer 25
3.2.6 Filtering 25
3.2.7 Traffic shaping 27
3.2.8 Transparency mode 28
3.2.9 Traffic injection 29
3.2.10 Status messages 30
3.3 Technical data 31
3.4 Scope of Delivery 31
4 Hardware 32
4.1 Connectors and controls 32
4.1.1 Power connector (1) 33

4.1.2 1000BASE-T (RJ45) connectors (2) 35
4.1.3 SFP+ slot (8) 35
4.1.4 IVN connectors (3) 36
4.1.5 Debug connector (4) 37
4.1.6 Rotary switch (5) 38
4.1.7 Status LEDs (6 & 7) 38
4.1.7.1 System LEDs (6): 38
4.1.7.2 Link LEDs (7): 38
5 Configuration 39
5.1 Home Screen 40
5.2 System Information tab 42
5.2.1 License Information 42
5.2.2 Firmware Information: 43
5.2.3 Device Information 44
5.2.4 Device PHYs Information 47
5.3 Control Panel tab 47
5.3.1 Power 47
5.3.2 System parameters 47
5.3.3 Configuration 48
5.3.4 Sleep parameters 49
5.3.4.1 Keep-alive conditions 50
5.3.4.2 Wake-up conditions 50
5.3.5 Wake-up lines and actuator 52
5.3.6 Wake-Up lines signaling timeout 53
5.3.7 Sleep Timeout 53
5.4 Status tab 53
5.4.1 Host Port 55
5.4.1.1 Port name 55
5.4.1.2 100BASE-T1 mode 55
5.4.1.3 Keep awake reason channel 56
5.4.2 GB Ports 56
5.4.2.1 Port name 56
5.4.2.2 Detected Speed 57

5.4.2.3 Keep awake reason channel 57
5.4.2.4 Logging ports 57
5.4.2.5 Configuration port 57
5.4.2.6 Traffic injection port 58
5.4.2.7 Status messages port 58
5.4.2.8 Control message port 59
5.4.2.9 Cascade to 60
5.4.2.10 802.1AS Mode 60
5.4.3 SFP+ Port 61
5.4.3.1 Port name 62
5.4.3.2 Detected Speed 62
5.4.3.3 Keep awake reason channel 62
5.4.3.4 Logging ports 62
5.4.3.5 Traffic injection port 63
5.4.3.6 Status messages port 63
5.4.3.7 Control message port 64
5.4.3.8 Cascade to 64
5.4.3.9 802.1AS Mode 65
5.4.4 Spy 66
5.4.4.1 EtherType 66
5.4.4.2 Use Message Header 67
5.4.4.3 Enable Traffic Shaping, Traffic Bandwidth 67
5.4.4.4 Log through 68
5.4.4.5 Keep original VLANs in Header 68
5.4.4.6 Use Outer VLANs 69
5.4.4.7 Destination MAC 70
5.4.4.8 Delay for Logging 70
5.4.4.9 Injection Mode 72
5.4.4.10 Packetization mode 72
5.4.4.11 Logging Ports for packetization 73
5.4.4.12 Timeout for packetization (ms) 73
5.4.5 1000BASE-T1 Ports 75
5.4.5.1 Port name 75

5.4.5.2 BASE-T1 mode 75
5.4.5.3 Port Speed 76
5.4.5.4 PHY Mode 76
5.4.5.5 Output to 77
5.4.5.6 Interface ID 77
5.4.5.7 Transparency Mode 78
5.4.5.8 Transparency wake-up (ms) 80
5.4.5.9 Enable pair traffic forwarding 81
5.4.5.10 Keep awake reason channel 81
5.4.5.11 Enable recovery for this port 82
5.4.5.12 Enable injection for this port 82
5.4.5.13 MAC/IP for Injection 82
5.4.5.14 Constant delay in bytes for injection 82
5.4.5.15 Timestamp 83
5.4.5.16 Messages Transmitted 83
5.4.5.17 Messages Received 83
5.4.5.18 Messages with Error 84
5.5 Diagnostics tab 85
5.6 Basic Filters tab 85
5.7 Link Events tab 87
5.8 Sync Events tab 87
5.9 Advanced Filters tab 88
5.9.1 How to manage advanced filtering rules 89
6 Application examples 91
6.1 Advanced filtering 91
6.2 Traffic injection 92
6.3 Time synchronization using PTPv2 (IEEE 1588v2) 94
6.3.1 How to set up a synchronized network with Capture Modules and PTPv2 95
6.4 Time synchronization using gPTP (IEEE 802.1AS) 95
6.4.1 Transport of time synchronization information 97
6.4.2 Peer-to-peer propagation delay measurement 98

6.4.3 How to set up a synchronized network with Capture Modules and gPTP 99
6.5 Optimized packetization 99
7 Maintenance 102
7.1 Firmware update 102
7.2 Update from R18 (or lower) to R17 (or lower) 102
7.2.1 Preconditions and important information 102
7.2.2 Update via GB-A port 103
7.2.2.1 Update via 100BASE-T1 host port 104
7.3 Update from R17 (or lower) to R18 or higher 105
7.3.1 Preconditions and important information 106
7.3.1.1 Update via GB-A port 106
7.3.1.2 Update via 100BASE-T1 host port 108
7.4 Update of release 18 or higher 111
7.4.1 Preconditions and important information 111
7.4.1.1 Update via GB-A port 112
7.4.1.2 Update via 100BASE-T1 host port 115
7.5 Additional information about R18 118
7.6 Error messages during flashing 118
7.6.1 Error Codes 118
7.6.1.1 Miscellaneous errors 120
7.6.1.2 Recovery 121
8 Appendix 123
8.1 TECMP protocol 123
8.1.1 TECMP frame 123
8.1.2 TECMP header 124
8.1.3 Control Messages 127
8.1.4 Status Messages 128
8.1.4.1 Status Capture Module message 128
8.1.4.2 Status Configuration message 130
8.1.4.3 Status IVN message 132
8.1.5 UDP Discovery messages 135
8.1.5.1 Broadcast discovery request 136

Unicast discovery response 136
8.1.6 UDP Configuration messages 137
8.1.6.1 Set Configuration mode 138
8.1.6.2 Set IP address 140
8.1.6.3 Set device time 141
8.1.6.4 Get Capture Module information 142
8.1.6.5 Fan configuration 150
8.1.6.6 WUP lines configuration 151
9 Disclaimer 152
10 Changelog 153
11 Contact 154
12 Declaration of conformity 155

1 Introduction

1.1 About this document

This user manual contains important information for the proper and safe operation of Technica Engineering 's Capture Module 1000 High (hereinafter *CM 1000 High*).

Read this user manual completely before using the device for the first time and follow the instructions, paying special attention to the safety-related parts.

For the latest version of this document please check the customer portal.

©2025 Technica Engineering GmbH

The content in this document is copyrighted by Technica Engineering GmbH. Copying or reproducing its content in full or in parts is not allowed. Technical contents in this document are subjected to change without prior notice. Third-party contents and trademarks are the property of the corresponding vendors.

1.1.1 Scope

This document includes instructions about how to use properly Technica Engineering's Capture Module 1000 High, together with information about some of the complex use cases that can be covered by this device.

Troubleshooting is not part of the scope of this document. For support queries, please contact Technica Engineering via support@technica-engineering.de.

Please note that, due to the dynamic development of the Capture Module product family, the information in this document is subject to change without prior notice to adapt to technical changes and updates. Some features may not be described yet in the user manual or maybe described incompletely.

Make sure to always use the correct version of this document, matching the firmware and hardware versions in operation (see next section).

1.1.2 Validity

This user manual applies to CM 1000 High devices with:

- Firmware version: v021.002.XXX
- Hardware version: v5.4

- TECMP: Specification document v1.7 (TECMP v3.0)

For previous firmware versions, older versions of this document can be used as a reference. Please check CHAPTER 10 for details.

1.1.3 Glossary

Term	Definition / Explanation
AVB	Audio Video Bridging
AVTP	(Layer 2) Audio-Video Transport Protocol
Control Message	A method to trigger events. Events can be triggered by a Capture Module or can be received by a Capture Module. Some Control messages can be configured
DLM	Data Logger Module
DUT	Device Under Test
Eth	Ethernet
FCS	Frame Check Sequence. Error-detecting code is added at the end of an Ethernet frame
FPGA	Field Programmable Gate Array. A programmable logic device that can have its internal configuration set by firmware
FR	FlexRay
gPTP	Generic Precision Time Protocol, IEEE 802.1AS
HW	Hardware
IVN	In-Vehicle Network
IFG	Interframe gap between two ethernet frames
MAX10	Intel® FPGA device
PTP	Precision Time Protocol, IEEE 1588
Sink	Logging endpoint. A device that receives the captured data to check traffic *live* or to store in a file for later analysis
TAP	Test Access Point. Dedicated hardware device to monitor events on a local network
TECMP	Technically Enhanced Capture Module Protocol, used for encapsulation of captured in-vehicle network data
TSN	Time-Sensitive Networking

Table 1-1-1: Glossary

Additionally, the following symbols are used throughout the document to highlight important information:

 Information: Useful tips, hints, or background information.

 Warning: Important information to consider for the correct use of the device.

2 Safety and warranty

2.1 General operating and safety strategy

Technica Engineering products are designed for operation in automotive systems and for supply voltages of typical 12 V. The applicable limit values adhere to the standard norms for 12 V automotive onboard power systems correspondingly and can be found in the mentioned norms.

Should Technica Engineering products be operated in voltage ranges beyond those specified in the norms, which represents a breach of the conditions of operation, then this will void the product warranty and Technica Engineering will assume no liability whatsoever of the results and/or consequences thereof.

This is especially valid whenever the voltage level reaches or exceeds the limits of the low voltage directive. In this case, damage to the devices cannot be excluded. Due to the manufacturing characteristics of the devices, there is no imminent fire hazard from the device itself, if the devices are being operated in an environment according to the conditions of use. A secondary fire hazard cannot be excluded, should those conditions not be met. Protection against overvoltage cannot be provided in such a breach of the conditions of use.

2.2 General design rules for the power supply

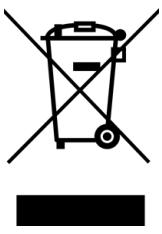
The power supply solution of Technica Engineering's products is equipped with a self-protection mechanism. This automatic function protects the internal electronic components against excessive temperature and too high supply-voltage by switching the device off. This automatic switch-off function is independent of any software function. Excessive temperatures in the device can be caused by high environmental temperatures or internal failures of the device. The device can protect itself by shutting down its own power supply avoiding further damage to itself.

The operator/user of this devices must take measures for the right protection of the power supply lines and the according connector. The maximum power consumption of the Technica Engineering products is written in this user manual.

Please take care that some of the products are working with a step-up converter. So, the current needed in case of undervoltage conditions must be considered as higher than at typical voltage level. The user must choose the right fuse or other over current protection measures.

2.3 Warranty and Safety Information

	Before operating the device, read this manual thoroughly and keep it for your reference. The latest documentation for the CM 1000 High can be obtained from the customer portal.
	As a user of this device, you may only perform tasks that are described in this document valid for the corresponding firmware version. Technica Engineering assumes no liability for any kind of damages arising from the use of or in relation with the use of our devices. Full responsibility of operation rests on the side of the end user. Manufacturer warranty of Technica Engineering products only applies as long as the user ensures proper installation, configuration and use of the device as described in this user manual. Use the device only as described in this manual. Use only in dry conditions. Do not insert any foreign object in the slots/openings of the housing. Do not apply power to a damaged device. The device is to be used only by specialists.
	Do not open the device. Otherwise, the warranty will be lost.

	This product is intended for use in automobiles or automotive-like environments. An automotive-like environment includes test setups or test benches in the office, laboratory, and workshop areas. In the test setups, the same environmental conditions apply as in-vehicle electrical systems. Technica Engineering products are not intended to be used as standard IT equipment. The test systems and products from Technica Engineering are designed as a customer and application-specific test modules that are only used by specialists for the development and test facilities. When integrating the modules in a vehicle or test setup, the user must ensure appropriate ventilation or air convection for the device(s). Additionally the user (Integrator) of this product is responsible for the external power supply of this module, assuring the appropriate cable protection (fuse). The maximal current load value for the product is described in the user manual. Technica Engineering products must not be considered as a safety element out of context when using safety-critical systems and must be included in the safety assessment when used. The development class in a safety system must be considered with standard QM referred to as ISO26262.
	 The device can get hot. Do not cover the device due to fire danger. Do not place the device near highly flammable materials due to fire danger. Do not use the device above the specified operating temperature. The operating temperature is the ambient temperature of the installation space.
	This symbol is only valid in the European Union. If you wish to discard this product, please contact your local authorities or dealer, and ask for the correct method of disposal. Technica Engineering GmbH is registered as a manufacturer of the brand "Technica Engineering" and the device type "Small devices of Information and Telecommunications technology for exclusive use in non-private households". WEEE Reg. No. DE 20776859

CE	Please refer to CHAPTER 12 for the EU Declaration of Conformity In accordance with directive 2014/30/EU.

2.4 RoHS Certificate of Compliance

Technica Engineering Capture Module 1000 High complies with RoHS (Restriction of Hazardous Substances Directive) Certificate of Compliance.

3 Product description

Automotive Ethernet has conquered the modern E/E vehicle network. The challenges that arise during test and validation include a reliable capture of AVB/TSN traffic, without interfering with the network timing. This challenge gets even bigger, with the need for diagnostics protocols to be sent to the DUTs during runtime.

Technica Engineering's Capture Modules have been specifically designed to solve these problems and many more, such as ensuring that it starts up faster than the DUTs whose traffic will capture and that the traffic is delivered with hardware timestamp.



Figure 3-1: Capture Module 1000 High

The **CM 1000 High** from Technica Engineering GmbH, part of our family of IVN2Eth Capture Modules, can be used to capture the traffic from up to 6 x 1000BASE-T1 link lines, each connecting two DUTs. The traffic is captured without causing interference on the network and delivered with a 40 ns resolution hardware timestamp, thus analyzing AVB/TSN traffic possible.

Several IVN2Eth Capture Modules can be used on the same setup in case more 1000BASE-T1 ports are needed or when other IVN technologies are present in the setup. The built-in synchronization using IEEE 802.1AS or IEEE 1588-2008 allows for simultaneous use and seamless integration of several IVN2Eth Capture Modules.

The CM 1000 High is optimized for automotive use.

3.1 Functionality

The main task of the Technica Engineering Capture Modules is capturing the data of in-vehicle (IVN) communication systems via their logging network based on Ethernet. Other features such as time synchronization of the capture modules to each other or the scalability of being able to connect other modules via the network are all used from the Ethernet solution space. In this way, a user can set up specific logging solutions fitting the user's needs. This is even possible in selected scenarios to use a regular computer instead of a data logger.

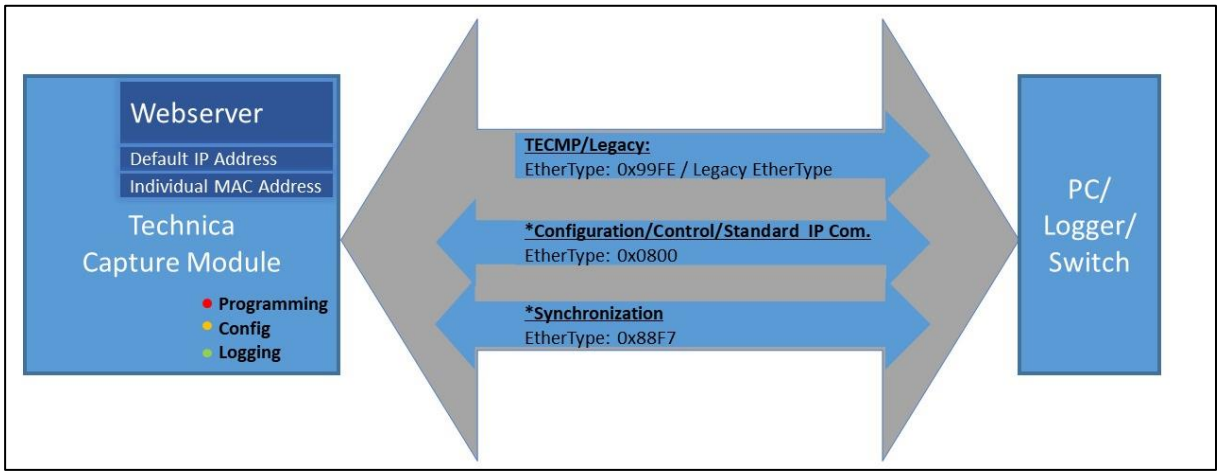


Figure 3-2 Capture Module 1000 High

The purpose of using Capture Modules is to convert and capture all IVN data to an Ethernet-based packet format. This enables the user to have a logging and tracing network that connects several capture modules and logging endpoints via a standard ethernet network. This logging network is scalable and based on IT standards. No additional transport protocol is required.

To be able to do that, each message captured from the in-vehicle network is encapsulated in a standard transport layer frame. This requires embedding in a standard Ethernet frame according to IEEE 802.3. Data of non-Ethernet communication systems, as well as Ethernet-based systems, are transported in Ethernet frames and identified unambiguously using the protocol TECMP. Ethernet frames are transported without changes to them or their headers. All messages are left intact. This ensures that analysis tools can easily access the unchanged data if support of TECMP was added or data was converted.

The main use case of the Capture Modules is to transmit the IVN data collected by the device(s) to a logging endpoint connected via Ethernet. This endpoint can be a computer, an Ethernet switch for a network connection, or a data logger module, and it is usually referred to as *sinks*. TECMP specification defines the data format in which

the collected data are transmitted to the configured sinks. Additional metadata like timestamping and other source data-related information are covered also by the TECMP protocol.

For more information about this protocol, please check the TECMP specification document.

3.2 Features

The main features of CM 1000 High are summarized below:

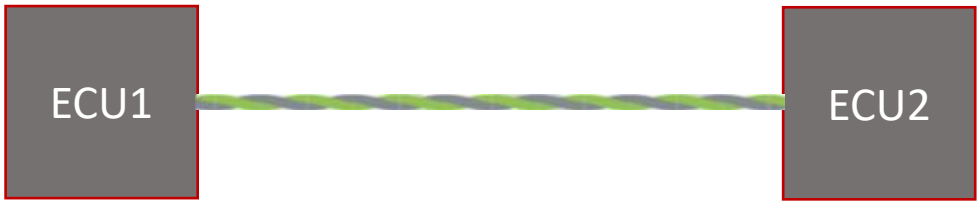
- 6 1000BASE-T1 link lines (12 ports)
- 1 SFP+ port for 10Gbps sinks
- Technically Enhanced Capture Module Protocol
- High-speed startup
- Startup buffer
- Output traffic shaping
- AVB/TSN-capture capable
- Time-aware injection
- Network time synchronization
- Cascading and synchronization of multiple devices
- Source timestamping with 40 ns resolution
- Wake-up capability
- 2 wake-up lines, configurable as input/output
- Extended power mode for car integration
- Easy configuration via webserver
- Additional configuration mode via UDP messages
- Rotary switch for manual configuration of the device IP address
- Robust stainless-steel case

The following features – and more – make this device appropriate for general-purpose testing:

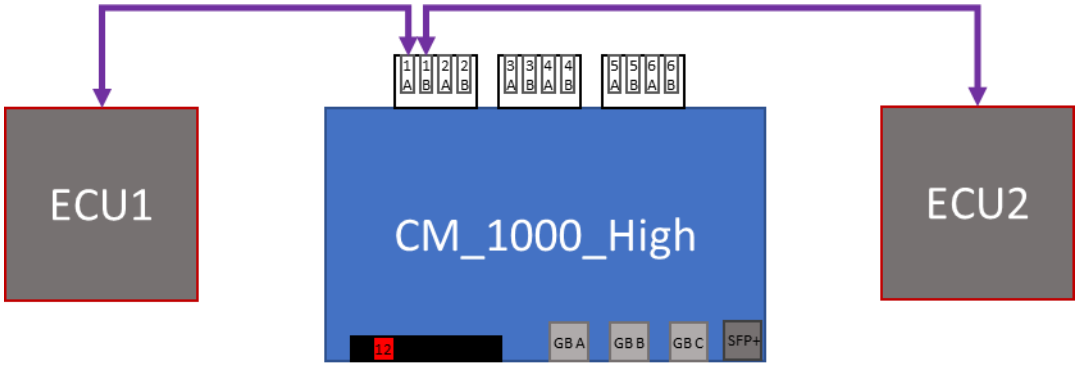
3.2.1 Logging

The CM 1000 High is provided with 6 1000BASE-T1 link lines, each with two ports, making it possible to connect up-to 12 ECUs.

Ethernet is a point-to-point technology, i.e., it allows the communications connection between two communication endpoints or nodes. In the vehicle, this means that two ECUs that exchange information over Ethernet are directly connected:



To capture the data that they are exchanging, it is necessary to put a device in the middle, creating a point-to-point connection with both ECUs. This is exactly how a Capture Module works:



Each of the 6 1000BASE-T1 link lines has two ports, marked as *A* and *B*, often referred to as *pairs*, to which – following the example above – ECU 1 and ECU 2 can be connected.

Capture Module is acting then as a so-called active TAP, physically linked to the monitored devices (ECUs), duplicating, and passing the captured data along to an endpoint such as a computer or a logging device.

This also makes it possible to offer other features like Traffic INJECTION, where control over the communication between ECUs is required.

Being an active TAP means that communication between the two ECUs connected to a pair is not possible if the Capture Module is off.

This way, the CM 1000 High can capture all the traffic coming from each ECU and forwarding it at the same time to the other ECU connected to the pair and the output ports for logging.

Note that the CM 1000 High is not accepting as input Ethernet frames not compliant with the standard IEEE 802.3:

- Frames with less than 42 bytes when an 802.1Q tag is present and 46 bytes when absent (total length under 64 bytes).
- Frames with more than 1500 bytes payload (non-standard jumbo frames).
    - The maximum frame size, including the payload, header and +CRC is 1518 bytes. If a VLAN tag is added, the maximum frame size increases to 1522 bytes.
- Frames without a correct preamble.

Simultaneously logging with several GB interfaces is fully filter dependent. This means different filters can be implemented on the traffic captured from IVN ports, and the traffic can be egressed using multiple interfaces. A maximum of 1Gbps can be logged in total, irrespective of the egress GB port(s) selected.

3.2.2 TECMP encapsulation

Captured data needs a context to be interpreted correctly: for a comprehensive analysis, it is essential to be able to identify when and where the data was sent. When the captured data is stored in raw format – as it is sent on the communication line –, this context is lost.

As a solution for this, the Capture Modules offer the possibility to encapsulate the raw data within Ethernet II frames that are provided with metadata that helps identify unequivocally the traffic, in terms of place and time. Among others, the following metadata is added to the original frame:

- Device ID: Identifier of the Capture Module that captured the data.
- Channel ID: Exact Capture Module port where the data has been captured.
- Timestamp: Time when the data has been captured.
- Data flags: Additional information about the captured data, like checksum error flag.

This information is provided to the user as part of the header of the TECMP frames used for encapsulating the captured data. The TECMP frames are then sent to the logging port and analyzed live with a traffic analysis tool or stored in a trace for later analysis.

In the case of a CM 1000 high, since the data captured consists also of Ethernet frames, the user can log the raw data directly as-is, without TECMP encapsulation. This implies though that all the benefits of this feature are not available.

3.2.2.1 Timestamping

All data coming from the in-vehicle network is hardware timestamped when it enters the device, and this timestamp is included in the header of the TECMP frame used for encapsulation. This way, how much time the data needs to reach its destination in the

logging network is irrelevant. The original timestamp remains untouched as part of the TECMP header and can be used later for an accurate analysis of the captured data.

During a trace analysis, the time when the TECMP frame arrives should not be considered; this will always include the time that the captured data requires from the input port (IVN port) to the logging device, which is irrelevant for the user and thus, should not be used as a reference. The timestamp that matters is the one part of the TECMP header, which indicates when the captured data was sent.

This feature is especially important when using several Capture Modules within the same logging network (see CHAPTER 3.2.3) and is tightly related to the time sync functionality explained in CHAPTER 3.2.4. A typical use-case can be when the Capture Module is used with a data logger. When the data logger is not ready to log data, and the Capture Module is logging the traffic. All the traffic stored in the internal memory of the Capture Module (before the first synchronization phase) must be correctly timestamped once the Capture Module gets the signal of LOGGER READY. Correction of timestamps is only done for the frames that are captured before the first synchronization phase.

 In scenarios where accurate timing is key for traffic analysis, TECMP encapsulation must always be activated.

 After first synchronization, if the Capture Module losses time synchronization again, the time correction is not done.

3.2.2.2 Interface identification

Incoming traffic is not only provided with a hardware timestamp, but also with an id that identifies unambiguously the IVN interface where the data was captured.

For communication buses like CAN or FlexRay, where the data does not contain any information about sender and receiver, the channel id helps to identify where exactly the data comes from.

In the case of Ethernet, the frames already contain information regarding source and destination, but in scenarios where the same ECU is connected via Ethernet with several other ECUs, the interface ID can be the only way to determine unequivocally the origin of the data.

3.2.2.3 Counter

Each TECMP frame is provided with a counter. This counter is incremented with each TECMP frame sent out by the Capture Module.

The purpose of the counter is to identify possible data loss in the path between the Capture Module and the logging endpoint.

The Capture Module offers the possibility to configure more than one output port to connect with the logging endpoint (also called *sink*) and forward the captured data to it. Thus, the Capture Module handles different counters, one per output port.

3.2.3 Cascading

Capture Modules have been developed with modularity in mind. Instead of having a big and cumbersome all-in-one device, with all possible kinds of IVN interfaces, Technica Engineering wants to give the users the possibility to define their logging network, adapted to their needs, including only those interfaces that are indeed required.

To make this possible, several hardware variants of Capture Modules have been designed, offering different sets of IVN interfaces:

- CM 100 High, with 12x 100BASE-T1 interfaces.
- CM 1000 High, with 12x 1000BASE-T1 interfaces.
- CM Ethernet Combo, with 2x 1000BASE-T1 and 4x 100BASE-T1 interfaces.
- CM CAN Combo, with CAN(-FD), FlexRay, and RS232 interfaces.
- CM LIN Combo, with LIN and analog interfaces.

With these hardware variants of Capture Modules, it is possible to create a flexible logging network containing as many devices as required. The different Capture Modules can be either connected to a time sync master (time-aware switch, data logger…) and synchronized through it or can be connected using the cascading functionality.

When several Capture Modules are interconnected using cascading, they can be seen as only one device from the point of view of the logging endpoint: the traffic captured from the last Capture Module in the cascade is forwarded to the next, from this to the next and so until it reaches the first Capture Module, which will be connected to the logging endpoint, either directly or via a switch.

In this case, time synchronization between the different Capture Modules (see CHAPTER 3.2.4) is of special importance, to guarantee that the data captured by all the devices share the same timing reference.

Cascading functionality is optimized to be used for connecting a high-speed Capture Module (Ethernet variants) with a low-speed one (CAN and LIN variants). Ideally not

more than 2 devices (one high speed variant and one low speed variant) should be cascaded. Cascading of Capture Modules must be designed carefully.

- If the logging port is cascaded to another logging port, the frames are exchanged bi-directionally with the following restrictions:
  - TECMP messages, excluding replay data and UE
  - PTP
- If the 100BASE-T1 port is cascaded to GB port, the user must take the change in speed into account. The 100Mbps traffic to 1Gbps traffic will flow without issues but traffic in other direction (that is 1Gbps to 100Mbps) will face the physical bottleneck and cause data loss. This will have an impact on flashing/ping etc. for the cascaded device.
- If the GB port is cascaded to 100BASE-T1, there is an obvious physical bottleneck from 1Gbps to 100Mbps, therefore this configuration must be avoided.

While cascading the following information needs to be considered that in the case of computers as logging endpoint, the performance in the processing of Ethernet data varies a lot depending on its characteristics: network card, CPU, memory, drivers, and OS.

- Using a low-performance computer, the probability of data loss is high. Choosing a computer with the right characteristics decreases this probability a lot, but rarely makes it disappear completely under any circumstances.

Data loggers are in most cases stable and performant enough to handle the data correctly. Only under very high-demanding situations like prolonged bursts, data loss may be observed.

For scenarios involving more devices, a connection via data logger or time-aware switch is preferred, to avoid possible bottlenecks due to high bandwidth.

3.2.4 Time synchronization

To have reliable logging data, time synchronization is required between the Capture Modules and other devices on the logging network.

Time synchronization allows two or more devices (one master and one or multiple slaves) to transmit/receive time-critical information through a physical link and assures that all the devices will handle the same time reference. In the case of the Capture Modules, this time is used for the timestamping of the captured data.

The Capture Modules offer two different mechanisms to perform synchronization between each other and other devices:

- IEEE 802.1AS (also called gPTP), automotive profile: In this mode, synchronization is achieved using peer-to-peer (P2P), 2 step synchronization mechanism.
- IEEE 1588-2008 (also called PTPv2): In this mode, synchronization is achieved using end to end(E2E), 2 step synchronization mechanism.

The CM 1000 High can take 3 different roles for time synchronization when using IEEE 802.1AS (gPTP):

- Grandmaster: An entity that provides a global time base for the whole-time aware network. A grandmaster sends information including the current time (also known as wall time) to all directly attached time-aware systems.
- Bridge master: An intermediate entity that receives its time from the grandmaster (directly or through intermediate devices) and acts as a time master for the succeeding device in the network chain. It forwards the corrected time information to all the succeeding attached time-aware systems.
    - Note that the ports configured as bridge master are masters, but not grandmasters since they are simply re-timing from an upstream grandmaster.
- Bridge slave: An entity that consumes its time from its master pair, connected directly to it.

Making use of these roles, several Capture Modules can be connected in cascade for time synchronization.

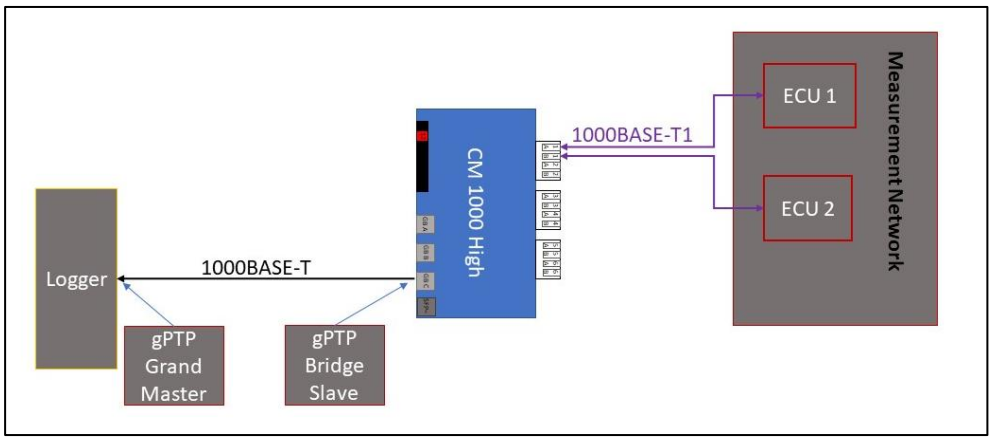


Figure 3-3 Example of Capture Module acting as gPTP slave for a gPTP Grand Master

On the other side, when configured to use IEEE 1588-2008 (PTPv2), the CM 1000 High can only act as a slave.

The time synchronization is provided via the 1000BASE-T (RJ45) ports (see CHAPTER 5.4.2) or the SFP+ port (see CHAPTER5.4.3). This bidirectional traffic (between masters and slaves) can be configured using the same ports as the ones used for logging,

coexisting then with the captured data, or can be performed using separate ports for time synchronization only.

The *Host* port is only for configuration purposes and does not support time synchronization.

3.2.5 Startup buffer

Being able to log all the traffic, including the very first messages sent by the ECU(s) after a startup, is critical for a successful analysis of lifecycle issues. Capture Modules have been designed to have a fast startup that allows capturing the initial traffic between ECUs.

The start-up time varies depending on the mode of operation of CM 1000 High

- In normal operation mode, it takes around 180ms for the first frame to be checked on the received port
- In transparency mode (See CHAPTER 3.2.8), it takes around 160ms for the first frame to be checked on the received port provided that the partner port also has a linkup.

Logging *sinks* like computers or data loggers have a slower startup. In a logging network where all devices are powered on at the same time, the Capture Modules would be up and running, capturing traffic, before the logging sinks can receive this traffic, resulting in data loss. To avoid this, Capture Modules are provided with an internal memory that buffers the startup traffic until the logging sink connected to the device is available and ready for logging the captured data.

The user can specify one or several conditions that the Capture Module will use to identify whether the logging sink is available or not, as described in CHAPTER 5.4.4.8. Once the selected conditions are met, all the buffered traffic is sent out to the logging sink.

All traffic captured from then onwards is sent out *live* to the logging sink, without being buffered internally.

If TECMP encapsulation is not activated in the CM 1000 High, all traffic buffered during the startup phase will be received on the logging sink with almost the same timestamp, matching the time where the device's internal buffer is emptied.

3.2.6 Filtering

Capture Modules offer the possibility to apply filters on the captured data before sending it out for logging. This way, the data can be pre-processed and sorted directly in the device, simplifying the later processing of the generated traces.

Filtering rules can be configured for each IVN port. A filtering rule is nothing but an action that will be applied to the captured frames whenever a specified condition is met.

All captured frames will go through all the rules and, when a frame matches with a filtering rule, the corresponding action will be applied to that frame.
Two types of filters are available on CM 1000 High, based on their scope of action:

- Basic filters are applied right after the data is captured, and before any further processing.
- Advanced filters are applied after taking the data out of the device's internal memory.

Currently, the following actions are offered as part of the basic filtering:

- Drop: this action can be used to drop frames matching a certain condition to not be captured, discarding unneeded data, and reducing the amount of data to be processed by the device. Frames are dropped right after being captured, not taking memory space, or requiring any further processing within the device.

And these actions are available in case of advanced filtering:

- Log without Message Header: with this action, the captured frames matching the filtering condition will be sent out without TECMP encapsulation. All other frames, not matching the filtering condition, will be encapsulated in TECMP frames.
- Drop: this action can be used to drop certain frames to not be captured, reducing the amount of data to be processed later by the logging sink(s). Opposite to the *drop* option from the basic filtering, all frames captured will take up memory space inside the device before being dropped.
- Mark: with this action, the user can mark frames with a so-called SPY flag (included in the TECMP header) to identify them later in the toolchain for further processing.
- Change Interface ID: this action gives the user the possibility to modify the original interface ID, which is configured in the incoming port, to separate the traffic in streams based on certain conditions. This can be used to log together all DLT traffic, traffic coming from port X (independently of the source address), etc.
- Control Message: this filtering action can be used to configure the sending of a control message whenever a certain condition is met. To control the recording system and the devices connected in the logging network (consisting of several loggers, Capture Modules, and instrumentation adapters) and signaling events in the recorded trace, the so-called *control messages* can be used. For more information about control messages, please check the TECMP specification.

- Multiple Log Interfaces: this action allows the user to define, for specific captured data, which output ports to use as logging port(s).

The use of filtering rules adds a high level of control over the captured traffic, simplifying and reducing a lot the post-processing of the data on the logging sinks.

3.2.7 Traffic shaping

Traffic shaping, a bandwidth management technique, is implemented in all Capture Modules variants, making it possible to control the way the captured data is transferred to the logging sinks. Thanks to this functionality, the CM 1000 High can process all incoming data, even when unexpected peaks or sporadic bursts are detected on the line.

When traffic shaping is activated, the output traffic sent by the CM 1000 High to the logging sink through the logging port(s) might be delayed, if necessary, to bring it into compliance with the desired traffic profile, configurable via device settings, specifying the maximum bandwidth to be used.

The higher the overall bandwidth of the captured data, the more important the use of traffic shaping is. Let's think of the worst-case scenario in the case of a CM 1000 High:

- All 12 1000BASE-T1 input channels of the device in use.
- Maximum bandwidth (i.e., 1000 Mbps) on all input channels.
- Only the SFP+ 10Gbps logging port is configured.
- All traffic captured redirected to that logging port.

In this case, we would be dealing with a situation where a theoretical maximum of 12 Gbps could be captured and forwarded to one port that allows a maximum of 10Gbps. That would lead inevitably to a traffic bottleneck and subsequent packet loss. Moreover, the traffic distribution will not be always constant, presenting peaks or small bursts that need to be handled.

With traffic shaping, the CM 1000 High can be configured to send out the captured traffic at a slower pace, avoiding exceeding the 10 Gbps of maximum bandwidth available in the logging port. To be able to do that, the captured traffic is buffered internally and sent out little by little according to the configured traffic bandwidth. This is especially important in the case of bursts, which need to be handled to add an interframe gap that allows the logging sink to process the data successfully.

As the traffic shaping shall be usable independently from the logging sink, the percentage applied here is calculated on the link speed of one downstream link to the

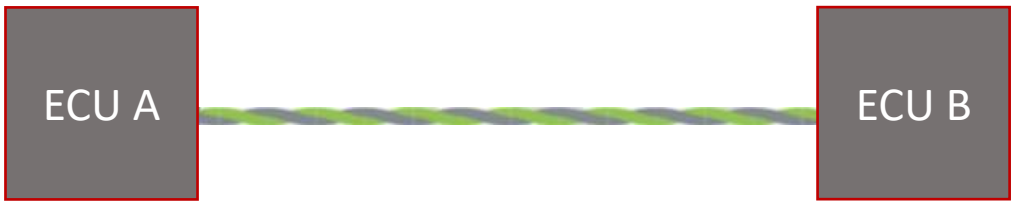
data logger. This results in the same overall output bandwidth towards the logging device, no matter how many output ports are defined.

Traffic shaping is based on the internal buffering of the data. Since memory is not an endless resource, please keep this in mind when configuring this feature. The more restrictive the settings (i.e., maximum bandwidth) are, the more memory is required, and the higher the risk of getting out of memory is.

Hardware limitations of the logging sinks, especially in the case of computers, are out of the scope of Capture Module performance and must be taken always into consideration.

3.2.8 Transparency mode

Automotive Ethernet is a point-to-point communication technology with a master/slave architecture:



When a Capture Module like CM 1000 High is placed between two ECUs, the original communication channel is split into two physical segments:

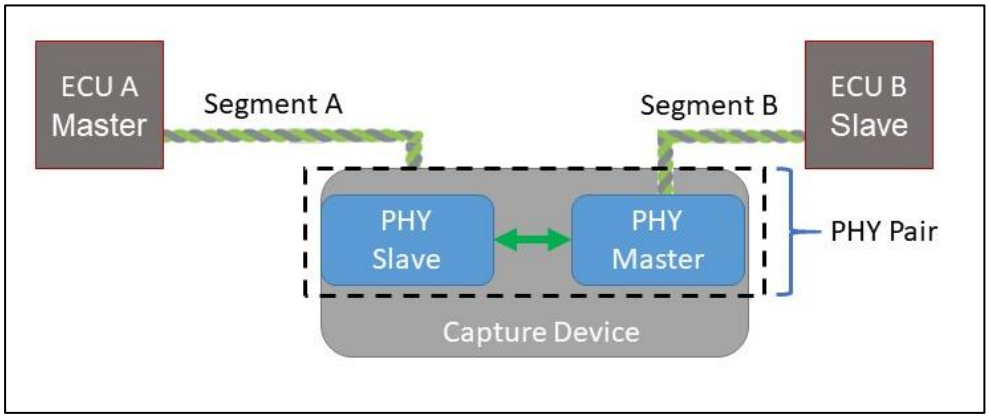


Figure 3-4 ECU connections with the capture device

This way, ECU A is not directly connected to ECU B and therefore, the ECUs are not capable of seeing each other's state. This can lead to data loss in the communication between the two ECUs in the case one of them goes to sleep mode, is turned off, or suffers link quality issues.

With transparency mode, the Capture Modules offer a mechanism to minimize the impact on the communication between 2 ECUs in such scenarios, taking into consideration the link state of both link partners.

Transparency mode replicates the status of ECU A in segment B and the status of ECU B in segment A. This way, communication is stopped on both sides as soon as one of the communication partners is not there and resumes it as soon as both are available again. See CHAPTER 5.4.3 for details about how to configure transparency mode for the 1000BASE-T1 ports.

3.2.9 Traffic injection

The main task of a Capture Module is to gather the traffic sent by the different DUTs connected to the device and distribute it into the logging network. This is a unidirectional communication in the direction of the logging sink. In certain scenarios though, it is necessary to send data to the DUT to activate debug options or simulate other network nodes. The CM 1000 High provides this functionality under the name of *traffic injection*.

This feature offers the user the possibility to send traffic in the opposite direction, from the logging sink (a computer or a data logger, with or without other devices in between) to an ECU through the CM 1000 High, without affecting the in-vehicle communication.

As mentioned already in this document, a Capture Module is placed always between two ECUs, splitting the original communication channel into two physical segments. The Capture Module must not interfere in the communication between the two ECUs. To make it possible to inject traffic in a transparent way, taking advantage of the idle time in the communication between the ECUs. This is possible thanks to the so-called injection window, which must be specified in the device configuration.

When the injection is enabled for a certain ECU connected to one of the 1000BASE-T1 channels of the CM 1000 High, the Capture Module will receive the data to inject, buffer it temporarily and look for a gap in the communication between that ECU and the one connected to the same link pair; a gap that is at least as big as the injection window configured. As soon as this gap is found, data will be injected.

With this mechanism, data can be injected without affecting the data exchange between the ECUs. It also guarantees a constant latency in each link pair, the result of the PHY latency plus the configured injection window:

Time in a pair = 5 µs + injection window * (0,08 µs or 0,008µs)

where

5 µs is the fixed latency of the PHY.
Injection window the time elapsed for the configured delay (in bytes).
0,08 µs is the time required in 100BASE-T1 to transmit one byte.
0,008 µs is the time required in 1000BASE-T1 to transmit one byte.

Currently, traffic injection can be done based on:

Layer 2 destination MAC address
Layer 3 destination IP address (IPv4)

Injection functionality is optimized for TCP traffic thanks to the retransmission mechanism that this protocol offers. This guarantees a retry whenever the traffic window is not found.

Details about how to configure the CM 1000 High for traffic injection can be found in CHAPTER 5.4.2.6

3.2.10 Status messages

For debug purposes, the Capture Modules can be configured to generate cyclically status messages that provide information about the device, its configuration, and the logging capability.

Three kinds of status messages are available:

1. Status Capture Module: With general information that helps identify the device, like HW variant, CM ID, or device IP address. Sent every second.
2. Status Configuration: Containing the whole configuration of the device in binary format. Configuration can be extracted using the offline configuration tool provided together with the device. Sent every 60 seconds.
3. Status Bus/IVN: Providing statistics about the captured data on each of the IVN channels. In the case of a CM 1000 High, stats about its 12 100BASE-T1 channels are included. Sent every second.

The generation of status messages can be enabled or disabled via device configuration.

Details of the layout and content of the status messages can be found in the TECMP specification document. An extract with the basic information related to status messages can be found at the end of this document for quick reference (see CHAPTER 8.1.4).

3.3 Technical data

Voltage requirement:	6.5 to 30 Volt DC (typical 12 Volt DC)
Power consumption in operation mode:	25 to 36 Watt
Power consumption in sleep mode:	Less than 0.01 Watt
Size:	186 x 130.4 x 32.5 mm
Weight:	0.8 kg
International protection:	IP 2 0
Operating temperature:	-40°Celsius to 75°Celsius
Storage temperature:	-40°Celsius to 50°Celsius
Maximum cable length (1000BASE-T1 segments):	15 m

Operating temperature is the guaranteed range of temperature under which all the device features are available and working. Outside this range, the user operates the device at their own risk.

An automatic shut-off of the device under too high a temperature is hardware-driven when the possibility of damage is detected, independently of the operating temperature range.

The temperature conditions where the device can be placed while not in use should not exceed 50° for the device to have consistent performance for longer periods of time.

3.4 Scope of Delivery

The delivery includes:

- 1x CM 1000 High

Cable sets can be ordered separately:

- 1x Cable set
- 1m Ethernet Cable
- Connectors for all interfaces
- Cabling for 1000BASE-T1 already crimped
  - Separate crimp contacts
  - Bunch plugs

4 Hardware

The CM 1000 High has been designed for multiple use-cases. Therefore, several hardware versions of CM 1000 High have been developed so far. Differences between the versions are collected in the table below:

HW variant	Properties
V5.0 (MATEnet version)	↻ Voltage requirement 6.5-18 Volt DC (typical 12 Volt DC) ↻ Operating temperature: -40°Celsius to 75°Celsius ↻ Changes in this version: ○ Less power consumption in sleep and standby mode
V5.1	↻ H-MTD and 24volts support
V5.3	↻ Minor improvements
V5.4 (H-MTD and MATEnet)	↻ Improved reliability over the input power range 6,5V to 30V (typ. 12V) ↻ Improvements to optimize product manufacturing and production process ↻ Minor changes

Please check the label at the bottom of your device to know what version of the hardware you have and pay attention to its characteristics and limitations for proper use.

Note that hardware >= v5.0 requires at least firmware v003.005.020. Older firmware versions are not compatible. For older hardware versions, no firmware restrictions apply.

4.1 Connectors and controls

The CM 1000 High is equipped with several connectors and control elements that are distributed between the front and back sides of the device.

On the front side (FIGURE 4-1) of the CM 1000 High, we can find:

- ↻ 1x Power connector
  - Including 1x 100BASE-T1 port

- 3x RJ45 Ethernet connectors
- 1x SFP+ interface
- 1x Rotary switch
- Several LEDs

The back side (FIGURE 3-2) of the device is reserved for the interfaces used to capture traffic from the in-vehicle network (IVN). In the case of a CM 1000 High, this means:

- 6x 1000BASE-T1 link lines (= 12x 1000BASE-T1 ports)



Figure 4-1 Capture Module 1000 High Front side



Figure 4-2 Capture Module 1000 High Back side

4.1.1 Power connector (1)

A black MQS connector on the front side (10 pins, one row) is used to supply the power to the CM 1000 High



Figure 4-3 Front connectors

- Power supply for the device is supplied by pin 1 (12 Volt) and pin 2 (Ground)
- Requirements for the device itself: 12 Volt DC up to 1 ampere (typically 600 mA).

Next to the pins for the power supply, the pins corresponding to two independent wake-up lines can be found, identified with the letter “W” in the device housing. For each of them:

Hardware up to v5.3:

- As a digital input, it is used to wake our device or stay awake. The wake-in voltage range is from 6 V to 30 V and the line consumes approximately 0.7 mA for the range of 6 V to 15 V and 16.5 mA for the range of 15 V to 30 V.
- As digital output, it is used to wake up external devices or keep them awake. When using the wake output feature, the CM Eth Combo device input supply (VBAT) range is limited from 12 V to 16 V. For this VBAT range, the wake-out line is able to supply a maximum output current of 25 mA and a minimum output voltage of 6V.

Hardware up to v5.4:

- As a digital input, it is used to wake our device or stay awake. The wake-in voltage range is from 6 to 30V and the line consumes approximately 0.7 mA for the range of 6 to 15V and 16.5 mA for the range of 15 to 30V.
- As a digital output, it is used to wake up external devices or keep them awake. When using the wake output feature, the CM 1000 High device input supply (VBAT) range is limited from 9 to 16V. The wake-out line is able to supply a maximum output current of 25 mA for the CM input supply range of 9 to 12V or up to 60 mA for CM input supply range of 12 to 16V. For the CM input supply range of 9 to 16V the minimum output voltage at the wake-out line is 6V.

Apart from that, in this connector, we can find two pins corresponding to a 100BASE-T1 port. Through this 100BASE-T1 port, it is possible to get access to the web server used for device configuration (see CHAPTER 5) using the default IP address (192.168.0.49, if not changed).

A converter between 100BASE-T1 (automotive Ethernet) and 1000BASE-T (Gigabit Ethernet, RJ45) is required to connect directly to the device from a computer.

The connector Tyco Electronics Micro Quad Lock System (MQS) is used, as described in the table below:

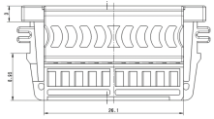
Name	Picture	Part Number
Tyco, MQS Socket Housing 10 Pol Black		1-929423-2
Tyco crimp contact		5-928999-1

Table 4-1: Parts of black MQS connector

The official Tyco tool can be used for these crimp contacts. A less expensive variant is the crimp tool for “PSK” contacts.

Pin	Function	Pin	Function
1	Battery +12 Volt Input	6	n.c.
2	Power Ground	7	n.c.
3	Wake-up 1	8	n.c.
4	Wake-up 2	9	100BASE-T1 Plus
5	n.c.	10	100BASE-T1 Minus

Table 4-2: Pinning of black MQS connector

4.1.2 1000BASE-T (RJ45) connectors (2)

There are three RJ45 Standard Ethernet connectors for Gigabit Ethernet on the front side, labeled as GB-A to GB-C, that can be used for configuration of the device and logging.

These ports are used as well for time synchronization between different IVN2Eth Capture Modules of Technica Engineering GmbH.

As explained later in the chapter related to configuration, each port can take several roles or functions:

- Configuration port
- Logging port
- Injection port
- Cascading port
- Status message port
- Control message port

Each port can have several functions. Out-of-the-box, GB-A port is set as configuration and status message port, while no functionality is configured for the rest of the ports.

4.1.3 SFP+ slot (8)

There is one SFP+ slot that can facilitate 10Gbps SFPs for configuration of the device and logging.

This port is used as well for time synchronization between different IVN2Eth Capture Modules of Technica Engineering GmbH.

As explained later in the chapter related to configuration, each port can take several roles or functions:

- Logging port
- Synchronization
- Status message port
- Control message port

Each port can have several functions. Out-of-the-box, GB-A port is set as configuration and status message port, while no functionality is configured for the rest of the ports.

Webpage access (for configuration) is via SFP+ port not possible

To detect a linkup for SFP+ module, Capture Module needs at least 600µs

4.1.4 IVN connectors (3)

The CM 1000 High is a device conceived to capture the traffic exchanged between Ethernet ECUs that communicate over 1000BASE-T1 automotive Ethernet.

With this goal in mind, the device has been provided with 12 1000BASE-T1 ports, distributed in green H-MTD connectors placed on the backside of the device (with 12 pins divided into 2 rows).

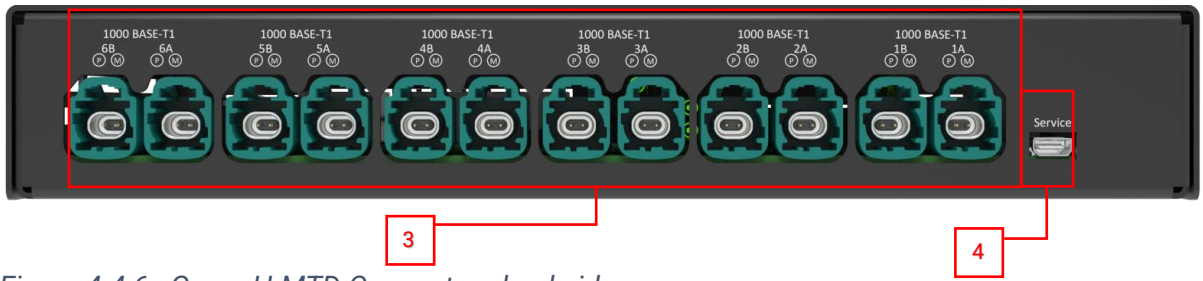


Figure 4-4 6x Green H-MTD Connectors backside

For cables connected correctly, as far as the polarity is concerned, the user must

- connect the (P) pin of the ECU to the (P) pin of the Capture Module.
- connect the (M) pin of the ECU to the (M) pin of the Capture Module.

where (P) stands for Plus/Positive (+) and (M) stands for Minus/Negative (-).

If the connections to positive and negative pins are interchanged, the corresponding LED of the 1000BASE-T1 port (1A, 1B, ..) may be lit indicating a link up but no data transfer will be possible due to incorrect polarity.

Name	Picture	Part Number
Rosenberger H-MTD Connector		E6K10A-1CAZ5-Z

Table 3: Parts of Rosenberger H-MTD Connector

Pinning:



Figure 4-5 H-MTD Connector

You can use the official Tyco tool for these crimp contacts. A less expensive variant is the crimp tool for PSK contacts.

Pin	Function	Pin	Function
1	1000BASE-T1 1A Minus	2	1000BASE-T1 1A Plus
3	1000BASE-T1 1B Minus	4	1000BASE-T1 1B Plus
5	1000BASE-T1 2A Minus	6	1000BASE-T1 2A Plus
7	1000BASE-T1 2B Minus	8	1000BASE-T1 2B Plus
9	1000BASE-T1 3A Minus	10	1000BASE-T1 3A Plus
11	1000BASE-T1 3B Minus	12	1000BASE-T1 3B Plus
13	1000BASE-T1 4A Minus	14	1000BASE-T1 4A Plus
15	1000BASE-T1 4B Minus	16	1000BASE-T1 4B Plus
17	1000BASE-T1 5A Minus	18	1000BASE-T1 5A Plus
19	1000BASE-T1 5B Minus	20	1000BASE-T1 5B Plus
21	1000BASE-T1 6A Minus	22	1000BASE-T1 6A Plus
23	1000BASE-T1 6B Minus	24	1000BASE-T1 6B Plus

Table 4-9: Pinning of Tyco H-MTD connector

4.1.5 Debug connector (4)

The CM 1000 High is provided with an interface for programming and debugging, of type micro-HDMI, whose use is reserved only for development.

4.1.6 Rotary switch (5)

A rotary switch is located on the left-most part of the front side of the CM 1000 High. This element allows the configuration of the IP address of the device associated with the GB ports (i.e., the 1000BASE-T ports, with RJ45 connector). If any of them is set as configuration ports. This is the IP address that will be used to access the web server for device configuration.

Each position of the rotary switch is associated with a predefined IP address, which will be the one set for the configuration port and thus, the one to use to access the web server.

4.1.7 Status LEDs (6 & 7)

The CM 1000 High has several status LEDs on the front side of the case.

4.1.7.1 System LEDs (6):

Information regarding the internal components of the device.

- LED 1: Called "100BT1" LED, it shows the linkup state of the 100BASE-T1 port connected directly to the microcontroller.
- LED 2: The LED labeled as "FPGA" shows the state of the FPGA, in charge of the processing of the captured data. It blinks with a frequency of approx. 2 sec during normal operation.
  - If it is not blinking or blinking faster, FPGA is damaged and must be repaired.
- LED 3: Named "Host" LED, it can blink at two different speeds:
  - Slow toggle (approx. 1 sec) during normal operation.
  - Fast toggle (approx. 0.2 sec) if the device is in Bootloader mode.

4.1.7.2 Link LEDs (7):

Information regarding the status of the devices connected to each 1000BASE-T1 port labeled 1A to 6B - on the vehicle side:

- LED permanently on indicates there is LinkUp in the corresponding port.
- LED blinking means communication on the corresponding port.

5 Configuration

The CM 1000 High is a highly configurable device, offering the user great flexibility to include it in many different scenarios.

This section describes the different ways of configuring the device. The user can configure the device through the following three methods:

- Online, connected directly to the device
- Offline, via a standalone configuration tool
- Remotely, using UDP configuration messages

For online configuration, the CM 1000 High is provided with a web server that allows the user to check and modify the device settings.

Out-of-the-box, access to the web server is possible by connecting the device via the GB-A port (RJ-45) to a computer. In this case, the IP address of the device is dependent on the position of the rotary switch (see CHAPTER 4.1.6)

Other GB ports can also be used to access the webpage after making them as “configuration port” in the web server.

Only one GB port can be used at a time to access the webpage.

Alternatively, the web server can also be accessed through the 100BASE-T1 port connected to the microcontroller. The default IP address, in this case, is 192.168.0.49 and the port is configured as 100BASE-T1 slave per default.

To configure your computer interface for communication with the Capture Module (CM), it must be set within the same subnet as the CM. For example, if using a CAN Combo evo with the Rotary switch in position 0, the IP address of the device will be 10.104.3.64. Therefore, a valid IP address for the network interface of your PC would be 10.104.3.XXX, where XXX is a value that depends on the netmask applied.

Once connected to the device:

1. Please go to the network adapter settings and adjust the IP in the same range of your CM 1000 High, depending on which port you are connected to.
2. Enter the IP address in your standard web browser. For details see CHAPTER 4.1.6 and CHAPTER 5.2.

Google Chrome is the recommended browser.

Capture Module offers 10/100/1000BASE-T mode on the RJ-45 port. This must be used in conjunction with 1000BASE-T. Please make sure the network adapter connected to the RJ-45 port supports 1000-BASE-T mode and communication is established at 1Gbps.

It is recommended to disable the Ultra-low power mode in the network card settings, if the connection with the Capture Module cannot be established or if there are issues in linkup when connecting to the RJ-45 port

For offline configuration, Technica Engineering GmbH offers a standalone configuration tool (ANDi Free) that allows the user to prepare a configuration without being connected to any device. The configuration, once ready, can be exported to a configuration file that can be afterward imported into a Capture Module.

ANDi Free features the offline configuration tool. Detailed information about this offline configuration tool can be found on the website of ANDi Free. ANDi Free can be used for additional features like:

1. Trance analysis of Capture Modules
2. Use of TECMP (Extracting configuration from the traces)
3. Conversion of logging file formats (pcap, pcapng, vpcap, mf4, asc etc)
4. Offline configuration of Capture Modules
5. Flashing of Capture Modules
6. Clock controller

Offline Configuration Tool shares the GUI structure with the device website. Thus, from now on, all references to the configuration will be done based on the online configuration via the website.

Additionally, the so-called remote configuration is available, to offer flexibility for automation tasks. It is possible to configure some of the settings of a CM 1000 High by sending UDP messages containing configuration commands.

For details about which commands are supported and how the layout of the UDP configuration messages is, please check CHAPTER 0.

5.1 Home Screen

The following home screen will welcome you when accessing the device website:

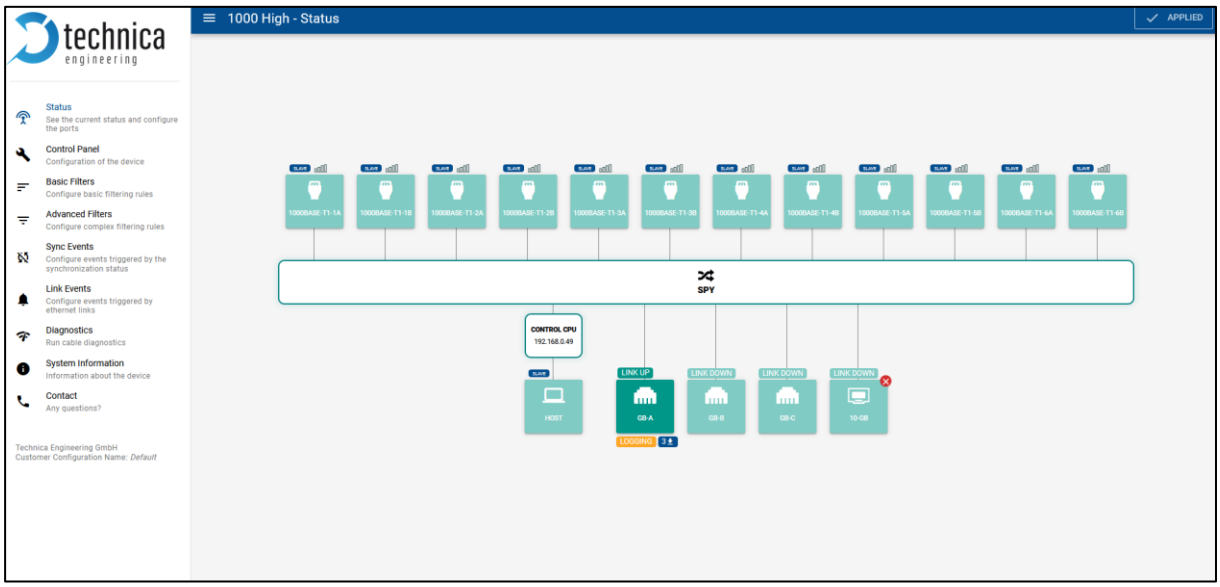


Figure 5-5-1: Home Screen

This home screen includes links to the different sections available, which allow the user to get information about the system and its configuration.

Please select one of the tabs for further configuration. Each tab is described in detail in the next sections of this document.

Any change in the configuration will lead to the following hints appearing at the top of the current web page:





This option sends data from the webpage to the memory of the device. There is no need to save every time this hint appears. All the desired changes can be done at once and the configuration can be saved at the end. To do that, please click "SAVE CONFIGURATION". Please note that the configuration will not be applied unless the device is restarted.



By clicking on this option, the user is prompted to introduce an optional name for the configuration file and to accept the changes. By selecting "OK" the configuration is saved similar to the option "SAVE CONFIGURATION". It also restarts the device, applying the saved configuration.

Note that without " SAVE & APPLY CONFIGURATION", the configuration of the device will not be fully updated. To be sure that all changes have been applied, a restart of the device is necessary.

5.2 System Information tab

On the *System Information* tab, the user has access to general device information that can be especially useful for debugging in case of faulty behavior. It is possible to find here, for example, the versions of application firmware and bootloader:

5.2.1 License Information

Serial number: Serial number that identifies the device unequivocally.

Maintenance date (DD-MM-YYYY): The date till when the license is valid. Till the expiry of this date, the user can update the firmware of the device to any firmware version released until the expiry date.
After expiry date, the user can either use the capture module with the same firmware version or any other firmware version that was released before the expiry date. If the user requires a newer firmware that was released after the expiry date, the user must purchase/renew a maintenance license to get newer updates.

Features: The list of all the features that are covered under the license are mentioned here. There can be several modes of the license.

1. Unlicensed: No license is available for this feature.



2. Temporal: The license is granted for the limited amount of time mentioned in the webpage.



3. Permanent: The license is granted permanently for this device



4. Expired: The license for the feature mentioned is expired and the feature cannot be used anymore



Currently Capture Module 1000 High has the following features that need a license.

100/1000BASE-T1 Mode

The variant of Capture Module 1000 High that operates in 100BASE-T1 mode can only be upgraded through a license such that the device can be used in both 100BASE-T1 and 1000BASE-T1 mode.

5.2.2 Firmware Information:

System Update: This option is used to update the firmware on the device. The process of updating a firmware/license is explained in CHAPTER 7.

⚠ When clicked on this option, the device goes into bootloader mode and loses its normal operation. The bootloader mode can also be verified by the fast toggle of the status LED 3 which in normal mode is in slow toggle mode. If this option is selected by mistake, please select “go back to application mode” to exit the bootloader mode.

MAX10 version: Additional FPGA needed for providing extended functionalities. The current version is shown here.

It is possible from Max10 version 4.0 onwards, to update the version via normal FW update process. To update to a newer version after version 4.0, at least version 4.0 must be already on the device. With an older version than version 4.0 the update via bootloader webpage will not work.

FPGA version: It specifies the version used by the FPGA module.

Application software version: Current firmware version installed on the microcontroller of the device.

Application build hash and date: This shows the internal reference to the repository build with the date.

Bootloader version: Current bootloader version of the microcontroller is used in the device.

5.2.3 Device Information

Rotary switch: This field indicates the current position of the rotary switch. By default, the rotary switch is on position 0.

With the rotary switch, it is possible to select an IP address to access the device. This is the so-called HW IP address, a shortcut for *HW-determined IP addresses*.

For each hardware variant of Capture Modules, a different IP address range has been defined. In the case of a CM 1000 High, the following IP addresses will be set:

Rotary switch on 0 -> IP: 10.104.3.144
Rotary switch on 1 -> IP: 10.104.3.145
Rotary switch on 2 -> IP: 10.104.3.146
...
Rotary switch on E -> IP: 10.104.3.158
Rotary switch on F -> IP: 10.104.3.159 / Reset

Rotary switch in position 0 to D

If the rotary switch is set to a position from *0* to *D*, the IP address of the device is not configurable via software and can only be changed using the rotary switch. The IP address will be configured according to the table above.

When changing the rotary position to any of these positions, the corresponding IP address is set after 500ms, after an automatic software reset. The IP address currently in use is shown in the *System Information* tab on the device website.

This IP address will determine the ID used in the TECMP header to identify unambiguously the messages generated by the device.

Access to the configuration website of the CM 1000 High is given through the GB port or the 100BASE-T1 Host port located on the power connector.

Rotary switch in position E

In this position the IP address is configurable. This configuration can be done either via the website, in the *System Information* tab, or using a UDP configuration message (see CHAPTER 5 for details).

When switching to this position, the IP address is set by default to 10.104.3.158, and an automatic soft reset is performed to make this change effective. The configured IP address is persistent, i.e., it remains even after a power reset.

This IP address will determine the ID used in the TECMP header to identify unambiguously the messages generated by the device.

The HW IP address can be configured using UDP configuration messages anytime, independently of the rotary switch position, but changes will only take effect when the rotary switch is switched to position E.

Rotary switch in position F

Position F offers the user the possibility to perform a factory reset of the device. To do this, the rotary switch must be changed to position F, followed by a power reset performed by the user. The **complete configuration** will be lost and set to default, including the IP addresses:

- Controller IP address: 192.168.0.49
- HW IP address: 10.104.3.159

Hardware version: Version of the hardware. Please check CHAPTER 4 for more details.

MAC address: This shows the physical address of the device

Controller IP address: IP address currently configured for the 100BASE-T1 config port located on the front side of the device.

HW IP address: IP address currently configured for the RJ45 port configured as configuration port.

As explained in CHAPTER 4.1, the CM 1000 High is provided with two different types of Ethernet interfaces used to connect the device with other test equipment, including a computer:

- 1x 100BASE-T1 interface, connected directly to the microcontroller hosting the website for device configuration

- 3x 1000BASE-T interfaces, connected to the FPGA of the device in charge of the actual logging capabilities.

These interfaces can be used to access the device for configuration if properly configured (check CHAPTER 4.1 for details):

- Access via the 100BASE-T1 port of the power connector at the front side is always possible.
  - The IP address for access to the webserver is the so-called controller IP address, which can be configured in the *System Information* tab.
    - To change the controller IP address, click ✎, type the new address in the field, and press Enter or move the cursor to another field. After that, change can be saved.
    - The Default IP address is 192.168.0.49
- Access via 1000BASE-T (also called GB port) is possible after configuring one of the three ports as a configuration port, being GB-A the default one.
  - The IP address for access to the webserver will be in this case the so-called HW IP address, defined by the position of the rotary switch.
    - By default, the rotary switch is set to position 0, being 10.104.3.144 the corresponding IP address.

The IP address can be changed in the field as shown in FIGURE 5-5-2:



Figure 5-5-2: IP addresses

The HW IP address can only be changed via a rotary switch, except if this is in position E. In that case, it is possible to configure it on the website, in a similar way for the controller IP address. The corresponding field, usually grayed out, will be editable in that rotary switch position.

The relationship between IP address and rotary switch is explained in more detail in the following subchapter, CHAPTER 4.1.

CM ID: A specific capture module ID can be assigned to the device to identify to which device is the traffic generation node connected. This value can be set either through the web page by adjusting the rotary position to E or by using the remote configuration method, which is explained in detail in subchapter, CHAPTER 0

5.2.4 Device PHYs Information

X Phy ID: The version number registers of the microcontroller and PHY chips of the device are displayed for information, where *X* is the descriptional name of the corresponding PHY. If these registers are displayed as “FFFFFFFF”, this element is defective.

5.3 Control Panel tab

In this tab, you can configure general settings that apply to the device itself and not to specific parts of it, plus configuration import/export options.

5.3.1 Power

Reset system: This button will perform a soft reset of the CM 1000 High. Any unsaved changes will be lost.

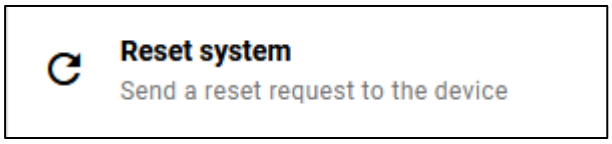


Figure 5-3 Reset device

5.3.2 System parameters

Date and Time: With this field, the time of the Capture Module can be set to be taken over as a reference for the timestamp in the TECMP protocol.

This setting can be useful in case the CM 1000 High is not connected to a time master, to have a timestamp close to real-time. For example, a standalone test bench with a gPTP Master. In this scenario, the timestamps are set based on the internal clock of the device and, if the *Date and Time* are not updated before making use of the device, timestamps will be calculated using the default date of 01.01.1970 00:00:00 as the basis.

To update the date and time of your device, select:

To update the date: When is selected, CM is set to the current date.

To update the time: When is selected, the CM is set to the current time.

Once the selection is made, click on APPLY

This field is neither included in a configuration export nor configurable offline.

Only positive values (values beyond 01.01.1970 00:00:00) are accepted. Any negative value would result in error. The user can either set positive values or operate the device with default value.

5.3.3 Configuration

With these options, it is possible to export/import the device configuration, as well as setting it back to the manufacturing state.

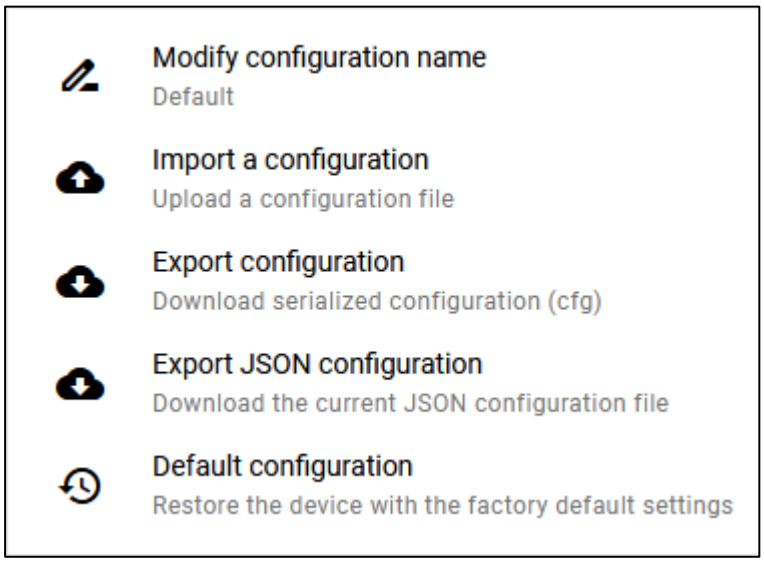


Figure 5-5-4: Handling of configuration files

The following actions are available:

Modify configuration name: The device configuration always receives a name, which will be used in case the configuration is exported, to make it easier for the user to identify it. This name is set by default to “Default” but can be changed anytime. Just replace “Default” with the name of your choice, considering the naming rules:

- Allowed characters are spaces, numbers, and the following symbols: + - _ ; : .

After saving, the new name is applied.

Import Configuration: This option allows the user to import a configuration created with another device or with the *CM Offline Configuration Tool*, loading the configuration

from a computer connected directly to the CM 1000 High. Follow these steps to import a configuration. The configuration file can either be in .cfg or .json format.

- Select the option “Import a configuration”.
- Choose the configuration file.
- Click the button  to apply the configuration to the device.
- Restart the device

Export configuration: Changes performed in the device configuration can be exported by selecting the “Export configuration” or “Export JSON configuration” options.

- Export configuration: This option can be used to export the configuration of the device in .cfg format. Configuration files with extension *.cfg correspond to a proprietary format that can be handled only by Technica Engineering’s *CM Offline Configuration Tool* if no Capture Module is available. Exporting the CM configuration as *.cfg file generates always only one file, containing the whole device configuration.
- Export JSON configuration: The *.json file can be opened and edited with a standard text editor. This option can be used to export the configuration of the device in .json format.

Default configuration: This option resets the CM 1000 High to the default configuration, except for the IP address settings (both controller and HW IP addresses).

5.3.4 Sleep parameters

The CM 1000 High can work in two different operation modes:

Normal operation mode: The CM 1000 High works at full capacity.

- Logging mode: Logging functionality is available. The device is permanently capturing traffic on the configured IVN interfaces via the 1000BASE-T1 ports and forwarding it after initial preprocessing to the logging network. This is the default mode of any Capture Module. Web configuration is available anytime in this mode.
- (Remote) configuration mode: Logging functionality is temporarily stopped. The device can be remotely configured via UDP configuration messages, described in detail in CHAPTER 0. The activation/deactivation of the configuration mode takes place also via UDP message, to be sent from the logging endpoint, which is described as well in CHAPTER 0. After deactivation of the configuration mode, the Capture Module goes back to logging mode automatically. Web configuration (via configuration port) is not possible in this mode.
- Sleep mode: The CM 1000 High works at full capacity if there is communication on any of the ports marked as the KEEP-AWAKE REASON CHANNEL (see CHAPTER 5.4.3). If there is no communication for a defined time in the field SLEEP TIMEOUT (IN SECONDS), the CM

1000 High goes into sleep mode. The consumption is reduced to zero and only the wake-up lines in the power connector at the front side can wake up the CM 1000 High.

The ports marked as the *keep-awake reason channel* are monitored by the Capture Module to detect traffic or the absence of it.

Sleep Mode: If required, it is possible to configure the device to enter the sleep mode using this option. By default, this option is not enabled.

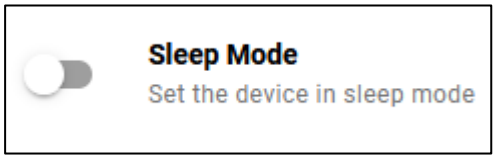


Figure 5-5 Prevent sleep

Not selected: The CM 1000 High works at full capacity and never enters the sleep mode

Selected: The CM 1000 High works at full capacity if there is communication on any of the monitored ports. If there is no communication for a defined time in the field SLEEP TIMEOUT (IN SECONDS) the CM 1000 High goes into sleep mode.

5.3.4.1 Keep-alive conditions

All 1000BASE-T1 ports of the CM 1000 High - host included - can be monitored to detect traffic, to determine whether it is necessary to keep the device in normal operation, capturing data, or not.

If there is communication on any of the monitored 1000BASE-T1 ports or a high level on one of the wake-up lines of the device is detected, the CM 1000 High will remain in a normal operation state.

As soon as all keep-alive reasons disappear and no wake-up line is activated, the device switches from normal operation to sleep mode, reducing the power consumption as described above.

As described in CHAPTER 4.1.1, there are two wake-up lines, which can be used anytime to wake up the CM 1000 High in case it is configured to allow sleep mode, and the device enters this state.

5.3.4.2 Wake-up conditions

If the device is in sleep mode, it can only be woken up by activating one of the two wake-up lines.

A power reset will not wake up the device from sleep mode if it is off for less than 15 seconds. Only the wake-up lines will have an immediate effect if activated below those 15 seconds.

5.3.5 Wake-up lines and actuator

There are two wake-up lines, referred to as Wake-Up line 1 actuator and Wake-Up line 2 actuator. Both wake-up lines can work as input and output, speaking from the point of view of the device:

- Input (a.k.a. WUP source): Activation of the wake-up line from the outside will wake up the CM 1000 High.
- Output (a.k.a. WUP actuator): The CM 1000 High can activate the wake-up line to keep other devices connected to it awake or wake them up if they were in sleep mode.

Using the wake-up lines as input lines is always possible and cannot be deactivated. Using them as output lines is configurable and can be activated/deactivated independently for each line using the setting *Wake-Up Line X actuator*, as described below:

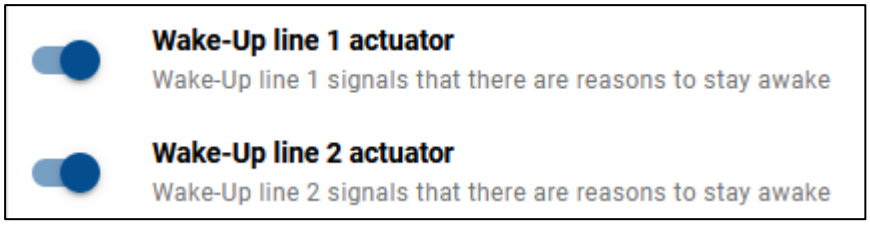


Figure 5-5-6: Wake-up lines

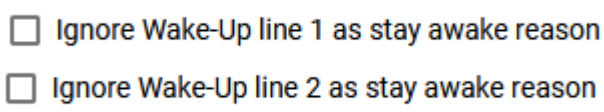


Wake-Up line X actuator selected: Wake-Up line X actuator is used also as output line and the corresponding pin (WUP*X*) provides the same voltage (vs. GND) as the supply voltage of the Capture Module if the device is awake.



Wake-Up line X actuator not selected: Wake-Up line X actuator is only input and its activation from the outside is used only to wake up the device if it was previously in sleep mode.

By default, both wake-up lines are enabled as the actuator. The user can also ignore wake-up lines acting as stay awake reason by checking the following box.



In case a wake-up line is configured as actuator - thus, it can work both as input and as output line – and the line should be at the same time controlled in both directions, in case of discrepancy (one tries to activate it, the other one to deactivate it), the line will be activated.

5.3.6 Wake-Up lines signaling timeout

This setting defines the time in seconds that must pass by before the wake-up lines if configured as actuators – according to *Wake-Up line X actuator* –, are deactivated after the reasons to keep the device awake have disappeared. This timer starts counting as soon as no wake-up and no keep-alive reason is present anymore.

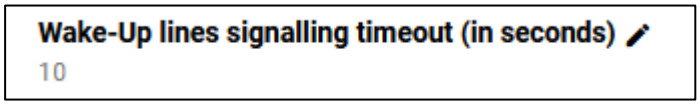


Figure 5-5-7: Wake up line signaling timeout

This setting only makes sense when at least one of the wake-up lines is configured as an actuator. In case none is configured, both lines should be already deactivated (since there is no wake-up reason present) and thus no timeout is required.

The default value for this setting is 10 seconds.

5.3.7 Sleep Timeout

This setting determines how much time must pass by with no communication on the monitored ports (i.e., ports configured as *keep-alive reason channel*, see CHAPTER 5.4.3) before the CM 1000 High enters the sleep mode.

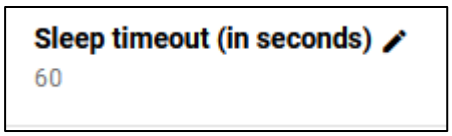


Figure 5-5-8: Sleep Timeout

Reasonable values for this setting are above 15s so that the user can get access to the webpage of the device.

If the field is left blank (i.e., not configured), the counter will start from power-up with default values.

5.4 Status tab

In this section of the website the user can find a graphical overview of the device, representing its main elements:

- Control CPU (microcontroller): Element that takes care of the configuration of the device, hosting the webserver.
- Host port: 100BASE-T1 interface directly connected internally to the microcontroller, granting access to the configuration website hosted there. For configuration and flashing purposes only.
- Spy (FPGA): Central component of the device, in charge of the capturing and processing of the data.
- Logging ports: SFP+ and 1000BASE-T interfaces used to connect the CM 1000 High with other testing equipment, including computers.
- IVN ports: 1000BASE-T1 interfaces used to connect the CM 1000 High with the different DUTs whose traffic needs to be captured for analysis.

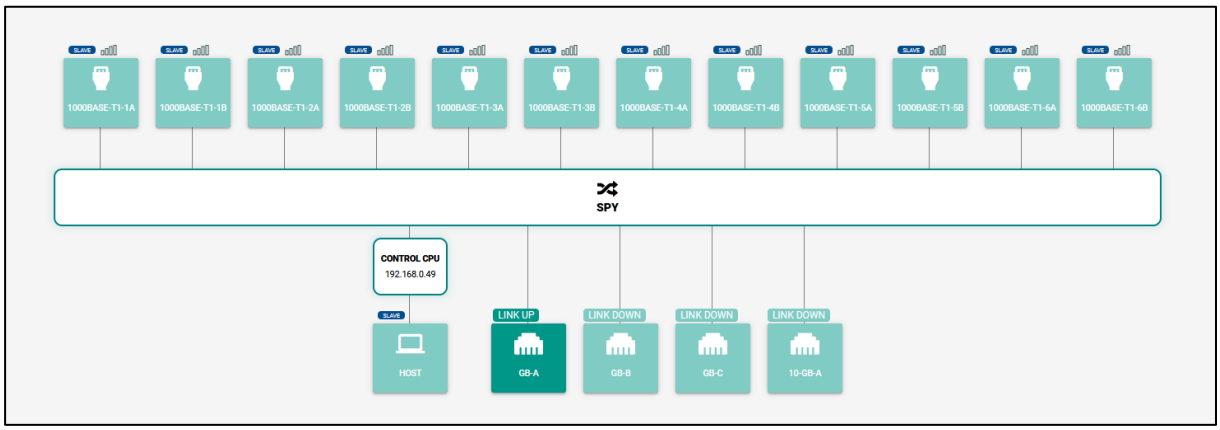
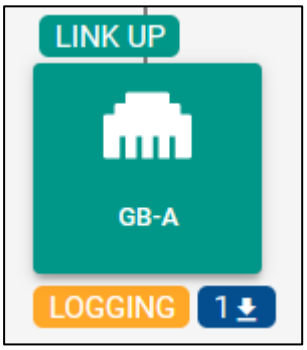


Figure 5-5-9: Status tab

Each of the elements offers several configuration possibilities to the user, as shown later in this chapter.

Color Indicators

Capture Module Ethernet ports (IVN as well as output ports) have color indicators in the web interface regarding:



- Link Up is indicated with an indication in solid green
- Logging output – only relevant for output ports – is marked with an orange bar. Each port where the option *Output to* is used to redirect the traffic received on 1000BASE-T1 ports will be marked this way. The number in solid blue in the bottom right specifies the number of ports that shall be captured on this Output port.

Figure 5-11 Port color indicators

5.4.1 Host Port

This is the 100BASE-T1 port located in the MQS connector on the front side of the CM 1000 High. It is directly connected to the microcontroller that hosts the webserver used for device configuration (hence the name).

Used as a fallback solution for flashing in case of the usual procedure via GB port is not possible.



Figure 5-5-12: Host Port

This port can be used for updating the firmware of the device as well, as described in CHAPTER 7.

The following configuration settings are available for this port:

5.4.1.1 Port name

This field lets the user specify a label to identify the port

5.4.1.2 100BASE-T1 mode

Automotive Ethernet requires one of the communication endpoints to act as a master and the other as a slave. This setting specifies the BASE-T1 operation mode the Capture Module will use on the host port:

- Master
- Slave

The configuration to choose will be determined by the device connected to the Capture Module on this port. Communication will only take place if each communication endpoint takes one of the roles.

5.4.1.3 Keep awake reason channel

The host port can be taken into consideration to decide whether the CM 1000 High can enter the sleep mode (if the device is configured for that). For that, the port will be monitored to detect incoming traffic.



Keep awake reason channel selected: Port is monitored. If traffic is detected on this port, the device must stay in full operation and cannot enter sleep mode.



Keep awake reason channel not selected: Port is not monitored. The device can enter sleep mode even if there is traffic on this port

5.4.2 GB Ports

The CM 1000 High is equipped with three GB ports. These ports implement the 1000BASE-T standard and can be used as the main interface to interact with the device and connect it to the logging network.

They are used mainly for logging and time synchronization.

The user can find in the *status tab* a graphical view of the device where the three GB ports are located on the front of the device, analog to their location in the physical device.

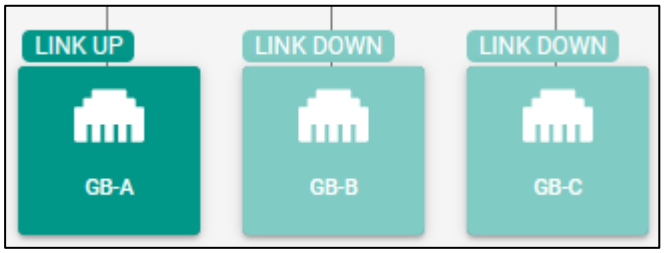


Figure 5-5-13: GB ports

5.4.2.1 Port name

This field lets the user specify a label to identify the port unequivocally. By default, the name of each port is of the form GB-*X*, where

- *X* can be A/B/C for port 1/2/3 respectively.

5.4.2.2 Detected Speed

This read-only field provides information about the detected speed in the link between the Capture Module and whichever device is connected to this port. Options are:

- No link
- 100 Mbps
- 1000 Mbps

5.4.2.3 Keep awake reason channel

The port can be taken into consideration to decide whether the CM 1000 High can enter the sleep mode (if the device is configured for that). For that, the port will be monitored to detect incoming traffic.



Keep awake reason channel selected: Port is monitored. If traffic is detected on this port, the device must stay in full operation and cannot enter sleep mode.



Keep awake reason channel not selected: Port is not monitored. The device can enter sleep mode even if there is traffic on this port.

5.4.2.4 Logging ports

This is a table that shows an overview of all 1000BASE-T1 ports used to connect to the IVN for capturing traffic, with an indication of whether each port is linked to the selected GB port or not.



1000BASE-T1-XX selected: 1000BASE-T1-XX sends its captured data to this GB port.



1000BASE-T1-XX not selected: 1000BASE-T1-XX does not send its captured information to this GB port.

5.4.2.5 Configuration port

This option lets the user select this port to access the webserver for device configuration.



Configuration port selected: The device can be configured by connecting a computer to this port and accessing from it the configuration website. Input control messages (sent from another device to the CM 1000 High) are only

recognized if sent to the configuration port: Incoming control messages on this port will be processed.



Configuration port not selected: Configuration website is not reachable from this port, thus not possible to use this port for configuration.
By default, the GB-A port is set as a configuration port. In case no port is selected as a configuration port, the GB-A port will be used as well.

See CHAPTER 5.3.2 for details about how to access the configuration website.

Only one port can be selected as a configuration port

5.4.2.6 Traffic injection port

Ethernet traffic can be injected into the in-vehicle network.



Traffic injection port selected: Traffic injection possible.



Traffic injection port not selected: Traffic injection not possible.

Both TCP and UDP traffic is supported. In the case of using TCP, this option will make it possible to establish a TCP connection with the devices connected to the IVN ports. Remember to also configure the IP for Injection on both sides of the communication.

For further information please see CHAPTER 6.2.

For UDP, please consider the limitations of this protocol: UDP neither offers windowing nor a function to ensure data is received in the same order as it was transmitted, and no error control is performed. By default, no port is configured for injection.

5.4.2.7 Status messages port

The CM 1000 High can send debug messages with information about the status of the device itself, its ports, and configuration. This can be configured by the user.



Status messages port selected: The status messages of the Capture Module are transmitted on this port



Status messages port not selected: The status messages of the Capture Module are not transmitted on this port

By default, the GB-A port is configured as a port for status messages. In case no port is selected as a status messages port, the GB-A port will be used as well.

Additionally, more than one GB port can be selected as an output for the status messages. In that case, the messages will be sent on both, with the same content, except for the TECMP counter (since this is specific for each output port).

Logging and status messages port can be different.

The temperature information in the status message is only sent by Capture Module hardware version 3.1 or greater

For details about status messages, please have a look at CHAPTER 8.1.4.

5.4.2.8 Control message port

The CM 1000 High can be configured to generate control messages when certain events take place.



Control message port selected: Control messages generated by the Capture Module are sent out to the logging network via this port.



Control message port not selected: Port is not used for sending control messages.

As of now, the generation of control messages can be configured in the LINK EVENTS TAB (link up/ link down events), SYNC EVENTS TAB (sync/lost sync events) and in the ADVANCED FILTERS TAB (events based on the incoming traffic). For more details about how to set filters for generating control messages, please check CHAPTER 5.7, CHAPTER 5.8 and CHAPTER 5.9

In the same way, as it is possible to select several GB ports as logging ports, with the option OUTPUT TO in the 1000BASE-T1 IVN interfaces, it is possible to select more than one GB port as a control message port.

Do not mix incoming and outgoing control messages. The CM 1000 High is capable of both processing incoming control messages and sending out control messages generated by itself. The control message port is only used for the CM 1000 High to send control messages, not to receive them. The reception of control messages from other devices is processed only through the configuration port. When control messages are sent to the Capture Module on ports other than the configuration port they will be ignored.

For details about control messages, please have a look at CHAPTER 8.1.3.

5.4.2.9 Cascade to

CM 1000 High, making use of the cascading functionality described in CHAPTER 3.2.3, can be connected to other Capture Modules to create a seamless network for capturing traffic from different interface types. The traffic collected with one device is forwarded to the next one in the cascading chain until it reaches the logging endpoint or *sink*.

With this option, it is possible to specify which port to forward:

- None: No forwarding of incoming traffic is done. Traffic captured by the next device in the cascading chain will then be discarded.
- GB-A: Forwarding of incoming traffic to GB-A.
- GB-B: Forwarding of incoming traffic to GB-B.
- GB-C: Forwarding of incoming traffic to GB-C.
- 10GB-A: Forwarding of incoming traffic to 10GB-A.

The port currently under configuration will not show up in the dropdown list with the ports to perform cascade to.

The option CASCADE TO can be used only once per device.

5.4.2.10 802.1AS Mode

The CM 1000 High offers the time master functionality for IEEE802.1AS. If decided to use the device as a time master in the logging network, then one and only one port can work as a time master.

In case IEEE 802.1AS is selected as time synchronization protocol in the system where the CM 1000 High will be used, then the user is requested to specify explicitly which of the ports will act as time master (if any) using this option to avoid possible misconfigurations.

Time synchronization between the CM 1000 High and the logging endpoint and/or other Capture Modules is possible by enabling this option in the SFP+ and GB port(s) where synchronization with the connected devices is required.

The SFP+ and GB ports can be configured as follows for time synchronization:

- Disabled: No time synchronization performed through this port.

- Slave: The CM 1000 High acts as an intermediate or end device that gets its time from another device, connected directly to it, which is acting as bridge master.
  - If there is already one port defined as SLAVE in the device, no other port can be configured as SLAVE, nor as 802.1AS MASTER.
  - Once Slave has selected the user has a choice to configure the port as:
    - gPTP Automotive slave: 802.1AS slave
    - PTPv2 Slave: IEEE 1588-2008 slave
- Master: The CM 1000 High acts as an intermediate device that gets its time from the grandmaster (directly or through other devices) and acts as a time master for the next device in the chain. It forwards the corrected time information to all the other attached time-aware systems. Sync / Follow up messages start to be sent only when a slave device is detected.
- Grandmaster: This port is generating the master clock used as a reference for the logging system. The device will be providing the time base for all directly attached time-aware systems.

Note that it is not possible to have two ports configured as grandmaster on the same Capture Module. If selected, nothing can be configured on the other SFP+/GB ports.

If no time synchronization is required, this setting should be set as *"Disabled"*. Otherwise, it is important to distinguish whether the synchronization will be done using gPTP or PTPv2 in the logging network.

By default, time synchronization is disabled on all ports.

Ports configured as Bridge Master or Bridge Slave can be cascaded as well.

5.4.3 SFP+ Port

The CM 1000 High is equipped with one SFP+ port. This port can be configured as 10G with a 10Gbps SFP and can connect to the logging network.

It is used mainly for logging and time synchronization.

The user can find in the switch status tab a graphical view of the device where the 10Gbps port is located on the right side of the device, analog to its location in the physical device. Clicking on the ports, a configuration page that looks as follows will be opened:

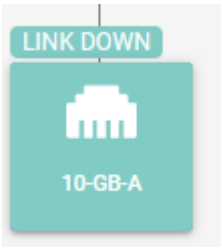


Figure 5-5-14: 10-GB-A port

5.4.3.1 Port name

This field lets the user specify a label to identify the port unequivocally. By default, the name is of the form 10-GB-*A*.

5.4.3.2 Detected Speed

This read-only field provides information about the detected speed in the link between the Capture Module and whichever device is connected to this port. Options are:

- No link
- 100 Mbps
- 1 Gbps
- 10 Gbps

5.4.3.3 Keep awake reason channel

Port can be taken into consideration to decide whether the CM 1000 High can enter the sleep mode (if the device is configured for that). For that, the port will be monitored to detect incoming traffic.

Keep awake reason channel selected: Port is monitored. If traffic is detected on this port, the device must stay in full operation and cannot enter sleep mode.

Keep awake reason channel not selected: Port is not monitored. The device can enter sleep mode even if there is traffic on this port.

5.4.3.4 Logging ports

This is a table that shows an overview of all 1000BASE-T1 ports used to connect to the IVN for capturing traffic, with an indication of whether each port is linked to the selected GB port or not.

☑ 1000BASE-T1-XX selected: 1000BASE-T1-XX sends its captured data to this SFP+ port

☐ 1000BASE-T1-XX not selected: 1000BASE-T1-XX does not send its captured information to this SFP+ port.

5.4.3.5 Traffic injection port

Ethernet traffic can be injected into the in-vehicle network.

 Traffic injection port selected: Traffic injection possible.

 Traffic injection port not selected: Traffic injection not possible.

Both TCP and UDP traffic is supported. In the case of using TCP, this option will make it possible to establish a TCP connection with the devices connected to the IVN ports. Remember to also configure the IP FOR INJECTION on both sides of the communication.

For further information please see CHAPTER 6.2.

For UDP, please consider the limitations of this protocol: UDP neither offers windowing nor a function to ensure data is received in the same order as it was transmitted, and no error control is performed. By default, no port is configured for injection.

5.4.3.6 Status messages port

The CM 1000 High can send debug messages with information about the status of the device itself, its ports, and configuration. This can be configured by the user.

 Status messages port selected: The status messages of the Capture Module are transmitted on this port

 Status messages port not selected: The status messages of the Capture Module are not transmitted on this port

By default, the GB-A port is configured as a port for status messages. In case no port is selected as a status messages port, the GB-A port will be used as well.

 Logging and status messages port can be different.

 The temperature information in the status message is only sent by Capture Module hardware version 3.1 or greater

For details about status messages, please have a look at CHAPTER 8.1.4.

5.4.3.7 Control message port

The CM 1000 High can be configured to generate control messages when certain events take place.



Control message port selected: Control messages generated by the Capture Module are sent out to the logging network via this port.



Control message port not selected: Port is not used for sending control messages.

As of now, the generation of control messages can be configured in the LINK EVENTS TAB (link up/ link down events), SYNC EVENTS TAB (sync/lost sync events) and in the ADVANCED FILTERS TAB (events based on the incoming traffic). For more details about how to set filters for generating control messages, please check CHAPTER 5.7, CHAPTER 5.8 and CHAPTER 5.9

In the same way, as it is possible to select several GB ports as logging ports, with the option OUTPUT TO in the 1000BASE-T1 IVN interfaces, it is possible to select more than one GB port as a control message port.



Do not mix incoming and outgoing control messages. The CM 1000 High is capable of both processing incoming control messages and sending out control messages generated by itself. The control message port is only used for the CM 1000 High to send control messages, not to receive them. The reception of control messages from other devices is processed only through the configuration port; when sent to the Capture Module on ports other than the configuration port will be ignored.

For details about control messages, please have a look at CHAPTER 8.1.3.

5.4.3.8 Cascade to

CM 1000 High, making use of the cascading functionality described in CHAPTER 3.2.3, can be connected to other Capture Modules to create a seamless network for capturing traffic from different interface types. The traffic collected with one device is forwarded to the next one in the cascading chain until it reaches the logging endpoint or *sink*.

With this option, it is possible to specify which port to forward:

- None: No forwarding of incoming traffic is done. Traffic captured by the next device in the cascading chain will be then discarded.
- GB-A: Forwarding of incoming traffic to GB-A.
- GB-B: Forwarding of incoming traffic to GB-B.
- GB-C: Forwarding of incoming traffic to GB-C.
- 10GB-A: Forwarding of incoming traffic to 10GB-A.

The port currently under configuration will not show up in the dropdown list with the ports to perform cascade to.

The option CASCADE TO can be used only once per device.

5.4.3.9 802.1AS Mode

The CM 1000 High offers the time master functionality for IEEE802.1AS. If decided to use the device as a time master in the logging network, then one and only one port can work as a time master.

In case IEEE 802.1AS is selected as time synchronization protocol in the system where the CM 1000 High will be used, then the user is requested to specify explicitly which of the ports will act as time master (if any) using this option to avoid possible misconfigurations.

Time synchronization between the CM 1000 High and the logging endpoint and/or other Capture Modules is possible by enabling this option in the SFP+ and GB port(s) where synchronization with the connected devices is required.

The SFP+ and GB ports can be configured as follows for time synchronization:

- Disabled: No time synchronization performed through this port.
- Slave: The CM 1000 High acts as an intermediate or end device that get its time from another device, connected directly to it, which is acting as bridge master.
  - If there is already one port defined as SLAVE in the device, no other port can be configured as SLAVE, neither as 802.1AS MASTER.
  - Once Slave has selected the user has a choice to configure the port as:
    - gPTP Automotive slave: 802.1AS slave
    - PTPv2 Slave: IEEE 1588-2008 slave
- Master: The CM 1000 High acts as an intermediate device that gets its time from the grandmaster (directly or through other devices) and acts as a time master for the next device in the chain. It forwards the corrected time information to all the other attached time-aware systems. Sync / Follow up messages start to be sent only when a slave device is detected.

- Grandmaster: This port is generating the master clock used as a reference for the logging system. The device will be providing the time base for all directly attached time-aware systems.

Note that it is not possible to have two ports configured as grandmaster on the same Capture Module. If selected, nothing can be configured on the other SFP+/GB ports.

If no time synchronization is required, this setting should be set as "DISABLED". Otherwise, it is important to distinguish whether the synchronization will be done using gPTP or PTPv2 in the logging network.

By default, time synchronization is disabled on all ports.

Ports configured as Bridge Master or Bridge Slave can be cascaded as well.

5.4.4 Spy

The Spy multiplexer is the part in charge of the logging activities. By clicking on the *Spy* element in the graphical overview, the user has access to the general logging configuration.

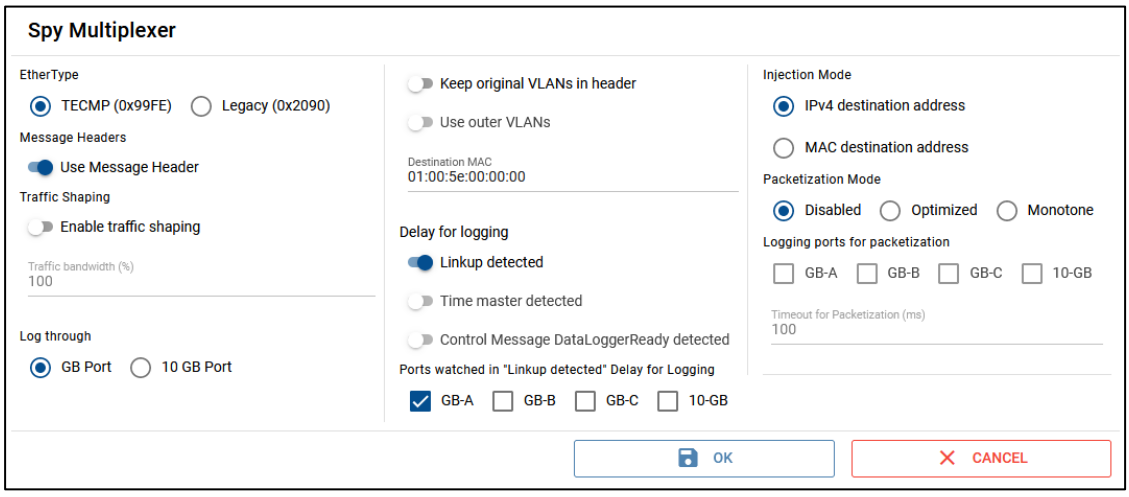


Figure 5-5-15: Spy multiplexer

5.4.4.1 EtherType

As explained in the introduction of this document (CHAPTER 3.2.2), the captured traffic is encapsulated in Ethernet frames following a specific layout and including metadata that guarantees that the data can be processed correctly on the computer side, identified unequivocally and without unwanted delays or data loss. This mechanism

to transport the captured data is defined in the TECMP protocol and the Ethernet frames generated following this protocol are referred to as TECMP frames.

This Ethernet frame is composed of an Ethernet header, a TECMP header, containing metadata about the captured frames, and the TECMP payload, which corresponds to the captured frames themselves, without modifications.

Two different protocol types for encapsulation are supported (TECMP and legacy mode), which can be distinguished by the EtherType:

- TECMP [0x99FE]: The generated Ethernet frames are identified with EtherType 0x99FE, following 100% the TECMP specification.
- Legacy Mode [0x2090]: The generated Ethernet frames are identified with EtherType 0x2090. The frame layout is the same, but some of the fields have a different name.

The device should always be used in TECMP (0x99FE) EtherType unless under special conditions

An equivalence table for the names used in TECMP and legacy mode is available in the TECMP specification document.

See CHAPTER 8.1 for more information about TECMP.

5.4.4.2 Use Message Header

In the case of the CM 1000 High, since the captured data is Ethernet traffic, the user has the possibility to decide whether to encapsulate the data in TECMP frames – with the benefits of the metadata about the captured traffic that they provide - or simply forward the traffic as is. This can be done with the option “Use message header”:



Use Message Header selected: TECMP protocol is activated, and captured data is encapsulated in TECMP frames, with the original frame as payload.



Use Message Header not selected: TECMP protocol is deactivated and captured data is sent to the logging port “raw”, without encapsulation and any kind of modifications.

5.4.4.3 Enable Traffic Shaping, Traffic Bandwidth

The captured traffic is forwarded to the 1000BASE-T port(s) configured as logging port(s), to be sent to a data logger, a computer, or any other device capable of storing the data in trace files. Each of these ports supports up to 1Gbps, but the network card

of the devices connected to them cannot always handle such an amount of data, especially in case of bursts, without packet loss.

To solve this issue, the CM 1000 High offers the user the possibility to activate traffic shaping to control the maximum bandwidth used on the logging port(s):



Enable Traffic Shaping selected: Traffic Shaping is enabled for all outgoing traffic, limiting the amount of bandwidth that can be consumed at most.

- Once enabled, the maximum TRAFFIC BANDWIDTH desired for the output through the logging port(s) must be selected.
- Default traffic bandwidth when traffic shaping enabled is 100%.



Enable Traffic Shaping not selected: No traffic shaping is applied, and no Traffic Bandwidth is configurable. The data will be sent as fast as possible, including bursts and eventual peaks.

There is a distinction between traffic shaping set to 100% and no traffic shaping. If the traffic shaping feature is disabled, that means the received frames are egressed out without any control over the bandwidth. The frames are egressed in an “as-is” way. However, when traffic shaping is enabled and set to 100% a minimum interframe gap (IFG) is maintained to make sure in case of non-continuous traffic (bursts/peaks etc.) frame loss and data corruption are avoided.

Traffic shaping will not be applied if logging is configured for two logging ports with different speed.

The output bandwidth will be different from the configured bandwidth when also packetization is enabled.

5.4.4.4 Log through

At one time, the CM 1000 High can be configured to log through either GB ports or the 10GB port. The selection between these ports can be made here.

5.4.4.5 Keep original VLANs in Header

When TECMP encapsulation is enabled, the TECMP frames do not use by default any VLAN tag, no matter whether the captured traffic uses VLANs or not. With this option, though, the user can make a setting such that the inner VLAN of the captured traffic is copied to the header of the TECMP frame.

This option can be useful to split the traffic on a switch or computer based on the VLAN tags, without having to look at the payload of the TECMP message (with the

extra processing it implies). For example, when an ethernet frame (original ethernet frame Figure 5-16) is received by CM 1000 High, it gets encapsulated into another ethernet frame (new ethernet frame). If the original ethernet frame had VLAN tags with it, this option will copy the VLAN tags to the new ethernet frame thus reducing extra processing of looking inside the payload of the TECMP message.

Keep original VLANs in Header selected: If there is an inner VLAN tag in the captured Ethernet frame, this VLAN ID is taken over to the TECMP frame.

- This is only possible when the *message header* is enabled (option USE MESSAGE HEADER)

Keep original VLANs in Header not selected: No extra handling of VLAN IDs.

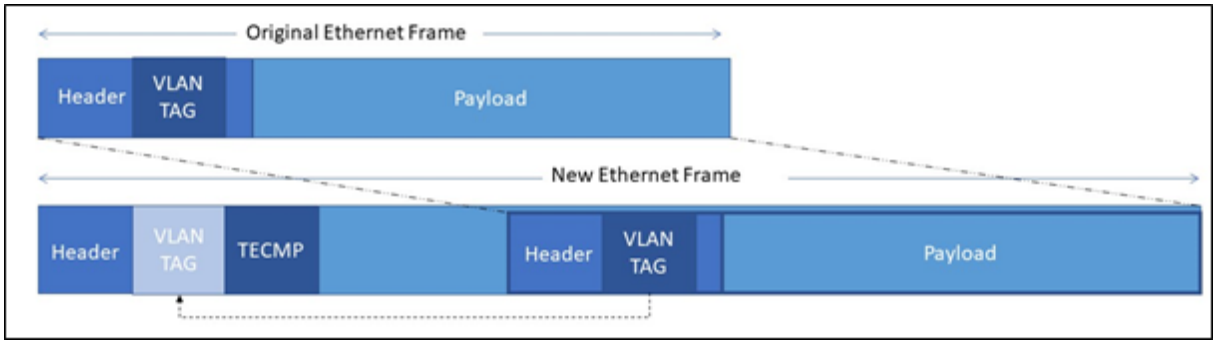


Figure 5-16 Showing keep original VLANs in Header

5.4.4.6 Use Outer VLANs

With this option, frames can be logged as received - without encapsulation in TECMP frames -, but with an outer VLAN tag added. With respect to the original frame, four bytes will be added. An important point to note here is the behavior of this field when different ethertypes are selected.

- If EtherType "Legacy" is selected: The corresponding IVN ports show a field (Section ID). The ID mentioned in the field Section ID will be placed as an outer VLAN ID for the captured frame
- If EtherType "TECMP" is selected: The corresponding IVN ports show a field (Interface ID). The first three digits of the Interface ID are selected as an outer VLAN ID for the captured frame

Please have a look at CHAPTER 5.4.3 about the configuration of 1000BASE-T1 ports regarding the use of the SECTION.

To allow OUTER VLANS, the option USE MESSAGE HEADER must be deactivated.

5.4.4.7 Destination MAC

For encapsulation, layer 2 (data link layer of the OSI model) is used. The encapsulated Ethernet frames are sent to a multicast MAC address. Default value is 01:00:5E:00:00:00.

When USE MESSAGE HEADER is enabled, with this option it is possible to modify the multicast address used as the destination MAC address of all TECMP frames.

5.4.4.8 Delay for Logging

Capture Modules must provide a very fast startup to guarantee that the devices are already up and running when the ECUs start sending traffic, but not all devices part of the test equipment share such a fast startup time. Thus, if the Capture Module starts sending out immediately the captured data to the device(s) connected to the output port(s), initial traffic will be lost.

To fill in this gap between the availability of one device and another, the CM 1000 High is provided with a functionality that makes it capable of buffering the initial traffic until the connected device (data logger, computer) is ready to receive the captured data.

To decide whether the Capture Module can start sending the buffered data or not, several conditions have been identified. The condition(s) to use can be configured with the option DELAY FOR LOGGING.

With this setting, the user can configure when the CM 1000 High starts sending captured data after power-on or wake-up through the configured logging ports. The conditions available for delay for logging are:

- Linkup detected: This option is coupled with the "ports watched in "Linkup detected" Delay for logging" feature. The selected ports here define the ports that are checked for linkup. By default GB-A is selected. If this option is selected, captured data will be sent to the logging port(s) when there is the first linkup after wake-up. If there is no linkup, at the selected output port(s), data will not be sent.
- Time Master detected: This option is dependent on the port configured as time sync slave port. If this option is selected, the captured data which was buffered on the CM will be sent out after a time sync master was detected on the configured port.
    i. The timestamp of the data buffered during the startup will be automatically corrected (only once) to the time provided by the time sync master. All the data received after this synchronization will be timestamp according to this time. (Note: If the time sync master is lost or changes its time, no second recalculation of the already buffered data will take place).

ii. This event can be combined with CONTROL MESSAGE DATALOGGERREADY DETECTED.
iii. If the logging port is different than the port connected to a PTP node, then data will be egressed out once a PTP node is detected and not when a Linkup is detected.

- Control Message DataLoggerReady detected: Unlike the linkup detected, this option is dependent on the config port selected (the control message must be detected on that port, not on others). If this option is selected, captured data from the IVN ports will be sent as soon as the mentioned control message is detected.
  i. This condition can be combined with TIME MASTER DETECTED.
  ii. The timestamp of the data buffered during the startup phase will be corrected only if the condition TIME MASTER DETECTED is also selected and is also fulfilled. Otherwise, the CM 1000 High will send the captured data with the original timestamp, without correction.
- Ports watched in "Linkup detected" delay for logging: The following ports can be checked for link up
  GB-A
  GB-B
  GB-C
  10-GB



No configuration can be saved without activating one of the options

All selected conditions must be fulfilled before the Capture Module starts sending out the captured data.

By default, LINKUP DETECTED is selected. That means that incoming data on the IVN ports will be directly passed to the configured output port as soon as the CM 1000 High is up and running.

During the startup phase, captured messages and status messages are stored together in the internal memory. When there is at least one port set as logging port (i.e., "OUTPUT TO" != none), the internal memory of the Capture Module will be emptied as soon as the DELAY FOR LOGGING condition is met, independently of whether there is linkup or not.

A warning message notifies the user when no port is selected as OUTPUT TO port. If no port is set as a logging port, the device will not log anything, and the only messages stored in the internal memory during startup will be the status messages. In that case, the only port that is monitored is the status port.



To protect the CM 1000 High from undesired side effects in case the condition or set of conditions is never fulfilled, the following protection mechanism has been implemented:

- If 95% (Limit one) of the internal memory is full, the data in the memory is sent out even if there is no control message detected.!
  - A hint in red text will show to the user on the website of the Capture Module.
  - This situation will also be reflected in the status messages (see CHAPTER 8.1.4).
- If 99% (Limit two) of the internal memory is full, because the average input is higher than the average output, the Capture Module stops capturing traffic. As soon as the memory is emptied (less than 10% of capacity), the CM 1000 High will start capturing live traffic again.
  - During this emptying process, the CM flag "Capture Module overflow" will be set.
  - Additionally, a hint in red text will be shown to the user on the website of Capture Module.

Please be aware that, if the port used for logging is different from the port(s) used for the delay logging conditions, traffic will start being forwarded to the logging port(s) as soon as the conditions are met, no matter what the state of the logging port(s) is

When buffered data is sent out, this is done always with maximum bandwidth, so please activate traffic shaping if needed to avoid data loss (see CHAPTER 5.4.1.3).

If timestamps are important for the analysis of the captured data, this setting makes only sense in case TECMP is used for encapsulating the captured traffic, where the timestamp when the data was captured is stored in the TECMP header and can be used to reconstruct the original communication. Otherwise, the user will receive the Ethernet frames captured during startup all at once, losing valuable information about timing.

5.4.4.9 Injection Mode

Two types of traffic injection are offered to the user, based on the destination address which is used to identify the target device for the injected data:

- MAC destination address: Any Ethernet frame from protocols corresponding to OSI layer 2 or above can be injected
  - The injection target device will be identified by its MAC address
- IPv4 destination address: Only possible to inject Ethernet frames from protocols corresponding to OSI layer 3 or above based on IPv4
  - The injection target device will be identified by its IPv4 address

5.4.4.10 Packetization mode

Packetization is a feature that allows encapsulation of more than one captured frame into one TECMP frame. This helps to increase the performance of the transmission of the captured data to a data logger or a computer, especially in case the traffic is composed mainly of small frames.

- Disabled: Each incoming Ethernet frame will be encapsulated in a separate TECMP frame.
- Optimized: Several Ethernet frames will be combined in a TECMP frame, optimizing as much as possible the bandwidth in the logging port.
  - The Capture Module will try to pack as many incoming Ethernet frames as possible in a TECMP frame. If a big frame is captured, it will be sent out directly, while smaller frames will be collected to be sent together in only one TECMP frame.
  - Thus, monotone timestamps are not guaranteed. The sequence in which the TECMP frames are sent out may not match the sequence of how the original frames were captured.
  - The timestamp in the TECMP frame helps find this out.
- Monotone: Several Ethernet frames will be combined in a TECMP frame, guaranteeing that the timestamp of the TECMP frames will increase always monotonously for a given interface ID.
  - For each interface ID, all the captured frames will be sent out encapsulated in TECMP frames in the same sequence as they were captured.

Messages with VLAN are not packetized. Messages, part of communication between DUT and logging sink (because of traffic injection) are also not packetized.



Details about optimized packetization can be found in CHAPTER 6.5.

5.4.4.11 Logging Ports for packetization

CM 1000 High can handle packetization in all the output ports at the same time. With this option, packetization can be enabled separately for each output port configured as a logging port.



When packetization is enabled, the option KEEP ORIGINAL VLANS IN THE HEADER is not available anymore.

5.4.4.12 Timeout for packetization (ms)

The amount of Ethernet frames included in a packetized TECMP frame is defined by two conditions:

- Length: Maximum TECMP payload length of 1500 bytes.

- Timeout for packetization: Maximum allowed time between two IVN frames to be packed into the same TECMP frame.
  - Allowed values for the timeout are:
    - Default: 100ms
    - Minimum: 20ms
    - Maximum: 1000ms

If one of the limitations is reached, the TECMP frame is sent out through the logging port.

5.4.5 1000BASE-T1 Ports

The CM 1000 High is provided with six link lines. Each of these lines is composed of two 1000BASE-T1 ports where two different Ethernet devices can be connected to capture the communication between them.

Each of the 1000BASE-T1 ports requires a basic configuration to determine how exactly the Capture Module will handle the traffic captured from the DUT connected to it if any.

The user can find in the switch status tab a graphical view of the device, with the twelve 100BASE-T1 ports represented on the left side, analog to their location in the physical device. Clicking on one of the ports, a configuration page that looks as follows will be opened:



Figure 5-5-17: 1000BASE-T1-1A-Port

Several configuration options are presented in the first place, followed by debug information that can be used to analyze possible issues in communication. A detailed description of each of the options is found in the subsections below.

The port currently selected for configuration is marked with text in blue.

5.4.5.1 Port name

This field lets the user specify a label to identify the 1000BASE-T1 port unequivocally. By default, it has the form P-*XY*, where

- X is the number of the link line, from 1 to 6.
- Y is A for the left side of the link line and B for the right side.

5.4.5.2 BASE-T1 mode

Automotive Ethernet requires one of the communication endpoints to act as master and the other as slave. This setting specifies the BASE-T1 operation mode the Capture Module will use on this port:

- Master
- Slave

The configuration to choose will be determined by the device connected to the Capture Module on this port. Communication will only take place if each communication endpoint takes one of the roles.

5.4.5.3 Port Speed

The 1000BASE-T1 ports can also be configured as 100BASE-T1, thus operating on 100Mbps. This option enables the user to choose compliance with 1000BASE-T1 or 100BASE-T1.

- 100BASE-T1: The selected port will act as 100BASE-T1 port
- 1000BASE-T1: The select port will act as 1000BASE-T1 port

The configured speed 100Mbps/1Gbps is always valid for a subset of 4 ports. This means that in Capture Module 1000 High it is possible to configure the speed in 3 blocks.

- Block 1: Port 1A to Port 2B
- Block 2: Port 3A to Port 4B
- Block 3: Port 5A to Port 6B

The "Port Speed" shall be the same for each pair of channels to have a link. This means the user cannot configure one port as a 100BASE-T1(1A) port and the other as a 1000BASE-T1 (1B) port in the port pair.

5.4.5.4 PHY Mode

1000BASE-T1 is an evolving technology. Marvell Phys have been used to provide the 1000BASE-T1 capability. These phys offer compliance with IEEE or A0 Phys from Marvell (See CHAPTER 8). Users can make the selection through this option.

- IEEE: The selected port expects a connected DUT with IEEE complaint Phy (See CHAPTER 8)
- Legacy: The select port expects a connected DUT with Phy in A0 mode

From firmware release 20 onwards, the 1000BASE-T1 ports of the Capture Module offer the option of selecting the PHY mode between SingleSendS and MultipleSendS per port. The reason for having these options is to improve interoperability of the system when starting up or establishing the linkup.

- Script A: With this selection SingleSendS is active (As in previous Capture Module versions)
- Script B: With this selection MultipleSendS is active (which can help in terms of IOP issues)

Script A is used by default. Before firmware release v021.002.090, Script A was MultiSendS and is now SingleSendS as described above.

5.4.5.5 Output to

The data captured by the device must be forwarded to the logging network so that it can be analyzed and stored, if needed, on the logging endpoint. The three GB ports or the one SFP+ port available in the CM 1000 High can be configured for this purpose.

Each IVN interface (another name for the 1000BASE-T1 ports) can be configured separately to be forwarded to any of the three GB ports or the SFP+ port. The GB port or ports configured here are called logging ports since they connect the Capture Module with the logging network.

The options available are:

- None: No logging port defined.
  - No logging possible. Traffic will be captured but discarded.
- GB-A: GB-A is defined as a logging port.
- GB-B: GB-B is defined as a logging port.
- GB-C: GB-C is defined as a logging port.
- 10-GB-A: 10Gb SFP+ port defined as a logging port

More than one port can be selected for each IVN interface. In this case, the traffic captured through this 100BASE-T1 port will be copied to each of the selected GB ports.

On the other side, the same GB port can be used to gather and forward to the logging network the traffic from several IVN interfaces.

It is up to the user to decide, based on the expected load on each IVN interface, how many of them to configure for the same GB port as the logging port. The maximum bandwidth forwarded to a logging port must be always kept below the 1 Gbps supported by a GB port or below 10 Gbps for the SFP+ port to avoid packet loss.

5.4.5.6 Interface ID

As explained previously in this document, the CM 1000 High can capture the traffic exchanged between two Ethernet devices by placing itself in the middle of the link that the two DUTs share, splitting this way the link into two segments.

This option can be used to specify an ID that uniquely identifies the DUT from which the data on this link has been captured, in hex format. Interface ID can be specified separately for each 1000BASE-T1 port.

5.4.5.7 Transparency Mode

The transparency mode will work the same way even if the device is configured in 100BASE-T1 mode. The following options can be found in transparency mode.

- Disabled: Transparency mode not active (default). The link state on one port does not affect the link state on the other port. This can result in packet loss in the communication path between the two DUTs connected.
- Enabled with event-based delay: In this option:
  - When a link down is detected on any port pair (e.g. 3A-3B), the paired port does not change its status until the time mentioned in Transparency sleep (MS).
    - If the original port is up before this time, both ports remain up.
    - If the original port stays down after this time, the paired port is also forced down.
  - When a link down is detected on any port pair (e.g. 3A-3B), the paired port is also muted for the time mentioned in TRANSPARENCY WAKE-UP DELAY (MS).
    - If the original port is up before this time, the paired port still waits for the time mentioned in TRANSPARENCY WAKE-UP DELAY (MS). to change its status to link up.
    - If the original port stays down after this time, the paired port remains down.
    - If the original port is up after the time mentioned in TRANSPARENCY WAKE-UP DELAY (MS), the paired port follows its original status (link up) immediately with no waiting time.

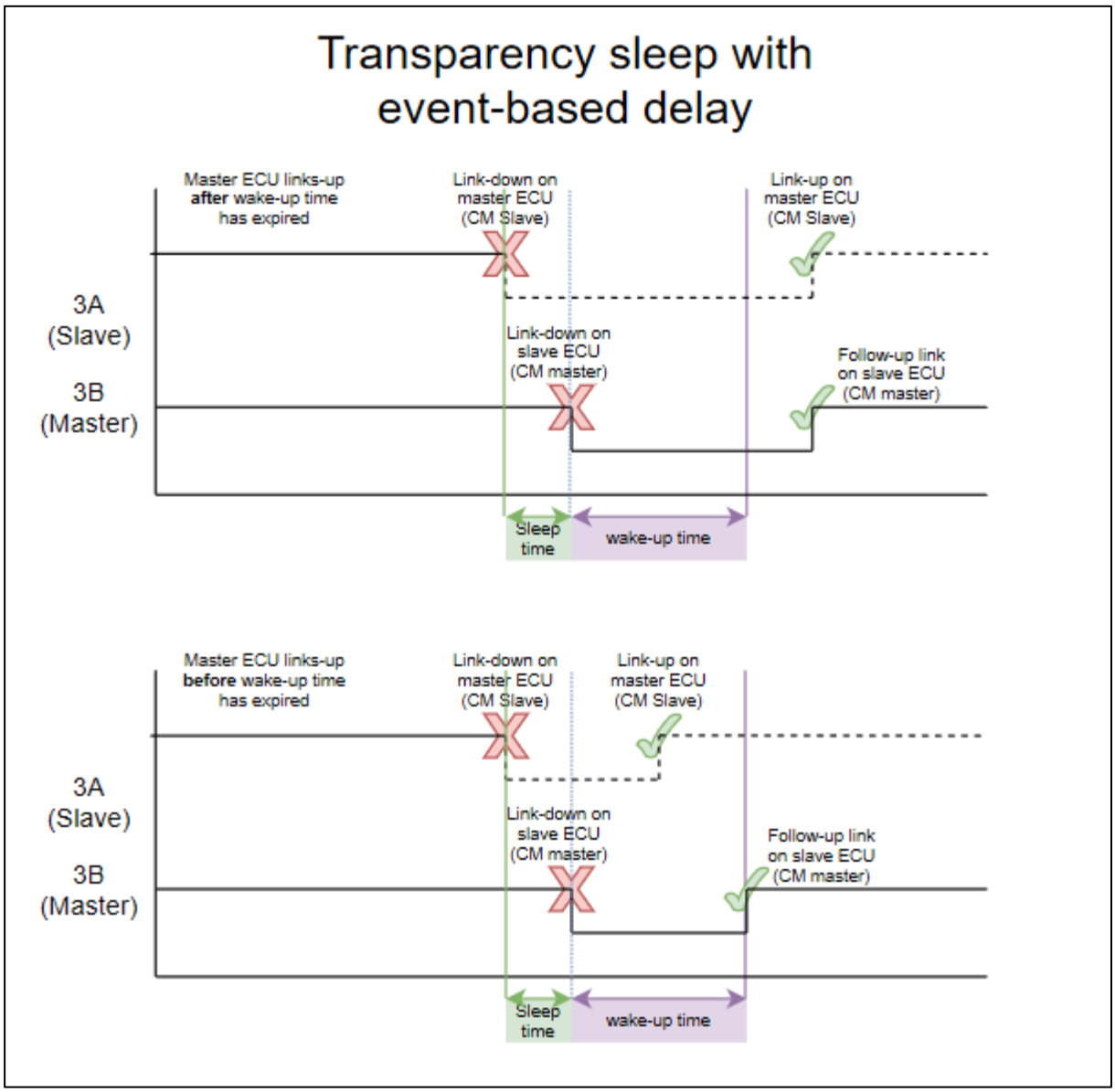


Figure 5-18 Transparency sleep with event-based delay

- Enabled with defined delay: In this option:
  - When a link down is detected on any port pair (e.g. 3A-3B), the paired port does not change its status until the time mentioned in Transparency sleep (MS).
    - If the original port is up before this time, both ports remain up.
    - If the original port stays down after this time, the paired port is also forced down.
  - When a link down is detected on any port pair (e.g. 3A-3B), the paired port is also muted for the time mentioned in TRANSPARENCY WAKE-UP DELAY (MS).
    - If the original port is up before this time, the paired port still waits for the time mentioned in TRANSPARENCY WAKE-UP DELAY (MS). to change its status to link up.

- If the original port stays down after this time, the paired port changes its status to link up behaving like an independent port.

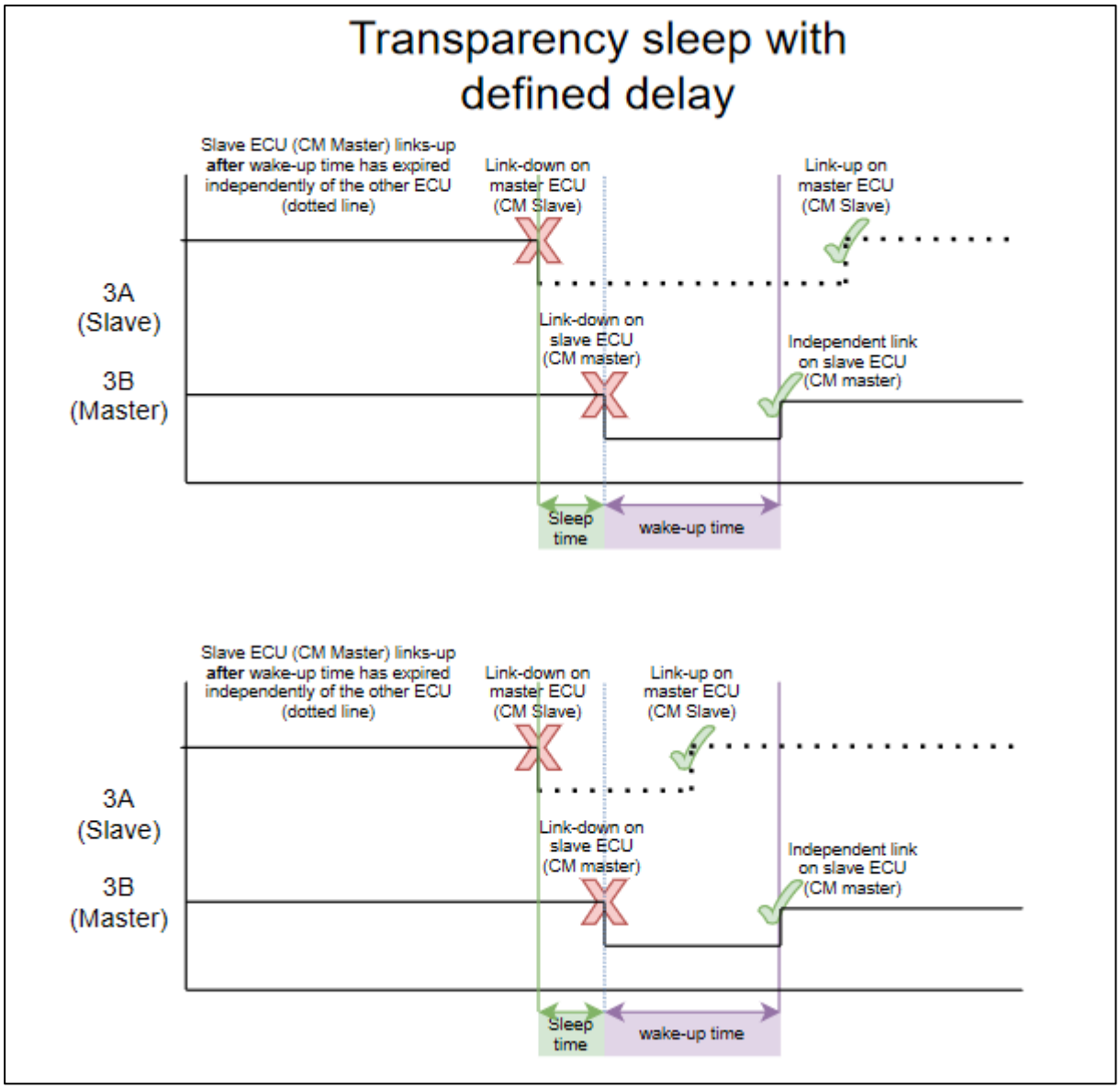


Figure 5-19 Transparency sleep with defined delay

The same functionality is observed when a 1000BASE-T1 port is configured in 100BASE-T1 mode.

5.4.5.8 Transparency wake-up (ms)

The time needed to unmute the PHY.

5.4.5.9 Enable pair traffic forwarding

As capture module is introduced between a source and sink, a link line is broken into to point to point links. This option enables/disables the source sink communication to be considered as point to point or as a complete link. By default the option is enabled.

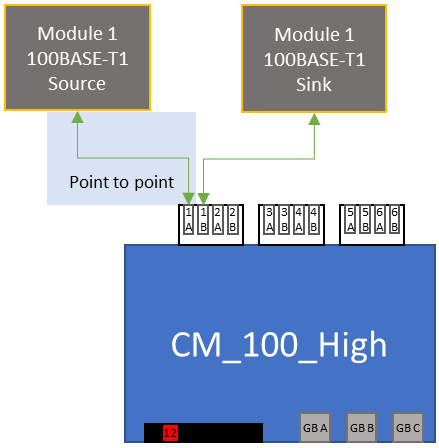


Figure 5-21 Disable pair traffic forwarding

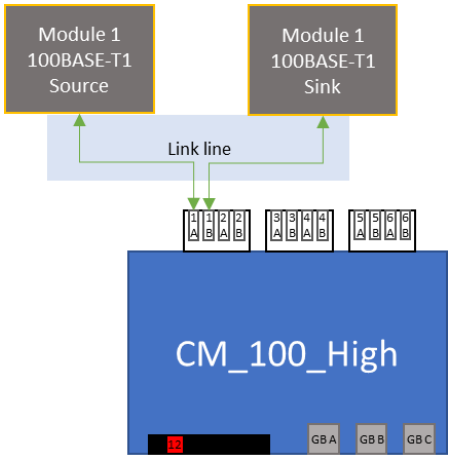


Figure 5-20 Enable pair traffic forwarding

Enable pair traffic forwarding When enabled traffic is forwarded between port pairs (1A-1B) so that traffic from source can be forwarded to the sink.

Enable pair traffic forwarding When disabled traffic is not forwarded between port pairs. Every port is treated as an independent point to point connection.

5.4.5.10 Keep awake reason channel

Port can be taken into consideration to decide whether the CM 100 High can enter the sleep mode (if the device is configured for that). For that, the port will be monitored to detect incoming traffic.

Keep awake reason channel selected: Port is monitored. If traffic is detected on this port, the device must stay in full operation and cannot enter sleep mode.

Keep awake reason channel not selected: Port is not monitored. The device can enter sleep mode even if there is traffic on this port

5.4.5.11 Enable recovery for this port

If the Capture Module detects a failure state in the PHY due to in-coherent frames, the capture module resets the PHY in order for it to recover.

5.4.5.12 Enable injection for this port

When activating this function, it is possible to inject data from the logging endpoint to the DUT directly connected to this 1000BASE-T1 port via the SFP+/GB port configured as a TRAFFIC INJECTION PORT

Traffic injection is a complex feature that requires the configuration of several settings to make it work properly:

Please also provide the IP FOR INJECTION and a valid injection window in the field CONSTANT DELAY (IN BYTES) FOR INJECTION for this port.

Do not forget to configure the IP FOR INJECTION on the GB port to which the logging endpoint in charge of the traffic injection is connected.

5.4.5.13 MAC/IP for Injection

When ENABLE INJECTION FOR THIS PORT is selected, this field must be filled in with the IP address of the DUT connected to this port.

This information will be used to guarantee that only traffic to the explicitly specified DUT will be injected through the Capture Module. If traffic with different target addresses is sent via this port, it will be discarded. A valid IP helps the user to avoid side effects like flooding another ECU with unwanted traffic. An invalid IP like 0.0.0.0 allows the traffic to be flooded to all the ports.

5.4.5.14 Constant delay in bytes for injection

When ENABLE INJECTION FOR THIS PORT is selected, this setting is available. The configuration of this field is mandatory to make the injection mechanism work properly.

The value configured in this field determines the injection window available, measured in bytes. When a frame is sent from the logging endpoint in direction DUT, the Capture Module will look for a gap in the IVN incoming traffic (from the DUT point of view) that is at least as big as the one configured as CONSTANT DELAY (IN BYTES) FOR INJECTION. If such a gap is found, the frame will be injected into the line; if no gap is found after several tries, the frame for injection will be discarded.

Possible values are:

- 512 bytes
- 1024 bytes
- 1536 bytes

Which value to choose depends on the size of the data to be injected: CONSTANT DELAY (IN BYTES) FOR INJECTION must be setup according to the formula:

Frame size + preamble + CRC

The value shall be at least that size. For example, 500 + 8 + 4 = 512. Otherwise, the Capture Module will not have enough space to inject the data between two consecutive frames of the IVN traffic.

For higher performance, the smallest value is recommended, if the data to be injected allows it.

A value of 0 for CONSTANT DELAY (IN BYTES) FOR INJECTION would mean that no injection window is required and thus, traffic injection will be immediately performed, no matter the traffic load on the line.

5.4.5.15 Timestamp

Read-only field that shows the current device timestamp, in nanoseconds (ns) resolution and accuracy of 40 nanoseconds (ns).

5.4.5.16 Messages Transmitted

Read-only field that shows a counter of the frames transmitted (also called Tx) by the DUT connected to this 1000BASE-T1 port.

The Tx counter of a port and the Rx counter of its paired port do not necessarily match. For example, injection data sent from the logging endpoint will be counted in the Rx counter of one port but have nothing to do with the Tx counter of the other port.

5.4.5.17 Messages Received

Read-only field that shows a counter of the frames received (also called Rx) by the DUT connected to this 1000BASE-T1 port.

In configuration mode, the count of messages received will not be updated, since the logging functionality is stopped. Messages sent by the DUT during this period will not be taken into consideration.

5.4.5.18 Messages with Error

Read-only field that shows a counter of the frames sent by the DUT connected to this 1000BASE-T1 port that contains errors like FCS (Frame Check Sequence).

5.5 Diagnostics tab

This tab is used to check the status of the cables connected to all 1000BASE-T1 ports. To trigger a diagnostic check is clicked which gives the status of the ports. The following status can be observed.

OK	Successful cable connection. The interface is ready
Open	The connection is open, i.e., there is no sink attached to the port
Busy	The PHY took more than several milliseconds to reply
Unknown	No cable diagnostics information has been collected
Short	There is a shortcut
Fail	Capture Module was not able to communicate with PHY
Not available	The browser was not able to communicate with the Capture Module

The cable diagnostic feature shall not be triggered during logging, as it can interfere with it. When the cable diagnostics is run while logging, traffic users might experience a link flicker (link down and link up again), thus resulting in a discontinuous log of traffic.

5.6 Basic Filters tab

As explained in CHAPTER 3.2.6, the CM 1000 High offers the possibility to apply filters on the captured data before sending it out for logging.

The FILTERS TABLE shows by default an overview of all the filters currently configured, in the form of a table in read-only mode.

The basic filters make it possible to discard frames based on criteria (called *filtering conditions*) set on the data included in the Ethernet header. This is very useful to reduce the amount of data captured, in scenarios where not all kinds of data is relevant.

Possible filtering conditions are:

- Source MAC address
- Destination MAC address
- EtherType
- Inner VLAN ID
- Outer VLAN ID

To modify, add or delete basic filters, the button CREATE NEW ENTRY will set this table in edit mode. Each table row represents a filtering rule.

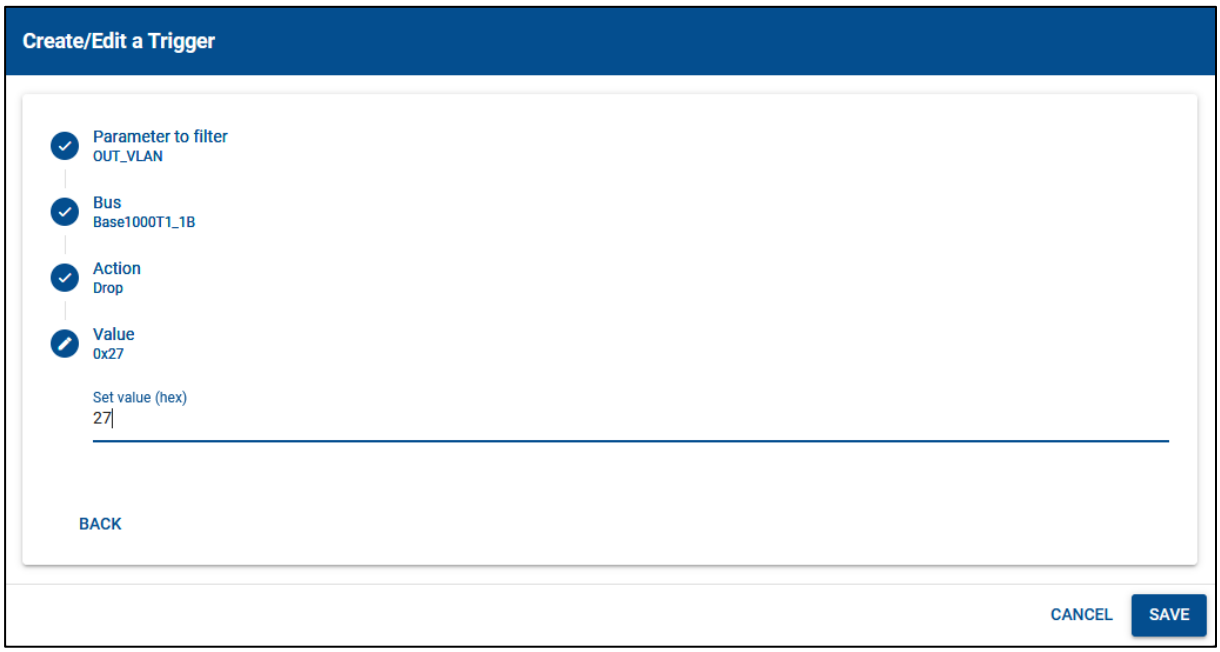


Figure 5-5-22: Close-up filters table

To manage these filters, the following options are available in edit mode:



Adds a line in the table that represents a new basic filtering rule



This button deletes all existing basic filter entries



Edit the created filter



Remove the created filter

After defining the filtering rules, the user must select 

to apply the filtering rules to the traffic

Parameter to filter: This option specifies the filtering condition, i.e. the field of the incoming data that will be used to decide whether the filter is applied or not.

- SOURCE MAC: Source MAC address
- DESTINATION MAC: Destination MAC address
- INNER VLAN: inner VLAN tag ID.
- OUTER VLAN: outer VLAN tag ID.
- ETHERNET TYPE: EtherType

Each pair *Port– Parameter to filter* is possible only once.

Port: This option defines the IVN port to which the filter will be applied. A dropdown list with ports *1000BASE-T1 1A* to *1000BASE-T1 6B* is available for the user to choose from.

Action: currently, the following action is available as part of the basic filters:

- DROP: Each frame that matches the filtering condition will not be forwarded to the configured logging port.

Value: Here the value of the field selected in PARAMETER TO FILTER that will be used for the actual filtering is specified, following the format defined below:

- Set MAC: 6 bytes in hexadecimal format, separated by a colon
  - e.g., 11:22:33:aa:bb:cc
- Set value(hex): 2 bytes in hexadecimal format.
  - e.g., 1FB

If the option GENERIC is selected in the field EtherType, this refers to any other EtherTypes than EtherTypes for ARP, IPv4 and AVTP.

5.7 Link Events tab

This feature enables the user to generate events on the Control Message Port (See CHAPTER 5.4.2.8) if the IVN port of the capture module has a link up or link down. The user can generate a control message with the desired event ID using this tab.
The link events tab has the following options:

LinkUp: Capture Module generates a control message with the configured Event ID when the selected interface port detects a link-up.
LinkDown: Capture Module generates a control message with the configured Event ID when the selected interface port detects a link down.

- Interface: The user can configure the port, which needs to be monitored for the link-up/down event
- Event ID: The control message generated from the capture module should contain the configure event ID, thereby making the event easy to filter while analyzing the traces

Link status is checked every 5 ms.

5.8 Sync Events tab

This feature enables the user to generate events on the Control Message Port (See CHAPTER 5.4.2.8) if the SFP+/GB ports of the capture module attain synchronization

or lose synchronization. The user can generate a control message with the desired event ID using this tab. Sync events tab has the following options:

Synced: Capture Module generates a control message with the configured Event ID when the GB port is time synchronized with the partner clock (master/slave)
Lost Sync: Capture Module generates a control message with the configured Event ID when the GB port losses time synchronization with the partner clock (master/slave)

- Event ID: The control message generated from the capture module should contain the configured event ID, thereby making the event easy to filter while analyzing the traces

5.9 Advanced Filters tab

This is an extended version of the filter functionality that offers more flexibility to the user on the captured data, with more actions to be performed on the incoming messages.

A maximum of 24 (0-23) filtering rules is possible.

Each advanced filtering rule has the following options:

Port: IVN port on which the filtering rule will be applied.

Action: Filtering action to be performed. As explained in CHAPTER 3.2.6, these are the actions available:

- Log without Message Header: In case the USE MESSAGE HEADER is enabled, with this action, the captured frames matching the filtering condition will be sent out without TECMP encapsulation.
- Drop: This action can be used to drop certain frames to not be captured, reducing the amount of data to be processed by discarding unneeded data.
- Mark: With this action, the user can mark frames with a so-called SPY flag (included in the TECMP header) to identify them later in the toolchain for further processing.
- Control Message: This action can be used to configure the sending of a control message whenever a certain condition is met.
- Change Interface ID: This action gives the user the possibility to modify the original interface ID, which is configured on the incoming port, to separate the traffic received on the port(s) based on certain conditions. Since the interface ID is 4 Bytes, the user can change every byte of this field.
- Not Packetize: This action is available when Packetization mode is enabled in the SPY Tab (CHAPTER 5.4.4.10). It gives the user the possibility not to packetize frames if
    - The frames that match the rule defined in "Not Packetize" advanced filter.
    - The frames that are Marked (i.e., frames that match the "Mark" rule)

- Multiple Log Interfaces: This action allows the user to define, for specific captured data, which output ports to use as logging port(s).
  - This is independent of the configured logging port (OUTPUT TO on the IVN port(s)). Captured data matching the condition set will be only redirected to the specified output ports.

Parameter: Additional information about the action to perform. Only applicable for the following actions:

- When ACTION: CONTROL MESSAGE is selected, it specifies the control message ID.
- When ACTION: CHANGE INTERFACE ID is selected, it is used to specify the value for the new INTERFACE ID to set.

Multilog Interfaces: Only available if ACTION: MULTIPLE LOG INTERFACES are selected. A selection between GB ports and SFP+ can be made here.

The frames matching the filtering condition are only sent out on the defined output ports. This overwrites, for the affected frames, the configuration set in field OUTPUT TO of the IVN port.

Ethernet Frame: If the original captured frame contains a particular Ethertype or a particular VLAN configuration (inner, outer, or both) then the condition of the filter should be applied

If the VLAN format is not correctly configured, the corresponding filter rule won't work as expected.

Condition: Here user can define the conditions of the filter along with the values. Users can combine multiple conditions by using the "AND" operator

5.9.1 How to manage advanced filtering rules

To manage these filters, the following options are available in edit mode:



Adds a line in the table that represents a new basic filtering rule



This button deletes all existing basic filter entries



Edit the created filter

 Remove the created filter

 Activate/Deactivate the filter. This can be seen in the “Enable” column of the filter if the filter is activated/deactivated



After defining the filtering rules, the user must select “SAVE & APPLY CONFIGURATION” to apply the filtering rules to the traffic

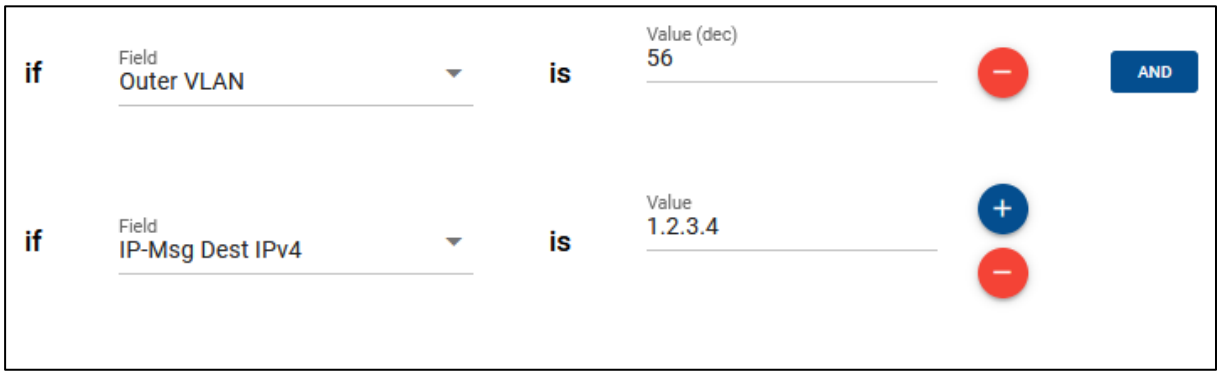
6 Application examples

6.1 Advanced filtering

Example 1: Drop Ethernet frames with outer VLAN (0x56) and IP destination address 1.2.3.4 received on port 1000BASE-T1-1B.

Steps:

1. Access the webpage using the IP address on CM 1000 High (default IP address: 10.104.3.144)
2. Navigate to the advanced filters tab
3. Click on the block labeled as “Create new entry”
4. Select port 1000BASE-T1-1B and click “Continue”
5. Select the action “Drop” and click “Continue”
6. Select the type of traffic, the filter should be implemented by specifying Ethertype and VLAN configuration. For this example, we would use “IPv4” in the Ethertype and “Outer VLAN” in the VLAN config field
7. Now to define values to filter we specify the following



8. Press the button “Save”
9. Press the button “Apply Configuration” to save the changes

➔ Finished

The output shall be visible in the panel



Figure 6-1 Example Advanced Filter Drop

Example 2: Change Interface ID for all Ethernet frames sent to UDP destination port 2020 on 1000BASE-T1-1B.

Steps:

1. Access the webpage using the IP address on CM 1000 High (default IP address: 10.104.3.144)
2. Navigate to the advanced filters tab
3. Click on the block labeled as “Create new entry”
4. Select port 1000BASE-T1-1B and click “Continue”
5. Select the action “Change Interface ID”, add the new hex value, and click “Continue”
6. Select the type of traffic, the filter should be implemented by specifying Ethertype and VLAN configuration. For this example, we would use “IPv4” in the Ethertype and “No VLAN” in the VLAN config field
7. Now to define values to filter we specify the following



8. Press the button “Save”
9. Press the button “Apply Configuration” to save the changes

➔ Finished

The output shall be visible in the panel

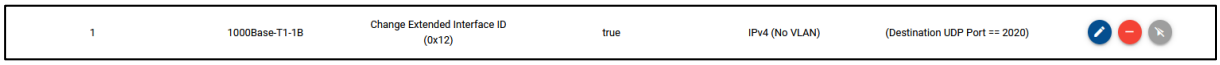


Figure 6-2 Advanced Filter Change Interface ID

6.2 Traffic injection

To interact with a DUT connected to one of the 1000BASE-T1 ports, some preconditions must be met:

- Traffic can be injected only to one DUT per injection-capable 1000BASE-T1 port.
- The target DUT must be connected directly to the injection-capable 1000BASE-T1 port, with no switches in between.

The example below shows how to configure the CM 1000 High for traffic injection. Having the following setup:

- Computer with IP address X.Y.Z.123, connected to GB-A port.
- DUT_1 with IP address X.Y.Z.10, connected to port 100BASE-T1-1A.
- DUT_2 with IP address X.Y.Z.20, connected to port 100BASE-T1-1B.
- Communication between all devices uses an inner VLAN tag.
- We want to establish a TCP connection between the computer and ECU 2 to exchange messages over TCP.

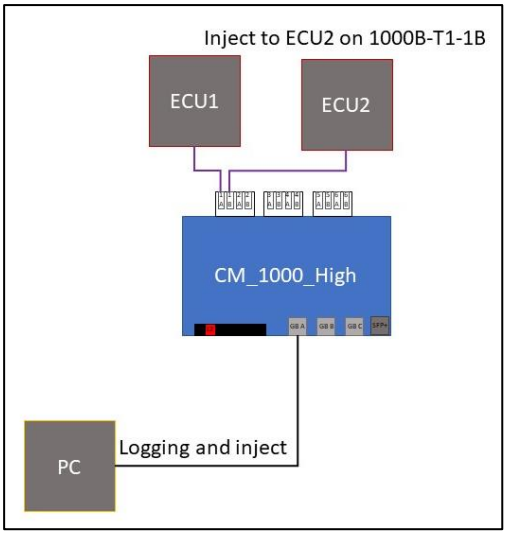


Figure 6-3 Injection setup

The steps to correctly configure the device for that are the following:

1. First the INJECTION TYPE must be set to value “IPv4 destination address”.
2. Then, port GB-A must be configured:
    a. Option TRAFFIC INJECTION PORT must be activated.
    b. Option Enable injection for this port must be activated.
    c. Field IP FOR INJECTION must be set to X.Y.Z.123.
3. Then, it is turn for port 1000BASE-T1-1B to be configured:
    a. Option Enable injection for this port must be activated.
    b. Field IP FOR INJECTION must be set to X.Y.Z.20.
    c. MAXIMAL DELAY (BYTES) FOR INJECTION must be configured with a value other than 0.
        i. The value to choose will depend on the input traffic.
4. Additionally, if TECMP encapsulation is enabled (see CHAPTER 5.4.4.2), filtering rules are required to allow bidirectional communication as required by protocols like TCP.
    a. One filtering rule with ARP target address as a condition pointed to the IP address of the computer.
    b. and another one with IP destination address as condition pointed to the IP address of the computer.

6.3 Time synchronization using PTPv2 (IEEE 1588v2)

IEEE 1588v2 (also known as PTP, Precision Time Protocol) is an industrial standard that allows electrical devices and/or systems to transfer time and frequency information to synchronize their clocks over packet-based Ethernet networks. This protocol synchronizes clock devices with a specific system Grandmaster, by means of exchanged timestamped messages (down to a sub-nanosecond precision), to deliver high accuracy in terms of both time and frequency synchronization.

Clock types

The PTP IEEE 1588v2 protocol differentiates between several kinds of clocks:

- Ordinary Clock (OC): it has only one port that can work as a Master, distributing IEEE 1588-time messages to Slaves, or Slave, synchronizing IEEE 1588 time with the Master.
- Boundary Clock (BC): it has one Slave clock port and multiple Master clock ports that need to communicate with each Slave individually when using the End-to-End delay mechanism. To help the Grandmaster reach a larger number of slaves, this kind of clock absorbs sync messages in the slave port, uses that port to set its clock, and generates new sync messages from this clock out of all its master ports
- Transparent Clock (TC): it assists in the delay measurement between Master and a Slave by including a Correction Factor (CF) that tells the slave how much delay the TC added. Two types are considered:
    - End-to-end transparent clocks.
    - Peer-to-peer transparent clocks.

The Capture Modules support only Ordinary Clocks in the role of slave.

Time Synchronization process

A Grandmaster sends information including the synchronized time to all time-aware systems. To make this work, two intervals must be precisely known: the forwarding delay in the intermediate devices (residence time) and the time taken for the information to transit the communication path between two devices.

6.3.1 How to set up a synchronized network with Capture Modules and PTPv2

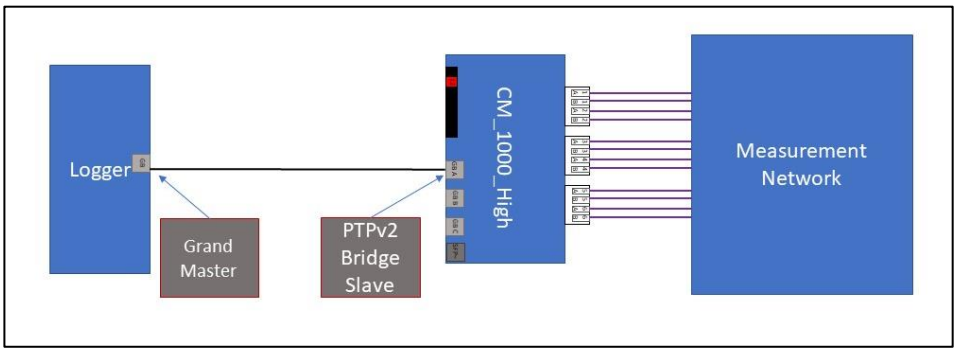


Figure 6-4 Example of PTPv2 with Logger as Grand Master and CM 1000 High as Bridge Slave

The CM 1000 High can be configured as IEEE 1588 Bridge slave. In Figure 6-4 Example of PTPv2 with Logger as Grand Master and CM 1000 High as Bridge Slave, the CM 1000 High is configured as the bridge slave. When the logger and CM 1000 High are synchronized. The traffic received from the measurement network will be timestamped according to the time received by the Grand Master (Logger).

Steps:

1. Configure the data logger as Grand Master
2. Access the CM 1000 High via a web browser through the assigned IP address (default IP address 10.104.3.144)
3. Navigate to the status tab
4. Select the GB port that is connected to the logger. This port needs to configure as a slave
5. According to Figure 6-4 Example of PTPv2 with Logger as Grand Master and CM 1000 High as Bridge Slave GB-A port of CM 1000 High is connected to the logger. Click on GB-A port for further option
6. In 802.1AS Mode, select “Slave”
7. In PTP Mode, select PTP v2

The GB-A port of CM 1000 High is acting as a slave now for the logger (Grandmaster). Similar steps can be repeated to configure other GB ports and the SFP+ port as well.

6.4 Time synchronization using gPTP (IEEE 802.1AS)

IEEE 802.1AS (also known as gPTP, generalized Precision Time Protocol) is an adaptation of Precision Time Protocol for use of Time-Sensitive Networking (TSN) and Audio Video Bridging (AVB). This protocol synchronizes clock devices with a specific system Grandmaster, by means of exchanged timestamped messages (down to a sub-

nanosecond precision), to deliver high accuracy in terms of both time and frequency synchronization.

In Ethernet systems, gPTP uses a complete IEEE 1588-2008 profile but with some features that go beyond PTP. gPTP is, therefore, more robust against delay variations since it demands that every switch in the network supports gPTP. It is for this reason why the delay mechanism used is the P2P mechanism.

Many implementations of Time Synchronization within the Automotive Industry relate to a specific Automotive Profile, which can be described as a simplified version of gPTP with faster start-up since the clock roles are initially established and hardcoded. This is because the automotive environment is a closed system (every device is known prior to startup and they do not enter or leave the network, except in the case of failures) so it is possible to simplify gPTP startup performance, the election of grandmaster and wire delay calculations. As the clock roles are already known beforehand, no Announce messages take place and the clock hierarchy (Grandmaster and slaves) is set from the beginning of the communication.

For startup performance, many implementations can preconfigure some gPTP values and make them persistent through the entire power cycle. The Automotive Ethernet AVB standard for gPTP by AVNU states this initial parameter configuration for implementing this gPTP profile in a testing environment.

Clock types

gPTP describes only three kinds of time-aware devices (less than PTP):

- **Grandmaster (GM):** It initializes the clock synchronization messages periodically from its only master port.
- **Bridges**: they comprise one slave port connected to the master port of another time-aware system and several master ports connected to different downstream paths. They would relate with the transparent and boundary clocks in the PTP protocol where its operation is very tightly defined, so much so that a time-aware bridge with Ethernet ports can be shown to be mathematically equivalent to a P2P transparent clock (TC) in terms of how synchronization is performed
- **End-stations**: last devices of the network that present only one port. Mostly the last slaves, even though the grandmaster could also fit in this description. In PTP IEEE 1588 they match the description of ordinary clocks (OC).

The Capture Modules support the three clock types.

Time Synchronization process

This process in gPTP is done the same way as in IEEE 1588-2008: A Grandmaster sends information including the synchronized time to all time-aware systems. To make

this work, there are two intervals that must be precisely known: the forwarding delay in the intermediate devices (residence time) and the time taken for the information to transit the communication path between two devices.

6.4.1 Transport of time synchronization information

Let's take as reference the case where communication is established only between two devices,

1. „Time-aware system i-1" configured as IEEE 802.1AS master
2. and „Time-aware system i+1" configured as IEEE 802.1AS slave.

The Capture Module will be placed between the two devices. The ports of the Capture Module („Time-aware system I") to which the two devices are connected take then an intermediate role, acting as counterpart element in the communication for each of the devices: it will behave as a slave where the master device is connected, and as a master where the slave device is connected.

This can be more easily understood with the following image:

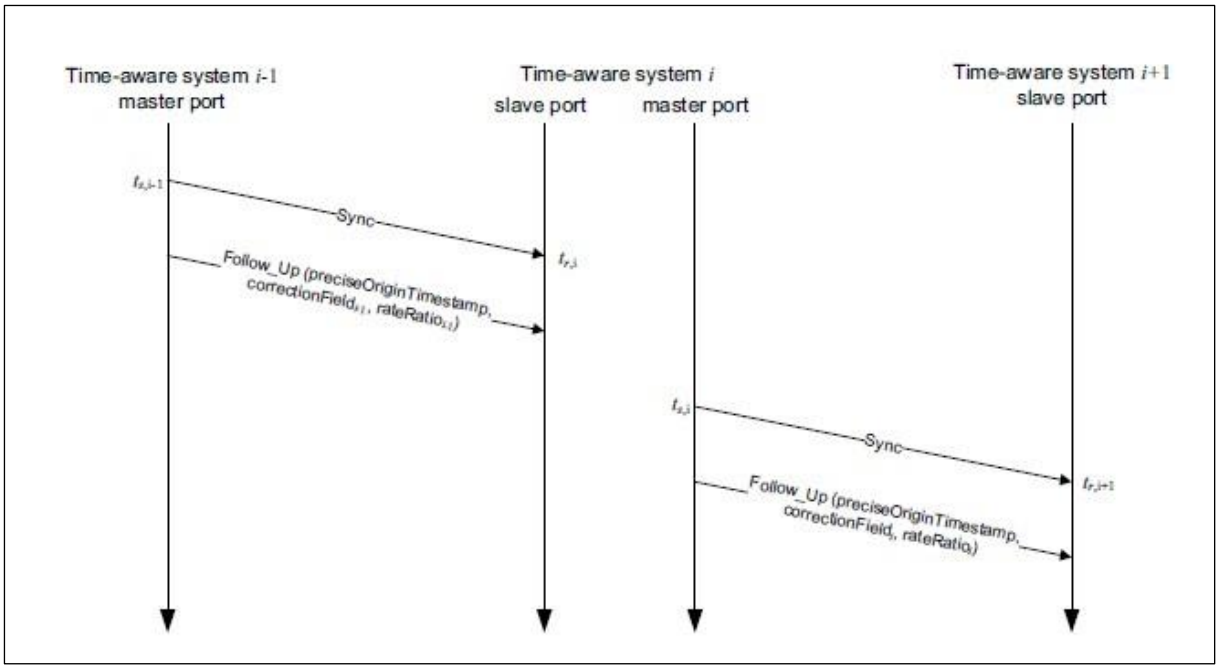


Figure 6-5: IEEE 802.1AS gPTP bridge

The device „Time-aware system i-1", which defines the time of the system, sends cyclically Sync and Follow-Up messages, containing time sync information, that is received by the Capture Module.

These messages are then interpreted by the Capture Module, and the timing information is extracted. This information is used later to generate, together with the information extracted from the propagation delay algorithm, the new corrected Sync and Follow-Up messages that will be sent to the „Time-aware system i+1“.

6.4.2 Peer-to-peer propagation delay measurement

For the correct adjustment of the time reference transmitted by the Sync and Follow-Up Packets, the delay between the initiator of the protocol „Time-aware system i-1“ and the first slave port of Capture Module „Time-aware system i“, should be measured

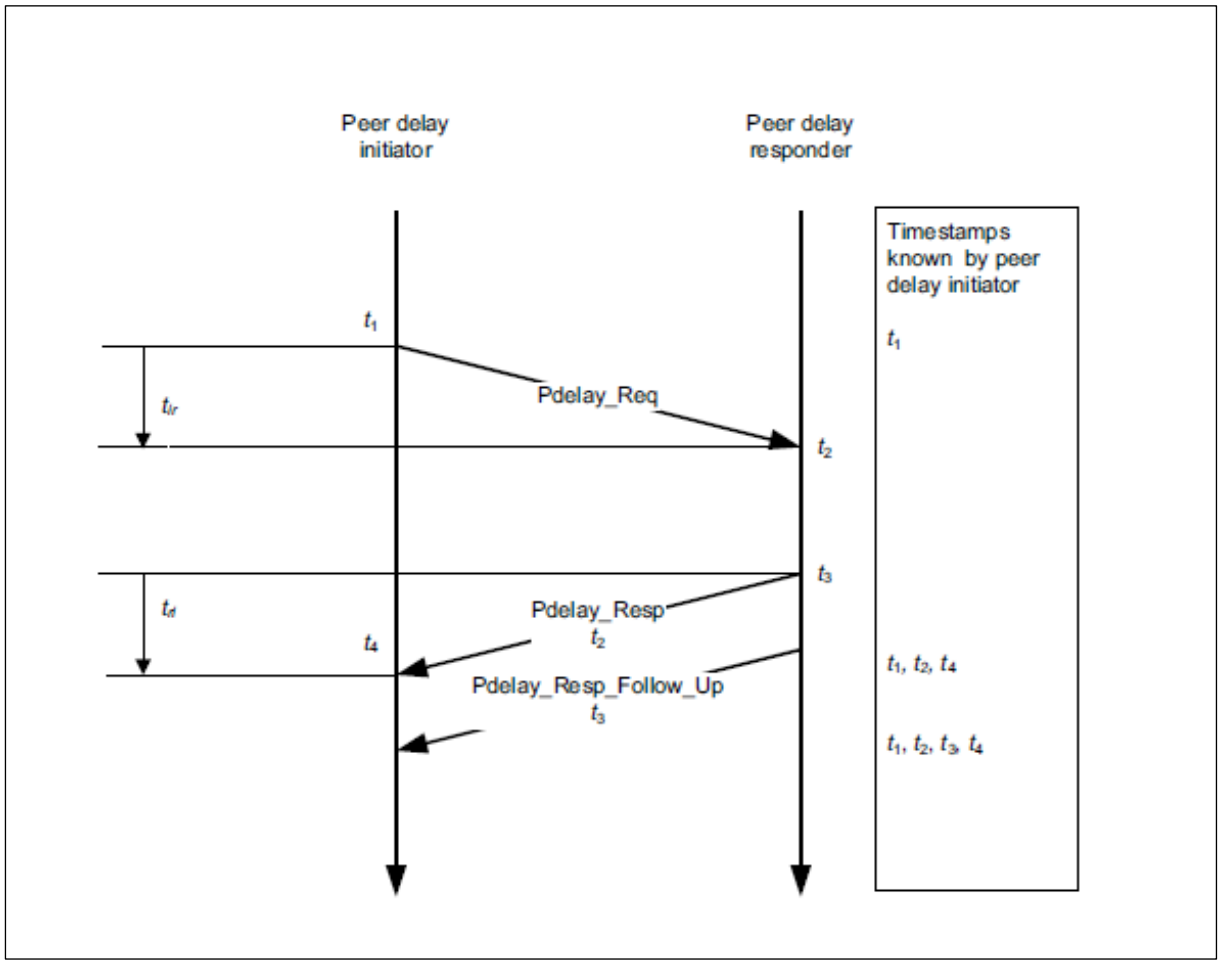


Figure 6-6: P-Delay

The measures of the propagation delays are performed by each attached link of an IEEE 802.1AS communication, which means each side of the link will generate and respond to these packets, called:

- Pdelay_Request

- Pdelay_Response
- Pdelay_Response_Follow_Up

6.4.3 How to set up a synchronized network with Capture Modules and gPTP

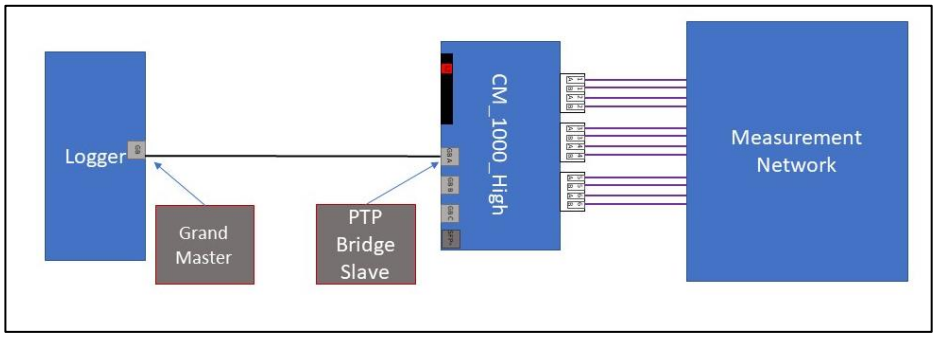


Figure 6-7 Example of gPTP with Logger as Grand Master and CM 1000 High as Bridge Slave

The CM 1000 High can be configured as 802.1AS Grand Master, Bridge Master, or Bridge slave. In Figure 6-7 the CM 1000 High is configured as the bridge slave. When the logger and CM 1000 High are synchronized. The traffic received from the measurement network will be timestamped according to the time received by the 802.1AS Grand Master (Logger).

Steps:

1. Configure the data logger as Grand Master
2. Access the CM 1000 High via a web browser through the assigned IP address (default IP address 10.104.3.144)
3. Navigate to the status tab
4. Select the GB port that is connected to the logger. This port needs to configure as a Bridge slave
5. According to the Figure 6-7 GB-A port of CM 1000 High is connected to the logger. Click on the GB-A port for further options
6. In 802.1AS Mode, select “Slave”
7. In PTP Mode, select gPTP Automotive

The GB-A port of CM 1000 High is acting as a Bridge slave now for the logger (802.1AS Grandmaster). Similar steps can be repeated to configure other GB ports and the SFP+ port as well.

6.5 Optimized packetization

The CM 1000 High offers two different algorithms for packetization. One of them is labeled as *optimized*. To understand better why this option increases efficiency in terms of bandwidth, several examples have been collected here.

Case 1: CM receives 5 ethernet frames with lengths less than MAX_PACKETIZATION_LENGTH

Received frames by CM:

- Frame 1, with payload length 500 bytes.
- Frame 2, with payload length 70 bytes.
- Frame 3, with payload length 800 bytes.
- Frame 4, with payload length 1000 bytes.
- Frame 5, with payload length 64 bytes.

CM behavior:

- Frame 1, payload length is 500 bytes so CM will try to packetize it (frame stored, and it is not sent out). Waiting for the next frame.
- Frame 2, payload length is 70 bytes. CM will try to packetize it. Frame 1+2 < 1250 bytes. Waiting for the next frame.
- Frame 3, payload length is 800 bytes. CM will try to packetize it. Frame 1+2 + 3 **>** 1250 bytes **(!)**. CM sends out packetized packets (1+2) and frame 3 is stored to be packetized.
- Frame 4, payload length is 1000 bytes. CM will try to packetize it. Frame 3+4 **>** 1250 bytes **(!)**. CM sends out packetized packet (3) and frame 4 is stored to be packetized.
- Frame 5, payload length is 64 bytes. CM will try to packetize it. Frame 4+5 < 1250 bytes. Waiting for the next frame.
- TimeOut expired. CM sends packetized packets (4+5).

Output order:

- Packet number 1: Frame 1 + 2.
- Packet number 2: Frame 3.
- Packet number 3: Frame 4 + 5.

Case 2: CM receives 5 Ethernet frames with random length

Received frames by CM:

- Frame 1, with payload length 300 bytes.
- Frame 2, with payload length 1400 bytes.
- Frame 3, with payload length 800 bytes.
- Frame 4, with payload length 1000 bytes.
- Frame 5, with payload length 1500 bytes.

CM behavior:

- Frame 1, payload length is 300 bytes so CM will try to packetize it (frame stored, and it is not sent out). Waiting for the next frame.
- Frame 2, payload length is 1400 bytes. Length bigger than MAX_PACKETIZE_LENGTH so CM will **not** try to packetize it. It will be directly forwarded. A packet with frame 2 inside is sent out and frame 1 is still waiting to be packetized.
- Frame 3, payload length is 800 bytes. CM will try to packetize it. Frame 1+3 < 1250 bytes. Waiting for the next frame.
- Frame 4, payload length is 1000 bytes. CM will try to packetize it. Frame 1+3+4 **>** 1250 bytes**(!)**. CM sends out packetized packets (1+3) and frame 4 is stored to be packetized.
- Frame 5, payload length is 1500 bytes. Length bigger than MAX_PACKETIZE_LENGTH so CM will **not** try to packetize it. It will be directly forwarded. A packet with frame 5 inside is sent out and frame 4 is still waiting to be packetized.
- TimeOut expired. CM sends packetized packets (4).

Output order:

- Packet number 1: Frame 2.
- Packet number 2: Frame 1 + 3.
- Packet number 3: Frame 5.
- Packet number 4: Frame 4.

7 Maintenance

7.1 Firmware update

The latest firmware and documentation for the CM 1000 High can be obtained from the customer portal.

The following sub-chapters highlight the necessary information required for flashing and bootloader updates for the corresponding releases. As with RELEASE 18, there is a big change in the software structure so, please check the following table to find the right chapter for the update process of the Capture Module:

Current release	Update to R17 or lower	Update to R18 or higher
R17 or lower	CHAPTER 7.2	CHAPTER 7.3
R18 or higher	Not possible!	CHAPTER 7.4

Please note: As soon as the device is updated to R18 or higher, a downgrade to R17 or lower is not possible.

7.2 Update from R18 (or lower) to R17 (or lower)

The following sections explain the pre-conditions and update process for firmware release 17 or lower. This section explains how a Capture Module can be upgraded/downgraded from one release to another before release 18.

7.2.1 Preconditions and important information

If an error occurs during the update, please perform a power reset, and try again

Make sure your network adapter is configured in the correct IP address range

Make sure you have installed the latest Java 8 version (e.g., Java 8 update 191).

The device can only be flashed/updated through GB-A port or 100BASE-T1 *Host* port (pin 9, pin 10)

Please follow these update instructions to avoid erroneous states of the device. Technica Engineering may charge support fees for repair services.

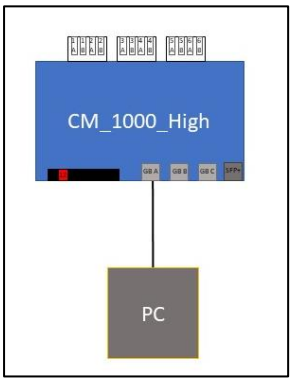
The firewall must be deactivated, or port 69 and port 9000 must be open in the firewall configuration so that the flashing process is successful.

A stable 12 Volt DC power supply is required. Do not switch off the power supply during the update process

Administration privileges on the Windows system are required to perform the firmware update.

If the device is flashed with firmware release 18 or higher, it cannot be flashed back to the releases older than firmware releases 18 anymore

7.2.2 Update via GB-A port



Pre-requisites:

- Please set the device into default settings
- Default flashing script expects the device to be configured with (HW) IP address 10.104.3.144 for CM 1000 High, corresponding to position 0 of the rotary switch
- Before flashing adjust the rotary switch on position “0”
- The CM 1000 High should be directly connected to the flashing PC. No intermediate device like a switch should be present in between

Figure 7-1 Update via GB Port

Steps:

1. User must configure the IP address on the network adapter connected to the CM 1000 High in the same subnet range as the default IP address of CM 1000 High
2. Power up the device
3. Connect the Windows PC with an RJ45 cable directly to the GB-A-Port of the CM 1000 High
4. Make sure there is a link, this can be verified by a simple ping request through the command line or by verifying the LEDs on the GB-A RJ45 interface. If the LEDs are blinking, this suggests an established link
5. Start “Flash ALL_10.104.3.XX.bat” from the same folder as the firmware files are stored
6. Wait until “Press any key ...“ appears in the command window and press any key to exit the command prompt
7. Perform a power reset
8. Check whether the Host-LED is blinking in normal mode.

a. YES => The device has successfully been flashed. The user can access the webpage through the default IP address and check the version of the flashed firmware on the web page under the SYSTEM INFORMATION tab
b. NO => repeat from point 4

➔ Finished

Flashing is also possible via GB-A port with the rotary switch of the device in a position other than 0, but this would imply a different (HW) IP address and the flashing script should be updated then to match the new IP address.

7.2.2.1 Update via 100BASE-T1 host port

This option requires a 100BASE-T1 MediaConverter or a MediaGateway to get access to the device by the 100BASE-T1 config port. The pictures below show the connection of the devices for both versions.
For the version with MediaGateway, please use the provided *.cfg-file. This configuration for MediaGateway is prepared for the update for the exact shown setup.

Pre-requisites:

- Default flashing script expects the device to be configured with (HW) IP address 192.168.0.49.
- Before flashing, the device must be powered up and adjusted to the rotary switch position "F"
- The CM 1000 High should be directly connected to the flashing PC. No intermediate device like a switch should be present in between.

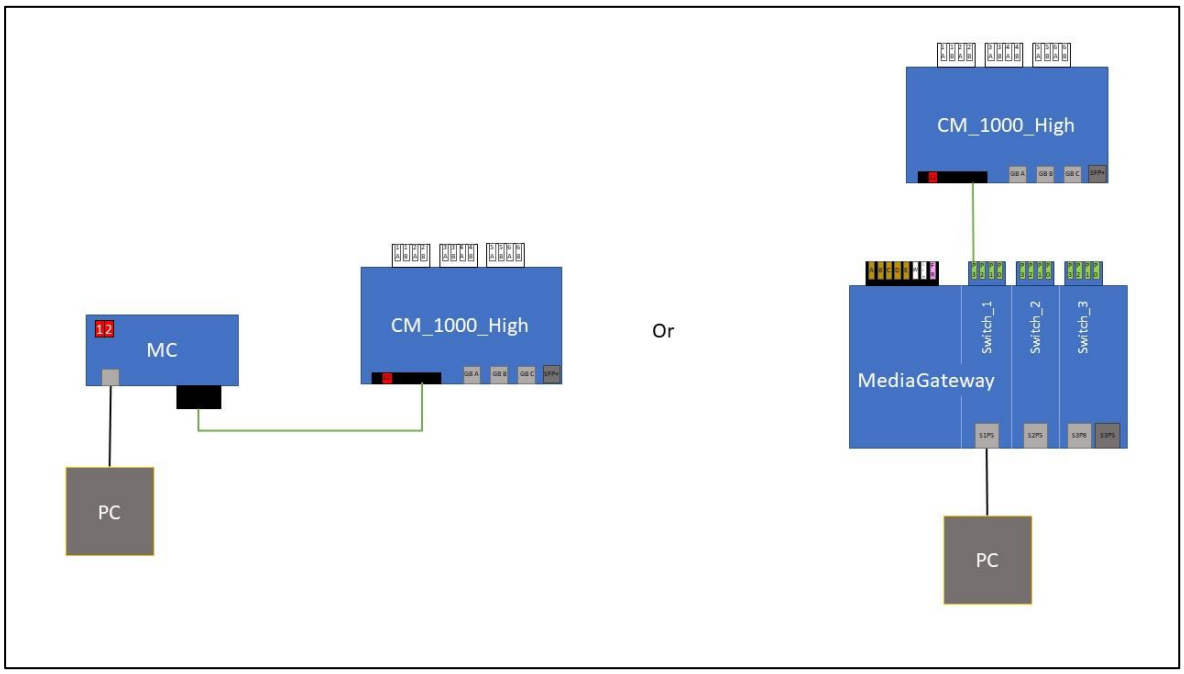


Figure 7-7-2: left side update with MediaConverter, right side update with MediaGateway

Steps:

1. User must configure the IP address on the network adapter connected to the Media Converter or Media Gateway in the same subnet range as the default CONTROLLER IP ADDRESS of the CM (192.168.0.49)
2. Connect a Windows PC with an RJ45 cable directly to the MediaConverter or MediaGateway.
3. Make sure there is a link by checking the LED's status. If the LEDs are blinking normally, this suggests an established link.
4. Start "Flash ALL_192.168.0.49.bat" from the same folder as the firmware files are stored.
9. Wait until "Press any key…" appears in the command window. press any key to exit the command prompt
5. Perform a power reset
6. Check whether the Host-LED of CM is blinking in normal mode.
    a. YES => The device has successfully been flashed. The user can access the webpage through the Controller IP address (192.168.0.49) and check the version of the flashed firmware on the web page under the SYSTEM INFORMATION tab
    b. NO => repeat from point 4

➔ Finished

7.3 Update from R17 (or lower) to R18 or higher

This section explains how a Capture Module can be upgraded from release 17 or any lower firmware release to release 18 and higher.

7.3.1 Preconditions and important information

Please update the device to R16 or R17 before starting the update process to R18.

If an error occurs during the update, please perform a power reset, and try again

Make sure your network adapter is configured in the correct IP address range

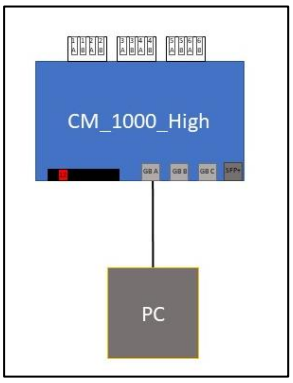
The device can only be flashed/updated through GB-A port or 100BASE-T1 *Host* port (pin 9, pin 10)

Please follow these update instructions to avoid erroneous states of the device. Technica Engineering may charge support fees for repair service.

A stable 12 Volt DC power supply is required. Do not switch off the power supply during the update process

If the device is flashed with firmware release 18 or higher, it cannot be flashed back to the releases older than firmware releases 18 anymore

7.3.1.1 Update via GB-A port



Prerequisites:

- Please set the device into default settings
- Default flashing script expects the device to be configured with (HW) IP address 10.104.3.144 for CM 1000 High, corresponding to position 0 of the rotary switch
- Before flashing adjust the rotary switch on position “0”
- The CM 1000 High should be directly connected to the flashing PC. No intermediate device like a switch should be present in between

Figure 7-7-3: Update on GB port

Steps:

Migration from old bootloader to the new bootloader:

1. User must configure the IP address on the network adapter connected to the CM in the same subnet range as the default IP address of CM
2. Power up the device
3. Connect the Windows PC with an RJ45 cable directly to the GB-A-Port of the CM
4. Make sure there is a link, this can be verified by a simple ping request through the command line or by verifying the LEDs on the GB-A RJ45 interface. If the LEDs are blinking, this suggests an established link
5. Start "MIGRATE_CM_XX_10.104.3.XX.BAT" from the same folder as the firmware files are stored
6. Wait until the command window closes by itself.
7. Perform a POWER RESET
    a. This must be performed
8. Please go to the web browser and type the IP address 10.104.3.XX

↻ The device is now in bootloader mode and the following website should appear:

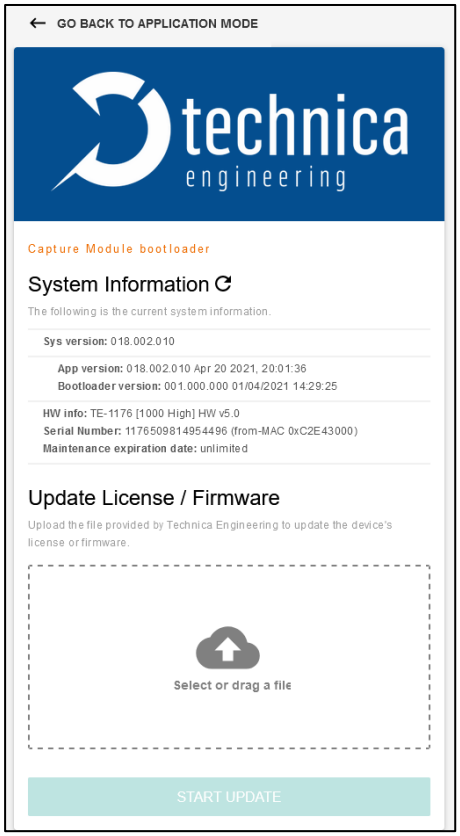


Figure 7-4 Firmware upload

Flashing new R18 or higher

9. Please click into the framed field and open the file FW_CM_XX_SIGNED.TEFW
   Alternatively, it is possible to drag and drop the mentioned file into the framed field

a. If no power rest was performed, a red pop-up appears, and the FW cannot be flashed.
    i. If this is the case, perform the reset, refresh the website, and try it again.

10. Press START UPDATE 

11. Wait until you are prompted by the following window:

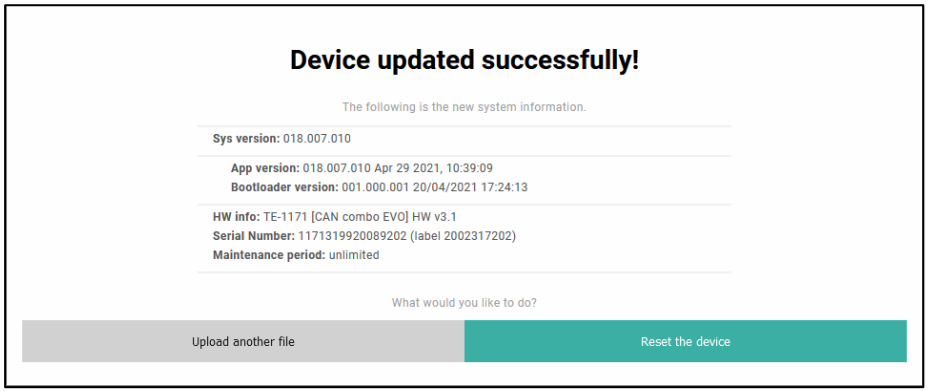


Figure 7-5 Firmware update successful

12. Press reset the device
13. The device will jump into application mode and show you now the new website for configuration.

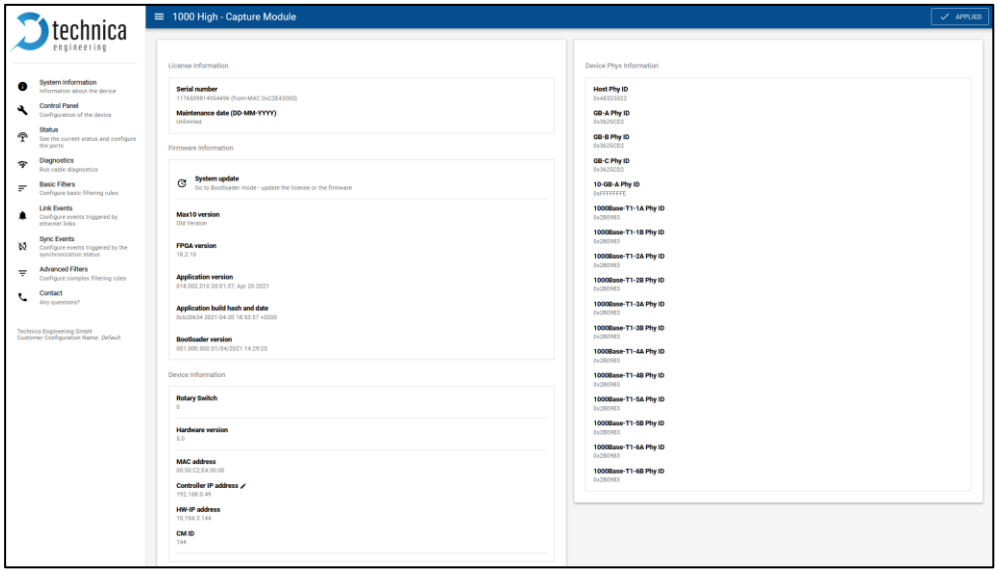


Figure 7-6 Home screen

➔ Finished

7.3.1.2 Update via 100BASE-T1 host port

This option requires a 100BASE-T1 MediaConverter or a MediaGateway to get access to the device by the 100BASE-T1 config port. The pictures below show the connection of the devices for both versions. For the version with MediaGateway, please use the provided *.cfg-file. This configuration for MediaGateway is prepared for the update for the exact shown setup.

Pre-requisites:

- Default flashing script expects the device to be configured with (HW) IP address 192.168.0.49.
- Before flashing, the device must be powered up and adjusted to the rotary switch position "F"
- The CM 1000 High should be directly connected to the flashing PC. No intermediate device like a switch should be present in between.

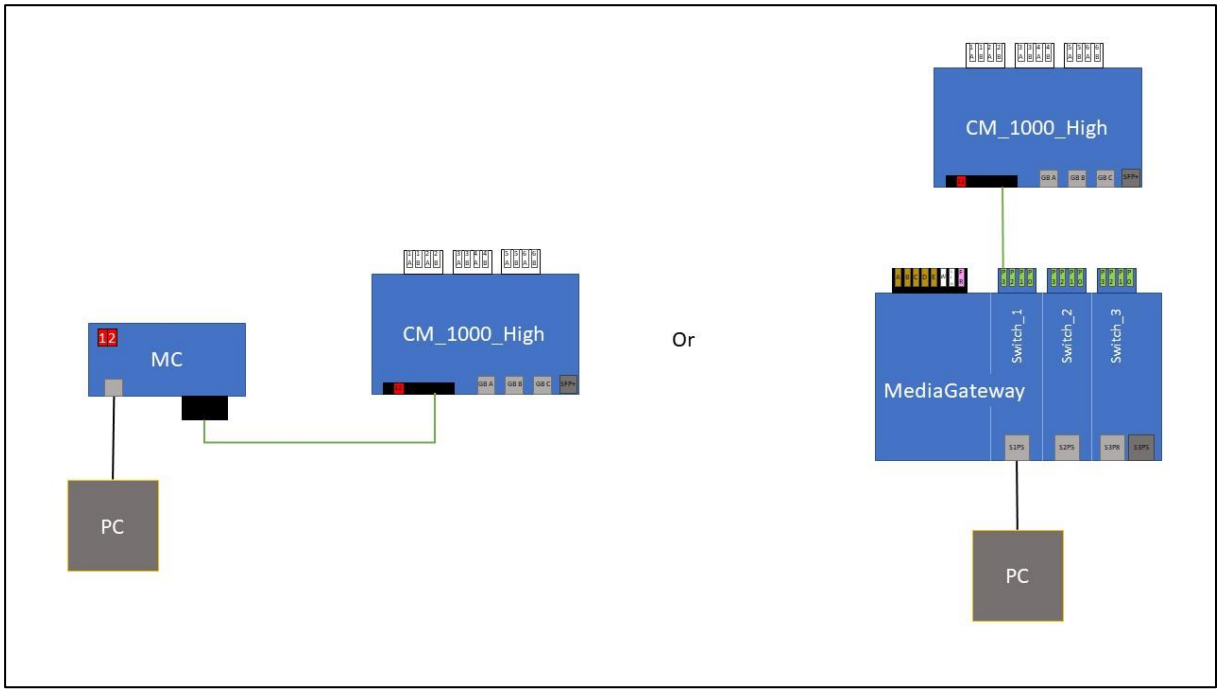


Figure 7-7 left side update with MediaConverter, right side update with MediaGateway

Steps:

Migration from old bootloader to the new bootloader:

1. User must configure the IP address on the network adapter connected to the Media Converter or Media Gateway in the same subnet range as the default Controller IP address of the CM (192.168.0.49)
2. Power up the device
3. Connect a Windows PC with an RJ45 cable directly to the MediaConverter or MediaGateway.

4. Make sure there is a link by checking the LED's status. If the LEDs are blinking normally, this suggests an established link.
5. Start "MIGRATE_CM_XX_192.168.0.49.BAT" from the same folder as the firmware files are stored
6. Wait until the command window closes by itself.
7. Perform a POWER RESET
    a. This must be performed
8. Please go to the web browser and type the IP address 192.168.0.49

The device is now in bootloader mode and the following website should appear:

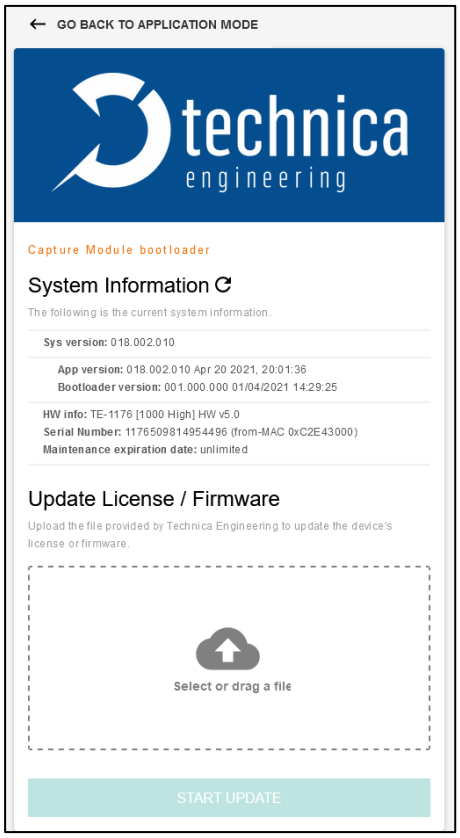


Figure 7-8 Firmware upload

Flashing new R18 or higher

9. Please click into the framed field and open the file FW_CM_XX_SIGNED.TEFW
   Alternatively, it is possible to drag and drop the mentioned file int to the framed field
    a. If no power rest was performed, a red pop-up appears, and the FW cannot be flashed.
        i. If this is the case, perform the reset, refresh the website, and try it again.
10. Press START UPDATE 

11. Wait until you are prompted by the following window:

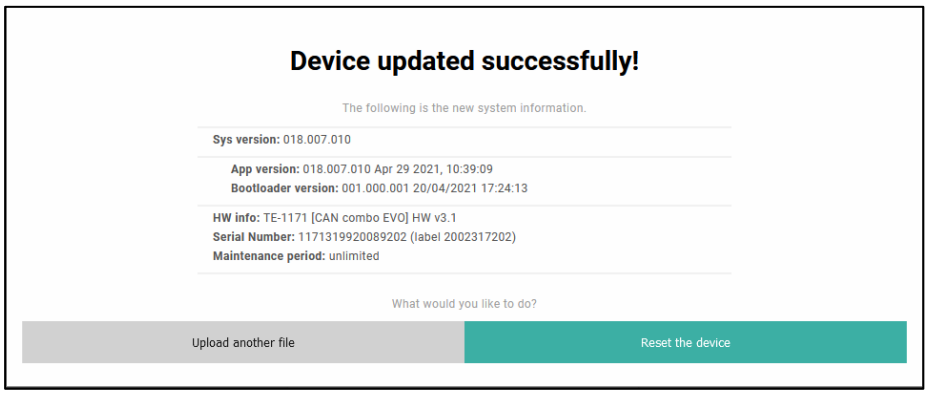


Figure 7-9 Firmware update successful

12. Press “Reset the device”
13. The device will jump into application mode and show you now the new website for configuration.

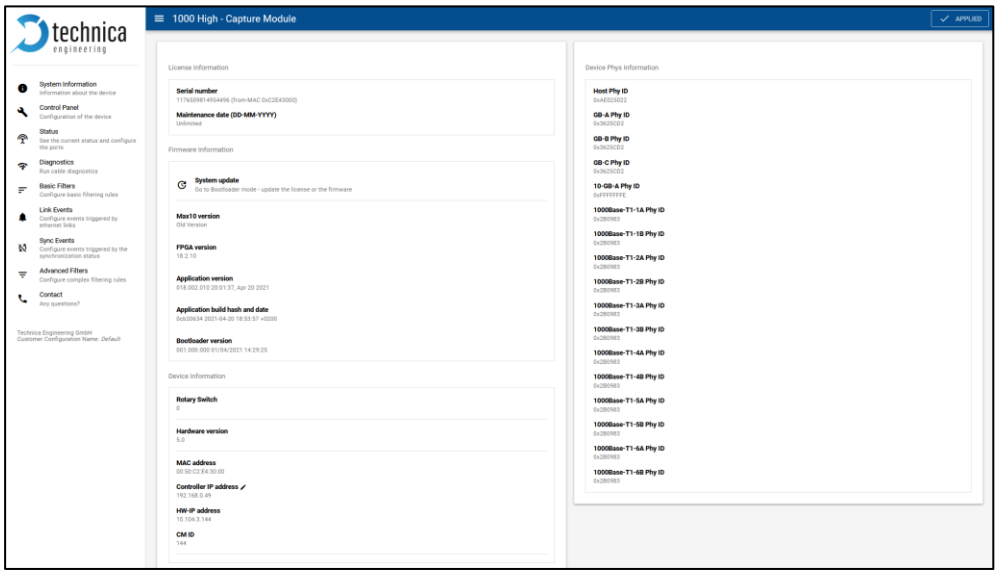


Figure 7-10 Home screen

➔ Finished

7.4 Update of release 18 or higher

This section explains how a Capture Module can be upgraded from release 18 to higher releases.

7.4.1 Preconditions and important information

Please refer to CHAPTER 7.3.1 for pre-conditions

7.4.1.1 Update via GB-A port

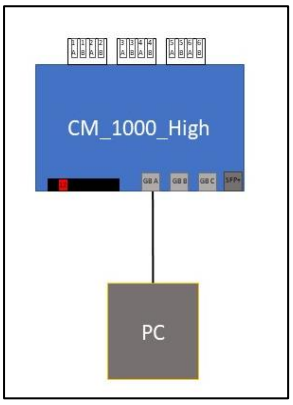


Figure 7-11 Update on GB Port

Pre-requisites:

- Please set the device into default settings

Steps:

1. Open the website through the configuration port and by typing the IP of the device defined by the rotary switch.
2. Go to the "System Information" tab and click System update

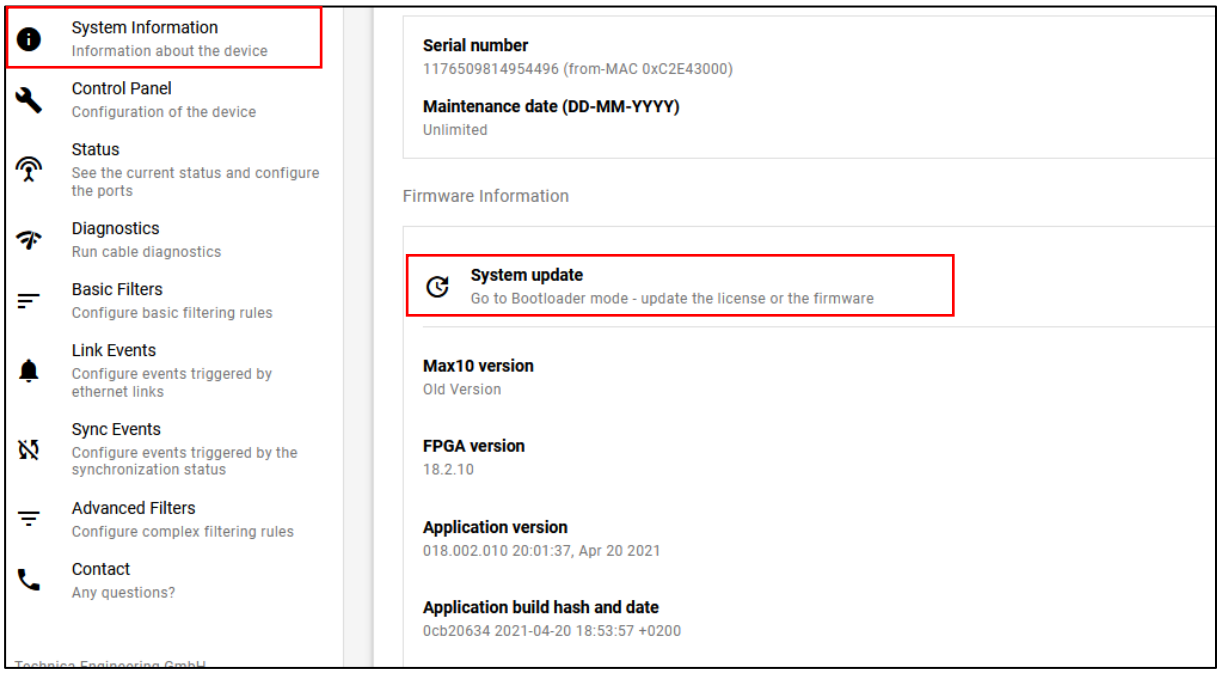


Figure 7-12 System information tab

- The device is now in bootloader mode and the following website should appear:

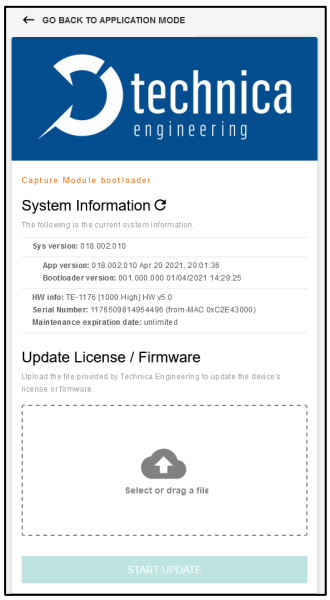


Figure 7-13 Firmware upload

3. Please click into the framed field and open the file FW_CM_XX_SIGNED.TEFW for new firmware or open new license files. Alternatively, it is possible to drag and drop the mentioned files int to the framed field
4. Press START UPDATE 

5. Wait until you are prompted by the following window:

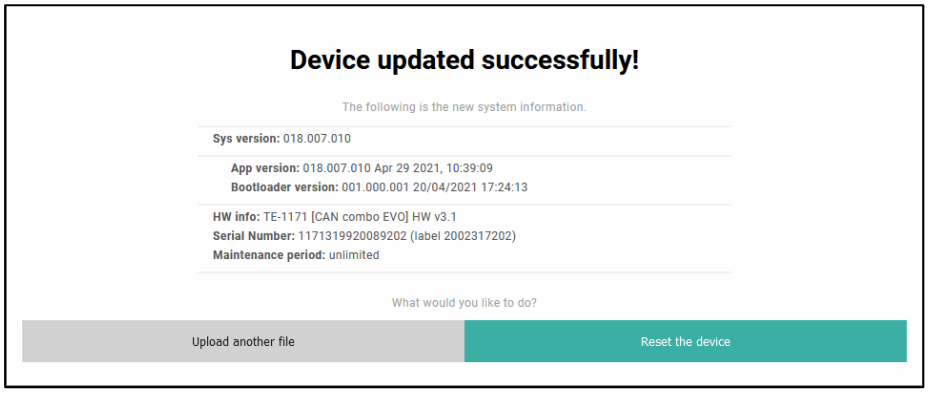


Figure 7-14 Firmware update successful

6. Press “Reset the device” if you are finished with updating.
7. The device will jump into application mode and show you now the website is again for configuration.

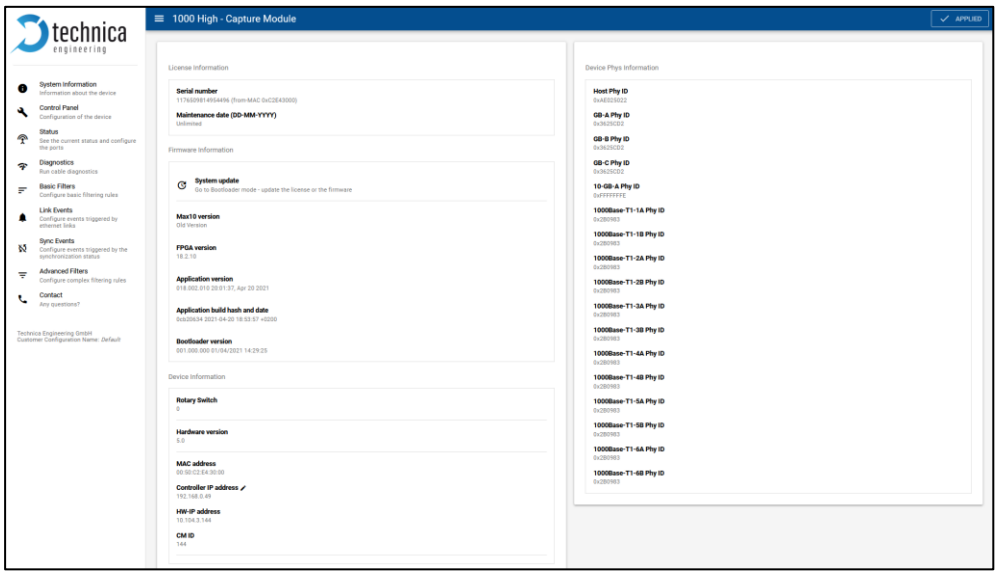


Figure 7-15 Home screen

➔ Finished

7.4.1.2 Update via 100BASE-T1 host port

This option requires a 100BASE-T1 MediaConverter or a MediaGateway to get access to the device by the 100BASE-T1 config port. The pictures below show the connection of the devices for both versions. For the version with MediaGateway, please use the provided *.cfg-file. This configuration for MediaGateway is prepared for the update for the exact shown setup.

Pre-requisites:

- Please set the device into default settings.

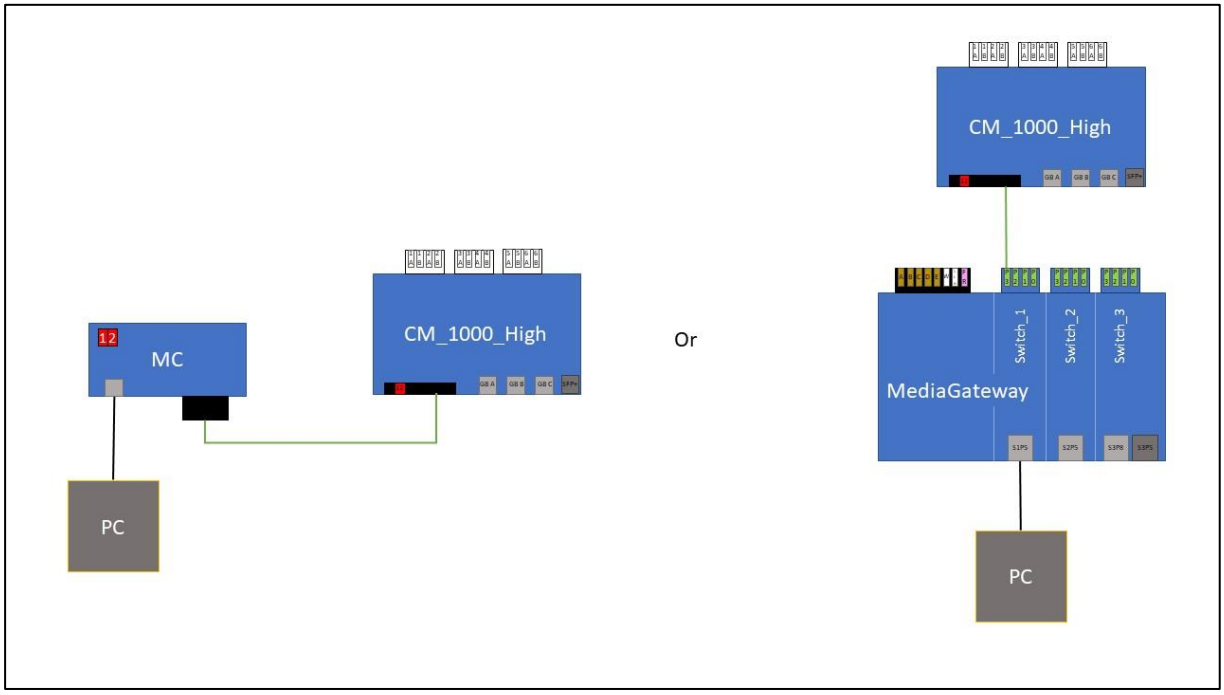


Figure 7-16 Left side update with MediaConverter, right side update with MediaGateway

Steps:

1. Open the website of the device the typing into the browser 192.168.0.49
2. Go to the “System Information” tab and click System update

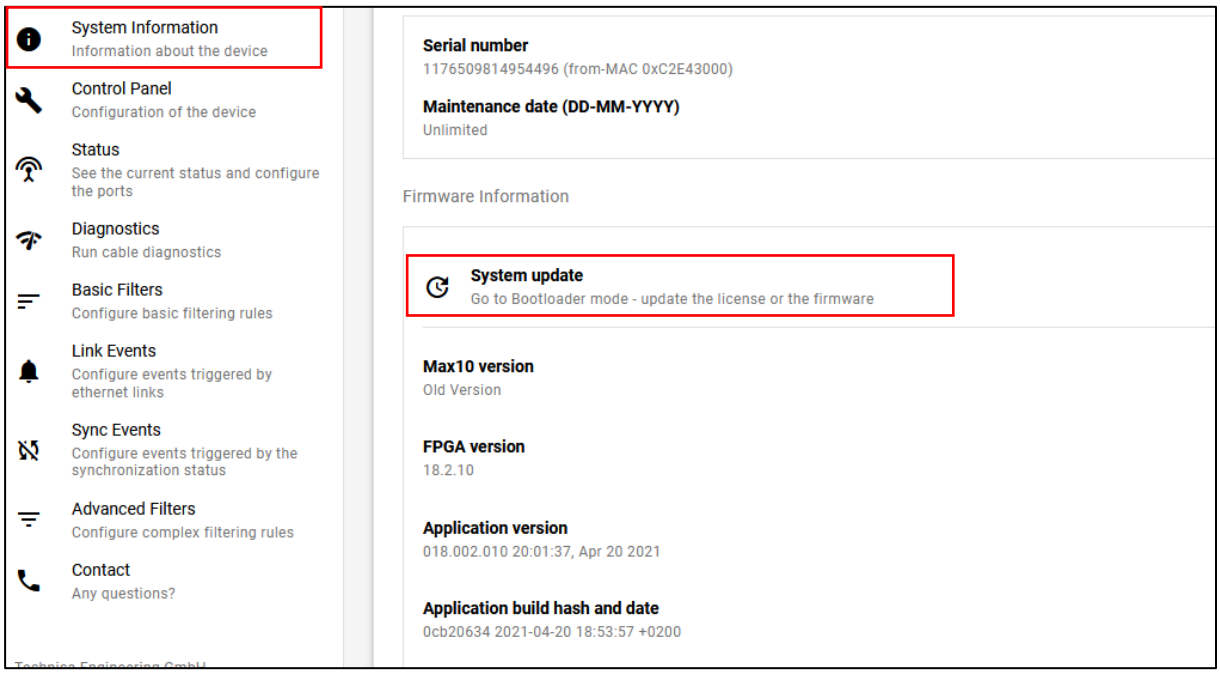


Figure 7-17 System information tab

- The device is now in bootloader mode and the following website should appear:

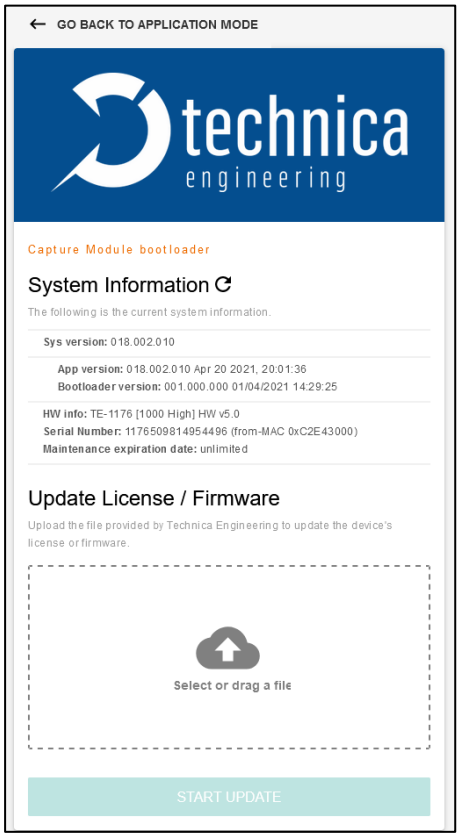


Figure 7-18 Firmware upload

3. Please click into the framed field and open the file FW_CM_XX_SIGNED.TEFW for new firmware or open new license files. Alternatively, it is possible to drag and drop the mentioned files int to the framed field
4. Press START UPDATE 

5. Wait until you are prompted by the following window:

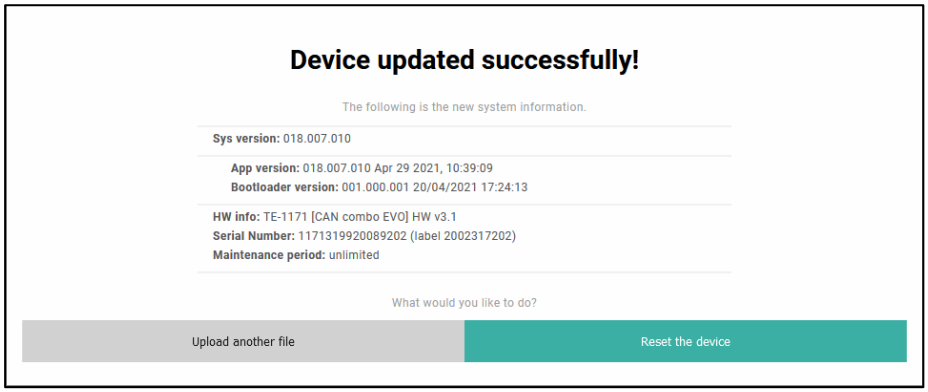


Figure 7-19 Firmware update successful

6. Press “Reset the device” if you are finished with updating.
7. The device will jump into application mode and show you now the website is again for configuration.

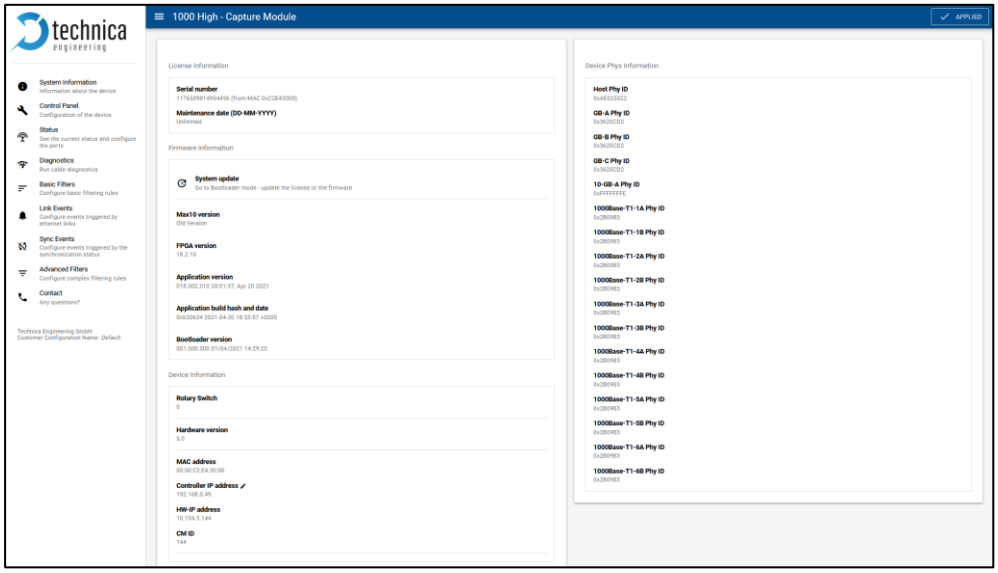


Figure 7-20 Home screen

➔ Finished

7.5 Additional information about R18

- Device config will be lost during the migration/update to R18.
- Do not use old configurations! Please create a new one.
- If the migration to the new bootloader is done, NO downgrade to an old release (lower than R18) is possible!
  - A new serial number (16 digits) will appear on the webpage.
  - The previous SN is still shown.
- The serial number (SN) for status messages will be the same (previous one) as before.

7.6 Error messages during flashing

During the flashing of license or firmware files using the bootloader, several different error messages can appear. These errors appear below the progress bar and consist of a result line (first line) and a description line (second or below line). Additionally, the same text as in the Result also appears in a red popup at the bottom of the website.

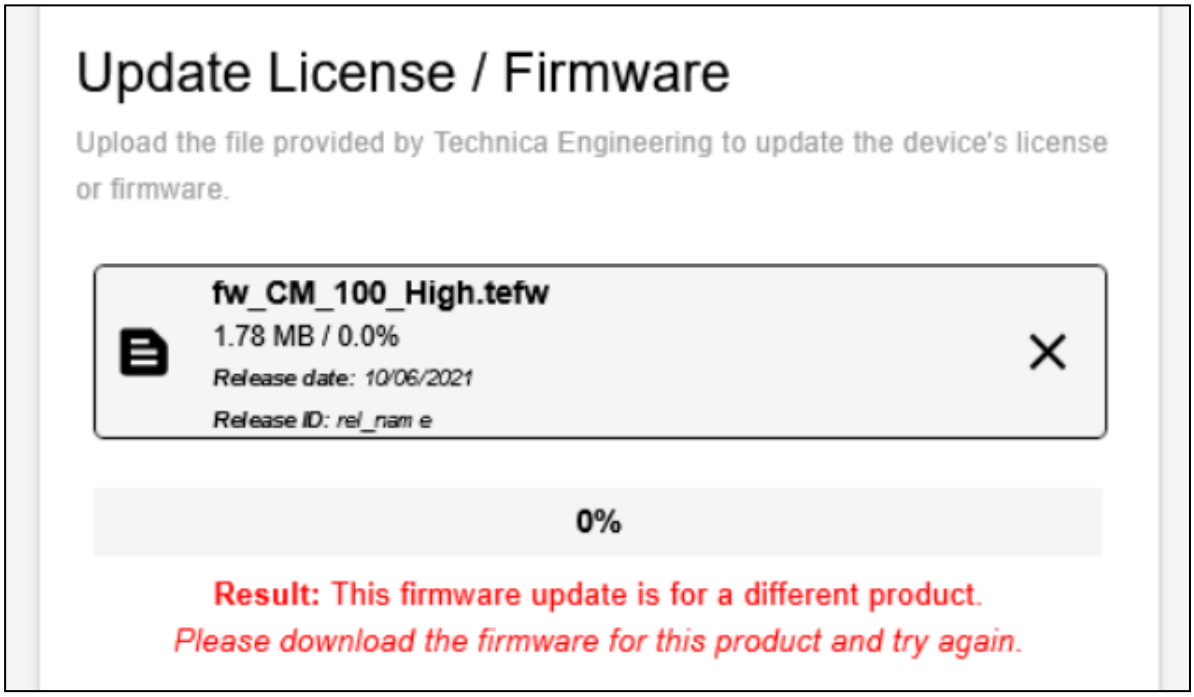


Figure 21 Update license error message

7.6.1 Error Codes

Here is a list of possible results and their description:

1. Result: A firmware update is already running.
   A firmware update is already running. Wait for it to finish and then try again.

2. Result: Internal device error

   An internal device error occurred. Try the process again, if the error persists, contact Technica Engineering for support at support@technica-engineering.de. This error is accompanied by one of the following error codes after the result message:

   - FW_ERR_DEVICE_PREPARE,
   - FW_ERR_DEVICE_ERASE,
   - FW_ERR_DEVICE_PROGRAM,
   - FW_ERR_DEVICE_VALIDATE

3. Result: Input file is not flashable

   There is an error with the supplied file. Please download the firmware for this product again and retry. If the error persists, contact Technica Engineering at support@technica-engineering.de.

4. Result: This firmware update is for a different product.

   The supplied file is for a different Technica product. Download the correct firmware file for your device and try again.

5. Result: Input file is corrupted

   There is an error with the supplied file. Please download the firmware for this product again and retry. If the error persists, contact Technica Engineering at support@technica-engineering.de. This error is accompanied by one of the following error codes after the result message:

   - FW_ERR_IN
   - PUT_TRUNCATED,
   - FW_ERR_INPUT_EXTRA_BYTES,
   - FW_ERR_ZLIB_INFLATE,
   - FW_ERR_INCONSISTENCY_ERROR,
   - FW_ERR_INVALID_ELEMENT_FORMAT,
   - FW_ERR_SIGNATURE_INVALID

6. Result: Input file is corrupted or MAX10 error (FW_ERR_INVALID_TARGET)

   There is an error with the supplied file or the MAX10 firmware is corrupted. Please download the firmware for this product again and retry. If the error persists, contact Technica Engineering at support@technica-engineering.de

7. Result: Internal device error (FW_ERR_NO_KEY).

There is an error with the supplied file. Please download the firmware for this product again and retry. If the error persists, contact Technica Engineering at support@technica-engineering.de.

8. Result: Firmware update is outside of the contracted maintenance license period.

   The device license maintenance date does not allow to flash this firmware release. Upload a license with a later maintenance date and try again.

9. Result: License file is for a different device.

   The supplied file is for a different Technica product. Download the correct license file for your device and try again.

10. Result: Input firmware is too old to be flashed in this unit.

    The firmware file is too old to flash with the current firmware. Please download the latest firmware and try again.

11. Result: Current bootloader is too old to flash this firmware.

    The current bootloader does not support the flashing of this firmware file. Please update the bootloader using the bootloader_updater firmware file and try again.

12. Result: Internal device error (FW_ERR_INTERNAL_ERROR)

    An internal device error occurred. Please try again. If the error persists, contact Technica Engineering at support@technica-engineering.de

7.6.1.1 Miscellaneous errors

Additionally, these error messages can appear in a red popup field at the bottom of the website:

1. Only .tefw and .telc formats supported:

   This error appears if a file is uploaded that doesnot correspond to .telc or .tefw format

2. Error reading the file:

   An unknown error occurred. Try reloading the website and/or redownloading the file and try again.

3. In case of a network error, one of the following messages may appear. In this case, check the network connection, reload the website, and try again.
   - Error getting status.
   - Error getting system information.
   - Error sending the binary file to the device. Please try again.
   - Something wrong happened. Please reload the website.

7.6.1.2 Recovery

In case of a flashing error(e.g., power shortage) the device can go into recovery mode after a restart. The status of the recovery is displayed using the Host LED pattern. During recovery the current state can also be seen on the bootloader webpage (only accessible through HOST port).

1. Recovery ongoing: The LED is alternating between 1 second of fast blinking and 1 second of permanently on status.



2. Recovery successful: The LED resumes to the normal bootloader pattern (fast blinking).



3. Recovery failed: The LED is permanently on.



After the recovery is finished (success or failure) the device must be restarted.

8 Appendix

8.1 TECMP protocol

This section includes an excerpt of the TECMP protocol for reference, with information about the frame layout and content.

Please check the TECMP specification for a detailed description of the protocol.

8.1.1 TECMP frame

The captured data is inserted into a TECMP frame as a payload. Please check the following table:

PRE/ SFD	DST MAC	SRC MAC	Outer 802.1Q Header (optional)	Inner 802.1Q Header (optional)	Ether Typ 0x99FE	TECMP Header	Source Data	FCS/ CRC
8 bytes	6 bytes	6 bytes	4 bytes	4 bytes	2 bytes	28 bytes	Up to n bytes (n <= 1500 bytes)	4 byte

Table 8-4: Complete frame

Field	Description
PRE / SFD	Preamble and Start Frame Delimiter according to IEEE Standard 802.3.
DST MAC	The destination MAC address must be assigned to the multicast address: 01:00:5E:00:00:00
SRC MAC	The source MAC address carries the MAC address of the Capture Modules.
VLAN 802.1Q	optional VLAN fields are provided for possible future extensions.
EtherType	0x99FE: TECMP - Identifies this packet as TECMP packet
TECMP header	See following chapter
Source Data	See following chapter
FSC - CRC	CRC checksum, calculated according to IEEE Standard 802.3

8.1.2 TECMP header

In general, all unspecified types are UINT. The header structure is identical for all messages that are captured using a Capture Module.

4 bytes			
1	2	3	4
CM ID		Counter	
Version	Message Type	Data Type	
Reserved		CM Flags	
Channel ID			
Timestamp Bit 63..32			
Timestamp Bit 31..0			
Length		Data Flags	
Source Data			
...			

Table 8-5: A vehicle data frame in a TECMP frame

4 bytes			
1	2	3	4
CM ID		Counter	
Version	Message Type	Data Type	
Reserved		CM Flags	
Channel ID			
Timestamp Bit 63..32			
Timestamp Bit 31..0			
Length		Data Flags	
Source Data			
...			
Channel ID			
Timestamp Bit 63..32			
Timestamp Bit 31..0			
Length		Data Flags	
Source Data			
...			

Table 8-6: Several vehicle data frames in a TECMP frame

Packing multiple vehicle frames is only done for the same IVN type.

Field Name	Data Type	Description
CM ID	UINT16	Each Capture Module can be configured with a unique ID. This ID can be used to identify different Modules if they are used in the same network
Counter	UINT16	The counter is incremented for each TECMP frame sent from a Capture Module. The counter starts at 0x0001 when the first TECMP frame after power-up is transmitted and increases by 1 at every new transmission. The counter starts again from 0x0000 when reaching the maximum value 0xFFFF. If a Capture Module has several physical interfaces to the sink (Ethernet links), a separate counter is implemented on each interface. The CM ID field is the same for all interfaces in this case. **When replaying data** (Message Type =" Reply data"): This field is set to: "0x000" for every message.
Version	UINT8	The used version of the TECMP protocol; This UINT8 uniquely identifies the version. Technica Engineering will try to synchronize its versions with Legacy versions to establish compatibility.
Message Type	UINT8	Description of the payload function, whether status, configuration, or bus data
Data Type	UINT16	Description of the Type of Source Data (e.g., CAN, LIN, Flexray, Ethernet, Undefined, Voltage, Current, …). These types are preconfigured and fixed
Reserved	UINT16	Reserved, pre-filled with 0x0000
CM Flags	UINT16	0b0000 0000 0000 0001 EoS (End of Segment; used to signal a the last segment of segmented information.) 0b0000 0000 0000 0010 SoS (Start of Segment: used to signal the first segment of segmented information Note: For a single segment transmitted in a TECMP frame, the CM Flags EoS and SoS are always set to '1'. 0b0000 0000 0000 0100 SPY (SPY flag can be used to trigger additional actions on the sink.) Note: For messages from recording on-board communication, the value for the flag is always '1'.) 0b0000 0000 0000 1000 Multi-Frame (a Capture Module that can send multiple Channel IDs in a TECMP frame must set this flag) 0b0000 0000 0001 0000 - reserved 0b0000 0000 0010 0000 - reserved 0b0000 0000 0100 0000 - reserved 0b0000 0000 1000 0000 - reserved

Field Name	Data Type	Description
		0b0000 0001 0000 0000 - reserved 0b0000 0010 0000 0000 - reserved 0b0000 0100 0000 0000 - reserved 0b0000 1000 0000 0000 - reserved 0b0001 0000 0000 0000 - reserved 0b0010 0000 0000 0000 - reserved 0b0100 0000 0000 0000 - reserved 0b1000 0000 0000 0000 Capture Module overflow (Module has lost data)
Channel ID	UINT32	ID that uniquely identifies the log data/bus/link on the IVN.
Timestamp	UINT64	The timestamp is the time in the past 1ns since Thursday, January 1, 1970, 00:00 UTC. The resolution is 1ns. Bit 63: - 1: time is asynchronous, synchronization with the timing master failed - 0: time is synchronous, synchronization with the timing master successful Bit 62: Reserved The timestamp value 0x0000000000000000 is not allowed.
Length	UINT16	Length of Source Data Note: bytes of the length and Data Flags fields are ignored.
Data Flags	16 Bitfield	The Data Flags refer to the messages transmitted on the IVN (100Base-T1, 1000Base-T1, FlexRay, CAN, CAN-FD, LIN). 0b0000 0000 0000 0001 IVN specific please see in the corr. IVN section 0b0000 0000 0000 0010 IVN specific please see in the corr. IVN section 0b0000 0000 0000 0100 IVN specific please see in the corr. IVN section 0b0000 0000 0000 1000 IVN specific please see in the corr. IVN section 0b0000 0000 0001 0000 IVN specific please see in the corresponding. IVN section 0b0000 0000 0010 0000 IVN specific please see in the corr. IVN section 0b0000 0000 0100 0000 IVN specific please see in the corr. IVN section

Field Name	Data Type	Description
		0b0000 0000 1000 0000 IVN specific please see in the corr. IVN section 0b0000 0001 0000 0000 TBD 0b0000 0010 0000 0000 TBD 0b0000 0100 0000 0000 TBD 0b0000 1000 0000 0000 TBD 0b0001 0000 0000 0000 TBD 0b0010 0000 0000 0000 CRC error (CAN, Flexray Frame CRC (trailer Segment: 24bits - 15), Ethernet). This bit indicates that a CRC error was found during logging. 0b0100 0000 0000 0000 TX (Capture Module as sender). This bit indicates a request to the CM to transmit the current message on the specified interface. This bit will not be set in Status messages nor captured messages. 0b1000 0000 0000 0000 Overflow (loss of previous message (s) per IVN)
Source Data (optional)		Data transferred from the vehicle IVN (source data).

Table 7: Description of TECMP header

8.1.3 Control Messages

A control message is a special type of TECMP frame that represents an event triggered.

To control the recording system and all possible connected systems on the logging network (consisting of several loggers, Captures Module, and instrumentation adapters) and signaling events in the recorded trace, control messages are configured in the measurement components. All Capture Modules can be informed about the occurrence of a configurable Control message. Received Control messages must be passed on by all components.

4 bytes			
0	1	2	3
CM ID		Control Message ID	

Table 8: Control message payload overview

The fields of a control message are:

- CM ID: identifier of the device sending the control message.
- Control Message ID: identifier assigned to the specific control message.

CM ID	Counter	Version	Message Type	Data Type	Reserved	CM Flags	Channel ID	Time-stamp	Length	Data Flags	CM ID	Control Message ID

Table 9 Control message overview

8.1.4 Status Messages

Three different status messages are currently supported by the Capture Modules family:

- Status IVN
- Status Capture Module
- Status Configuration

The status messages use the following TECMP header:

CM ID	Counter	Version	Message Type	Data Type	Reserved	CM Flags	Channel ID	Time stamp	Length	Data Flags	Status Message

Table 10: Status messages TECMP overview

The maximal length of status messages is 1400 bytes. If the payload to be transferred does not fit in a single status message, the payload will be segmented and transferred in multiple and successive status messages of the same type, each carrying one segment.

8.1.4.1 Status Capture Module message

Status Capture Module messages are sent cyclically every second after the logging link has been established.

The payload is depicted in the next illustration.

4 bytes			
0	1	2	3
Vendor ID	CM Version	CM Type	Reserved
Vendor Data length (VDL)		CM ID	
Serial Number			
Vendor Data			

Table 11: Status CM Message payload overview

Field Name	Length	Description
Vendor ID	1 byte	Vendor ID – Technica Engineering: 0x0C
CM Version	1 byte	Versioning of the CM
CM Type	1 byte	Type of Capture Module (eg: CAN-Combo
Reserved	1 byte	For future extensions
Vendor Data Length	2 byte	Length of the "Vendor Data" field (in Bytes)
CM ID	2 byte	The configured ID of the Capture Module
Serial Number	4 byte	Serial number of the Capture Module
Vendor Data	n byte	Specific status information (e.g.: temperature processor, or CM's error conditions)

Table 12: Description of the Status CM message payload fields

Vendor Data - for all CM Variants:

Vendor Data << all variants >>			
4 bytes			
0	1	2	3
Reserved	SW Version		
HW Version		Buffer fill level	Buffer overflow
Buffer Size			
Lifecycle			
Lifecycle			
Voltage Integer part	Voltage fractional part	Chassis Temperature	Silicon Temperature

Table 13: Status CM message Vendor Data overview

Field Name	Data Type	Description
Reserved	UINT8	Reserved
SW Version	3x UINT8	vX.Y.Z, one byte for each part. E.g. v8.6.40 à 0x08 0x06 0x28
HW Version	2x UINT8	vX.Y, one byte for each part. E.g. v3.1 à 0x03 0x01
Buffer fill level	UINT8	Percentage of internal memory currently used. Possible values: 0, 10, 20, 30, 40, 50, 60, 70, 80, 85, 90, 95 100% is never possible because the memory is emptied before it reaches that level.
Buffer overflow	UINT8	Flag to indicate whether there is a memory overflow or not. x00: No, x01: Yes
Buffer Size	UINT32	Size of the internal memory in gigabits.

Field Name	Data Type	Description
Lifecycle	UINT64	Time since startup, in ns.
Voltage integer part	UINT8	Voltage value, the part before the comma. E.g: 12,30 V à 12 = 0x0C
Voltage fractional part	UINT8	Voltage value, the part after the comma. E.g: 12,30 V à 30 = 0x1E
Chassis Temperature	SINT8	The temperature of the Capture Module chassis is °C. • x00 up to 0x7E → 0°C up to 126°C • 0x7F → temp ≥ 127°C • 0x80 → Not Available • 0xF6 down to 0xFF -1°C down to -10°C
Silicon Temperature	SINT8	Internal temperature (junction temperature) of the Capture Module processing unit in °C. • x00 up to 0x7E → 0°C up to 126°C • 0x7F → temp ≥ 127°C • 0x80 Not Available • 0xF6 down to 0xFF -1°C down to -10°C

Table 14: Status CM message Vendor Data fields description

8.1.4.2 Status Configuration message

Status Configuration messages are sent cyclically after the logging link has been established. As opposed to the other status messages, which are sent every second, the Status Configuration messages are sent every 60 seconds to balance the number of status messages sent by the Capture Modules.

The payload is depicted in the next illustration.

4 bytes			
0	1	2	3
Vendor ID	CM Version	CM Type	Reserved
Vendor Data Length (VDL)		CM ID	
Serial Number			
Vendor Data			

Table 15: Status Configuration message payload

Field Name	Length	Description
Vendor ID	1 byte	Vendor ID - Technica Engineering: 0x0C

CM Version	1 byte	Versioning of the CM
CM Type	1 byte	Type of CM (e.g.: CAN,FR,Ethernet,LIN) , see Table 18
Reserved	1 byte	For future extensions
Vendor Data Length	2 byte	Length of the “Vendor Data” field (in Bytes)
CM ID	2 byte	Configured Capture Module ID
Serial Number	4 byte	Serial number of the CM
Vendor Data	n byte	Technica Engineering specific CM configuration

Table 17: Status Configuration message fields description

Any Capture Module HW variant can be easily identified based on the notified Serial Number and Capture Module version.

CM ID	Counter	Version	Message Type	Data Type	Reserved	CM Flags	Channel ID	Timestamp	Length	Data Flags
Vendor ID	CM Version	CM Type	Reserved	Vendor Data Length	CM ID	Serial Number	Vendor Data			

Table 18: Status Configuration message overview

CM Version: version of the Capture Module. Used to distinguish different Capture Module variants/revisions.

Vendor ID: set fixed to 0x0C, representing Technica Engineering.

CM Type: hardware variant of the Capture Module.

Hardware variant	CM Type
CM LIN Combo	0x02
CM CAN Combo	0x04
CM 100 High	0x06
CM Eth Combo	0x08
CM 1000 High	0x0A

Table 18: CM Type

Capture Module Serial Number: This field is 4 bytes long. In case that the correct serial number for the device is not saved in the memory, the last 4 bytes of the MAC address will be shown as the serial number.

Vendor Data: Multiple configuration parameters are exchanged with the Capture Module in the form of a serialized JSON configuration file. The format of the Vendor Data field is depicted and described in the next two illustrations.

4 bytes			
0	1	2	3
Version	Reserved	Config Message ID	
Total Length (Bytes)			
Total number of segments		Segment number	
Length of segment (Bytes)		Segment Data	...
...			

Table 20: Vendor Data overview

Field Name	Data Type	Description
Version	UINT8	The version number of the Status Configuration message vendor data format
Reserved	UINT8	Reserved field: 0x00
Config Message ID	UINT16	Identifier incremented by one at every new cyclic transmission
Total Length	UINT32	Sum of the lengths of the individual segments
Total number of segments	UINT16	Number of segments onto which the information is distributed. As many numbers of Status Configuration messages as segments will be generated per cyclic transmission
Segment number	UINT16	Identification of the present segment: between 0 and the value of the field “Total number of segments”-1
Length of segment	UINT16	Length (x) of the following “Segment Data” field in Bytes
Segment Data	x * UINT8	Serialized configuration

Table 21: Status Configuration message Vendor Data fields description

8.1.4.3 Status IVN message

Status IVN messages are sent cyclically every second after the logging link has been established. Capture Modules with multiple interfaces send the status of all connected IVN's. The payload is depicted in the next illustration.

4 bytes			
0	1	2	3
Vendor ID	CM Version	CM Type	Reserved
Vendor Data Length (VDL)		CM ID	
Serial Number			
Bus Data			

Table 22: Status bus/IVN message payload overview

4 bytes
Bus Data Entry 1
Bus Data Entry 2 (optional)
Bus Data Entry 3 (optional
Bus Data Entry 4 (optional)
Optional further Entries ...

Table 23: Bus Data

The format of the Bus Data Entry depends on the value of the "Version" field inside the header:

4 bytes
Channel ID
Messages Total
Errors Total
Vendor Data

Table 24: Bus Data Entry

Field Name	Length	Description
Vendor ID	1 byte	Vendor ID - Technica Engineering: 0x0C
CM Version	1 byte	Versioning of the CM
CM Type	1 byte	Type of CM (e.g.: CAN,FR,Ethernet,LIN)
Reserved	1 byte	For future extensions
Vendor Data Length	2 byte	Length of the "Vendor Data" field per Bus (in Bytes)
CM ID	2 byte	Configured Capture Module ID
Serial Number	4 byte	Serial number of the CM
Channel ID	4 byte	ID that uniquely identifies the log data/bus/link of the IVN on which the Messages/frames were received or sent
Messages Total	4 byte	Total number of messages received on the bus/IVN (Channel ID) since the startup of the CM
Errors Total	4 byte	erroneous messages received on the bus/IVN (Channel ID) since the start of the Capture Module

Vendor Data	n byte	See description below

Table 25: Description of the Status bus/IVN message payload fields

For one IVN interface:

CM ID	COUNTER	VERSION	Message Type	DATA TYPE	RESERVED	CM Flags	Channel ID	TIMESTAMP	LENGTH	Data Flags
VENDOR ID	CM VERSION	CM TYPE	RESERVED	LENGTH	CM ID	Serial Number	Channel ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA

Table 26: Packaging a single Status IVN in one Status IVN message

For multiple IVN interfaces:

CM ID	COUNTER	VERSION	Message Type	DATA TYPE	RESERVED	CM Flags	Channel ID	TIMESTAMP	LENGTH	Data Flags
VENDOR ID	CM VERSION	CM TYPE	RESERVED	VENDOR DATA LENGTH	CM ID	Serial Number	Channel ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA
							Channel ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA
							Channel ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA

Table 27: Packaging multiple Status IVN in one Status IVN message

Messages Total / Errors Total: Counted are the absolute received/sent messages and messages with errors. An error increases both the Messages Total and the Errors Total value. Further information can be transmitted in the Vendor Data.

Vendor Data: The Vendor Data field is used to convey specific information in every Capture Module variant. Consequently, the field has a different meaning in different Capture Module types. As an example, for both the CM Ethernet Combo and CM 100 High variants:

Vendor Data << Ethernet variants only >>	
Bytes	
0	1
Link Status	Link Quality
Linkup Time	

Table 28: Status IVN message Vendor Data example

Field Name	Data Type	Description
Link Status	UINT8	Flag to indicate whether there is linkUp or not in the corresponding channel: x00: No x01: Yes
Link Quality	UINT8	Quality of the link of the corresponding channel, from 0 to 5: 0x00: link down 0x01: Lowest Link quality … 0x05: Highest Link quality
Linkup Time	UINT16	Time (ms) measured between power-up of the board and link-up on this port. Done once at startup. Two default reference values are defined based on an internal reference timeout elapsing 500ms after power-up. • 0x0000: No linkup detected, and the reference timeout did not elapse yet. • 0xFFFF: No linkup detected, and the reference timeout already elapsed. For instance: x153 means 339 ms.

Table 29: Link Status, Link Quality, and Linkup Time fields description

8.1.5 UDP Discovery messages

A mechanism to make it possible to detect and find all the Capture Modules present in a logging network has been implemented. This discovery mechanism is based on UDP messages, comprising a multicast discovery request and a unicast discovery response.

This makes it possible to discover all Capture Modules connected and retrieved for each of the basic information such as CM ID, firmware version, hardware version, etc.

8.1.5.1 Broadcast discovery request

This message is a UDP broadcast message with the following parameters:

- Destination MAC address: FF-FF-FF-FF-FF-FF
- Destination IP address: 10.104.3.255
- UDP Src Port = UDP Dst Port = 9999

The following values are used for the UDP payload:

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0000
Length options	UINT32	0x00000000
Options	Variable	None
Reserved	UINT32	0x00000000

Unicast discovery response

The Capture Module responds to a multicast discovery request with a unicast response within a maximum time interval of 2 seconds. The response is only sent to the port of the Capture Module to which the request was addressed.

- Source MAC address: MAC address of the Capture Module
- Destination MAC address: MAC address of the logging endpoint
- Destination IP address: IP address of the logging endpoint
- Source IP address: configured IP address of the Capture Module
- UDP Src Port = UDP Dst Port = 9999

The following values are used for the UDP payload:

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0001
Length options	UINT32	0x0000001C

Options	Variable	parameter1 [6B]: Current MAC address parameter2 [4B]: Current source IP address (default 192.168.0.49): parameter3 [6B]: Destination multicast address (default 01-00-5E-00-00-00) parameter4 [4B]: Current destination IP address parameter5 [4B]: Capture Module ID parameter6 [4B]: Capture Module firmware version
Reserved	UINT32	0x00000000

Example for options:
0xC47D46155922 C0A80033 01005E000000 008041AEFD7E 00112233 00000103.
The Capture Module has:

- MAC address C4:7D:46:15:59:22
- Source IP address 192.168.0.49
- Destination multicast address 01005E000000
- Destination IP address 192.168.0.255
- Capture Module ID 0x00112233 and
- Capture Module firmware version 1.03

8.1.6 UDP Configuration messages

The Capture Modules provide a specific mode for remote device configuration, without the need to access the built-in webserver.

By default, any Capture Module is in logging mode, ready to capture any traffic observed on the IVN interfaces. It only switches to remote configuration mode when it receives on the configuration port a UDP message with a specific layout and content that enables this mode.

The configuration mode is not enabled persistently. To keep the Capture Module in this mode, the UDP message for mode activation must be sent cyclically from the logging endpoint every 20 seconds. While the device is in configuration mode, it is ready to receive and process configuration requests, transmitted also via UDP messages.

As soon as the UDP messages for mode activation disappear, and after a short timeout, the Capture Module goes back into logging mode and logging functionality is reactivated.

Details about how exactly this remote configuration mode can be found below.

All configuration messages, including the one used for mode activation, are based on the IP/UDP protocol (EtherType: 0x0800) and share the same header data:

- Source MAC address: MAC address of the DLM
- Destination MAC address: MAC address of the Capture Module
- Source IP address: IP address of the DLM
- Destination IP address: IP address of the Capture Module
- UDP src port = UDP dst port = 9998

And the same layout for the UDP payload:

<table>
<tr><td colspan="4">Config Message</td></tr>
<tr><td colspan="4">4 bytes</td></tr>
<tr><td>0</td><td>1</td><td>2</td><td>3</td></tr>
<tr><td colspan="2">Version</td><td colspan="2">Operation</td></tr>
<tr><td colspan="4">Length Options</td></tr>
<tr><td colspan="4">Options
...</td></tr>
<tr><td colspan="4">Reserved</td></tr>
</table>

The meaning of each field is described in the table below:

Field name	Data type	Description
Version	UINT16	Indicates the version of the control API
Operation	UINT16	Indicates the operation to be performed
Length options	UINT32	Specifies the number of bytes to be read after this field in order to register and analyze all of the options required for the preceding operation.
Options	Variable	Options or parameters required for a particular operation.
Reserved	UINT32	Reserved for future use. If not used, it must be set to 0x00000000

8.1.6.1 Set Configuration mode

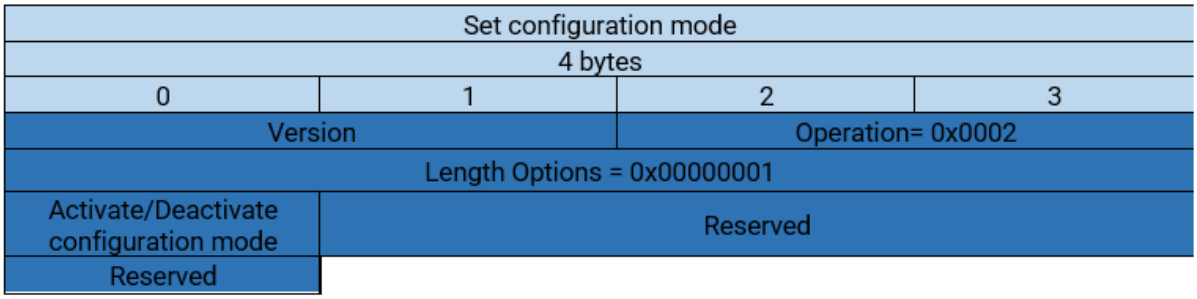
As explained at the beginning of this section, to set the Capture Module in configuration mode, an instance in the system (usually the logging endpoint) has to send cyclically a UDP message containing the command for mode activation.

This command is expected to be received every 20 seconds. If this message fails to arrive, the Capture Module switches back to its initial state on its own after 25 seconds.

In this configuration mode, the Capture Module deactivates logging and enables other functions that allow the device configuration.

Most of the configuration settings are only possible in this special configuration mode.

The following values are used for the mode activation command:



Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0002
Length options	UINT32	0x00000001
Activate/Deactivate configuration mode	UINT8	0x00 = Deactivate configuration mode 0x01 = Activate configuration mode
Reserved	UINT32	0x00000000

Figure 8-1 Set UDP Configuration mode

An example to set configuration mode with correct format:
UDP payload = 0x0001 0002 00000001 01 00000000

Payload can also be set without reserved bytes as below:
UDP payload = 0x0001 0002 00000001 01

8.1.6.2 Set IP address

In configuration mode only. A restart is necessary after performing this configuration. If there is no restart the IP value is old even though the website shows the new IP.

This command allows to change the HW IP address of the Capture Module, configurable when the rotary switch of the device is in position E.

Note that this command can be sent anytime while in configuration mode, independently of the position of the rotary switch. Changes won't be applied until this is switched to position E.

The following values must be used for the *Set IP address* command:

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0003
Length options	UINT32	0x00000004
Options	UINT32	IP address when rotary switch in position E e.g. 0xC0A80033 for 192.168.0.51
Reserved	UINT32	0x00000000

Description of discovery sequence:

1. Discover devices.
2. The configuration instance saves the list of Capture Modules, obtains the corresponding IP address, and uses that IP address to allocate a new IP address. No response is received for this command.
3. IP address requests may be tested by means of a new multicast request. Alternatively, a direct request may be used, e.g. Get Capture Module information.
4. If not OK, the error status shall be reported by the configuration instance.

Set multicast destination MAC address

In configuration mode only.

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0004
Length options	UINT32	0x00000006
Options	UINT48	MAC address e.g. 0x008041AEFD7E for 00-80-41-ae-fd-7e
Reserved	UINT32	0x00000000

It sets the multicast MAC address for a destination

8.1.6.3 Set device time

⚠ In configuration mode only.

It is the same as when configured via device website, set time will be lost after device reset.

Set device time			
4 bytes			
0	1	2	3
Version		Operation= 0x010B	
Length Options = 0x0008			
Ignore PTP Slaves	Unix Timestamp		
Unix Timestamp			
Unix Timestamp			

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x010B
Length options	UINT32	0x00000009
IgnorePTPslaves	UINT8	if greater than 0, only the time of device is not in a network (all ports configured as 802.1AS Mode == Disabled) or Grand Masters will be configured.
Unix Timestamp	UINT64	Unix Timestamp in nanoseconds

Example: February 3rd , 2023, 09:12:29 and ignore PTP slaves
Payload: 00 01 01 0b 00 00 00 09 00 17 40 44 7d b0 39 c0 00

8.1.6.4 Get Capture Module information

In the request, the user can ask for information regarding the CM HW or the Logging Interfaces.

Request

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0006
Length options	UINT32	0x00000001
Options	UINT08	0x01= Get CM logging interfaces 0x02 = Get CM HW information
Reserved	UINT32	0x00000000

Response (Get CM HW information)

The response message when information about CM HW is requested

Response time: no longer than 2 seconds.

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0008
Length options	UINT32	0x00000002 + 0x08 * IVN Interface Count
Options	Variable	parameter1[2B]: IVN_INTERFACE_COUNT. Total number of bus interfaces of the Capture Module. The IVN_INTERFACE_ID, MsgType and Channel ID fields are included in this message for every bus interface. parameter2 [2B]: INTERFACE_ID. Device internal ID for identification of the IVN interface. parameter3 [2B]: MsgType. The type of the IVN interface with the preceding IVN_INTERFACE_ID. parameter4 [4B]: Channel ID

		The configured channel ID of the bus interface with the preceding IVN_INTERFACE_ID *<< … Next IVN_INTERFACE_ID >>*
Reserved	UINT32	0x00000000

Response (Get CM logging interfaces)

The response message when information about CM logging interfaces is requested

Response time: no longer than 2 seconds.

Field name	Data type	Value
Version	UINT16	0x0001
Operation	UINT16	0x0007
Length options	UINT32	0x00000007
Options	Variable	parameter1 [1B]: CM vendor ID parameter2 [1B]: CM type parameter3 [4B]: CM serial number parameter4 [1B]: Total number of logging, synchronization, and configuration ports of a Capture Module
Reserved	UINT32	0x00000000

Import configuration

In configuration mode only.

It imports any .cfg file exported by the Capture Module.

Import Config			
4 bytes			
0	1	2	3
Version		Operation= 0x0101	
Length Options			
Seq. number	Total number	Length total	
Serialized config (n bytes)			

Field name	Data type	Description
Seq. Number	UINT8	Eth. Frame sequence out of the total number of frames (1 based)
Total Number	UINT8	Total Number of Eth. Frames needed to send the whole information

Length options	UINT32	Length of the payload of the current frame. Length options = 1 byte (seq.number) + 1 byte (TotalNumber) + 2Bytes (Length total) + nBytes (serialized config in this frame)
Length total	UINT16	total length in Bytes of the serialized config
Reserved	UINT32	0x00000000

Export configuration

In configuration mode only.

It pulls the current configuration from the Capture Module. Note that the configuration may be split into more than one ethernet frame.

Export Config Request			
4 bytes			
0	1	2	3
Version		Operation=0x0102	
Length Options=0x00000001			
Action=0x01	Reserved		

Field name	Data type	Description
Action	UINT8	Action 0x01 to pull (with any other the message will not trigger the export of the config)
Length Options	UINT32	The value of this field must be 0x01 (with any other the message will not trigger the export of the config)

Export Config Response			
4 bytes			
0	1	2	3
Version		Operation=0x0102	
Length Options			
Seq. number	Total number	Length total	
Serialized config (n bytes)			

Field name	Data type	Description

Length Options	UINT32	Length of the payload of the current frame Length options = 1 byte (seq. number) + 1 byte (total number) + 2 bytes (length total) + n bytes (serialized config in this frame)
Seq. Number	UINT8	Ethernet frame sequence out of the total number of frames (1-based)
Total Number	UINT8	Total number of Ethernet frames needed to send the whole information
Length total	UINT8	Total length in bytes of the serialized config

Restart device

It simply restarts the Capture Module.

Restart			
4 bytes			
0	1	2	3
Version		Operation= 0x0103	
Length Options = 0x00000001			
Restart=0x01	Reserved		

Field name	Data type	Description
Restart	UINT8	0x01 to perform a restart With any other value, the message will not trigger the restart
Length Options	UINT32	0x01 With any other value, the message will not trigger the restart
Reserved	UINT32	0x00000000

Mute/Unmute PHY

This instruction is meant to be used in a testing environment. These changes are applied immediately, no reset is needed.

If transparency mode is active for a pair, this message will not be considered for the pair. A negative acknowledge (NACK) message can be sent so that the user will know that the remote-controlling action was not executed. The PHY to be muted given its Interface ID is muted immediately.

For firmware release 021.005.XXX and onwards, the user must not fill the reserved bytes with 000000, otherwise the UDP message will be treated incomplete and will not Mute/Unmute the PHY.

Mute PHY			
4 bytes			
0	1	2	3
Version		Operation= 0x0104	
Length Options = 0x5			
Channel ID			
Mute	Reserved		

Field name	Data type	Description
Mute	UINT8	0: false 1: true_normal
Reserved	UINT12	Reserved for future use. If not used, it must be set to 0x000000

Set PHY transparency mode

It sets the transparency mode on the desired pair to none, transparency mode, or transparency delay mode.

A restart will be needed to apply the new value.

Set Transparency Mode			
4 bytes			
0	1	2	3
Version		Operation= 0x0105	
Length Options = 0x00000002			
Pair	Mode	Reserved	

Field name	Data type	Description
Version	UINT16	0x0001
Operation	UINT16	0x0105
Pair	UINT8	Pair of IVN interfaces to apply the new

		configuration on. The value goes from 1 to the maximum number of pairs depending on the variant. For example: For ETH Combo the possible pairs are 1, 2 and 3
Mode	UINT8	0x00: none 0x01: transparency normal 0x02. Transparency with delay 0x03 transparency with event-based delay 0x04 transparency with defined delay
Reserved	UINT8	Reserved for future use. If not used, it must be set to 0x0000

Set PHY delay time

Set the delay time for the transparency delay time on a pair

A restart will be needed to apply the new value.

Set Delay Time			
4 bytes			
0	1	2	3
Version		Operation= 0x0106	
Length Options = 0x00000005			
Delay time			
Pair	Reserved		

Field name	Data type	Description
Version	UINT16	0x0001
Operation	UINT16	0x0106
Length options	UINT32	0x00000005
Delay time	UINT32	Delay time for transparency mode in ms
Pair	UINT8	Pair of IVN interfaces to apply the new configuration on. The value goes from 1 to the maximum number of pairs depending on the variant. For example: For ETH Combo the possible pairs are 1, 2 and 3

Acknowledge

Sent by the Capture Module as a response to the requests listed above.

Some operations are not possible. E.g., trying to mute a PHY while its pair is in transparency mode. So, when we receive a valid message with an invalid configuration, we send these UDP messages informing the user that the operation was invalid.
See that the operation code of the acknowledged operation is sent in this message.
After looking at the code these operations are currently being acknowledged:

- Mute/Unmute PHY
- Set PHY Transparency Mode
- Set PHY Delay Time

Field name	Data type	Description
Acknowledged operation	UINT16	The operation is acknowledged with this ack_message
Error code	UINT8	• Only != 0x00 if Result == 0x01 • List of error codes defined until now 0x00: No Error 0x01: Not Permitted
Result	UINT8	• 0x00: OK • 0x01: Not_OK

Set PHY event-based delay time

Sets the event-based delay time for the transparency sleep mode on a pair

⚠ In configuration mode only.

A restart will be needed to apply the new value.

Set Event-based Delay Time			
4 bytes			
0	1	2	3
Version		Operation= 0x010C	
Length Options = 0x00000009			
Transparency wake-up Delay time			
Transparency sleep time			
Pair	Reserved		

Field name	Data type	Description
Version	UINT16	0x0001
Operation	UINT16	0x010C
Length options	UINT32	0x00000009

Transparency wake-up delay time	UINT32	Transparency wake-up delay time in ms
Transparency sleep delay time	UINT32	Transparency sleep delay time in ms
Pair	UINT8	Pair of IVN interfaces to apply the new configuration on. The value goes from 1 to the maximum number of pairs depending on the variant. For example: For ETH Combo the possible pairs are 1, 2 and 3

Set PHY defined delay time

Sets the defined delay time for the transparency sleep mode on a pair

In configuration mode only.

A restart will be needed to apply the new value.

Set Defined Delay Time			
4 bytes			
0	1	2	3
Version		Operation= 0x010D	
Length Options = 0x00000009			
Transparency wake-up Delay time			
Transparency sleep time			
Pair	Reserved		

Field name	Data type	Description
Version	UINT16	0x0001
Operation	UINT16	0x010D
Length options	UINT32	0x00000009
Transparency wake-up delay time	UINT32	Transparency wake-up delay time in ms
Transparency sleep delay time	UINT32	Transparency sleep delay time in ms
Pair	UINT8	Pair of IVN interfaces to apply the new configuration on. The value goes from 1 to the maximum number of pairs

		depending on the variant. For example: For ETH Combo the possible pairs are 1, 2 and 3

8.1.6.5 Fan configuration

In configuration mode only.

Configuration fan PWM

Config FAN PWM			
4 bytes			
0	1	2	3
Version		Operation= 0x0107	
Length Options = 0x00000006			
Enable	Polarity	Period	
Duty Cycle		Reserved	

Field name	Data type	Value
Enable	UINT8	Enable remote control: 0x00 = Remote control not enabled 0x01 = Remote control enabled
Polarity	UINT8	Enable or disable fan: 0x00 = Fan not enabled 0x01 = Fan enabled
Period	UINT16	Time in ms for the activation of the fan. Recommended value = 400ms Ranges (max, min): not specified
Duty cycle	UINT16	In percentage, from 0% to 100%
Reserved	UINT32	0x00000000

Enable Fan

Fan enable			
4 bytes			
0	1	2	3
Version		Operation= 0x01F0	
Length Options = 0x00000001			
Enable		Reserved	

Field name	Data type	Value
Enable	UINT8	0x00 = Not enabled 0x01 = Enabled
Reserved	UINT32	0x00000000

8.1.6.6 WUP lines configuration



In configuration mode only.

If leaving the configuration mode, selected state will be released

Wake-up lines control			
4 bytes			
0	1	2	3
Version		Operation= 0x01F0	
Length Options = 0x00000002			
Link	State	Reserved	

Field name	Data type	Value
Line	UINT8	0x00 = Wake-up line 1 0x01 = Wake-up line 2
State	UINT8	0x00 = Not forced (released) 0x1 = Force to enable (High) 0x2 = Force to disable (low)
Reserved	UINT32	0x00000000

9 Disclaimer

The **“IEEE Compliant mode”** is defined by the chip vendor. Technica Engineering cannot be held responsible for the interoperability of this chip and hence the complete product with other devices if the interoperability issues are caused by the **“IEEE Compliant mode”.**

10 Changelog

Version	Chapter	Description	Date
1.0	All	First release	Sept 2023
1.1	All	Updates about new features in release 21	Mar 2024

11 Contact

If you have any questions regarding this product, please feel free to contact us:

Technica Engineering GmbH
Leopoldstr. 236
80807 München
Germany

Technical support:
support@technica-engineering.de

General information:
info@technica-engineering.de

Most current user manuals, software releases and product information:
https://portal.technica-engineering.de/downloads

12 Declaration of conformity

Български

С настоящото Technica Engineering GmbH декларира, че продуктът Модул за улавяне Capture Module 1000 High (TE-1176), е в съответств ие с Директива 2014/30/EC. Цялостният текст на ЕС декларацията за съответствие може да се намери н а следния интернет адрес:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Čeština

Tímto Technica Engineering GmbH prohlašuje, že produkt Capture Module 1000 High (TE-1176), je v souladu se směrnicí 2014/30/EU. Úplné znění EU prohlášení o shodě je k dispozici na této internetové adrese:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Dansk

Hermed erklærer Technica Engineering GmbH, at produktet Capture Module 1000 High (TE-1176), er i overensstemmelse med Direktiv 2014/30/EU. EU-overensstemmelseserklæringens fulde tekst kan findes p følgende internetadresse:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Deutsch

Hiermit erklärt Technica Engineering GmbH, dass das Produkt Capture Module 1000 High (TE-1176) die Richtlinie 2014/30/EU entspricht. Der vollständige Text der EU-Konformitätserklärung ist unter der folgenden Internetadresse verfügbar:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Eesti

Käesolevaga deklareerib Technica Engineering GmbH, et toode hõivamismoodul Capture Module 1000 High (TE-1176), vastab direktiivi 2014/30/EL nuetele. ELi vastavusdeklaratsiooni tielik tekst on kttesaadav jrgmisel internetiaadressil:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

English

Hereby, Technica Engineering GmbH declares that the product Capture Module 1000 High (TE-1176), is in compliance with Directive 2014/30/EU. The full text of the EU declaration of conformity is available at the following internet address:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Español

Por la presente, Technica Engineering GmbH declara que el producto Capture Module 1000 High (TE-1176), es conforme con la Directiva 2014/30/UE. El texto completo de la declaración UE de conformidad está disponible en la página web siguiente:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Ελληνικά

Με την παρούσα ο/η Technica Engineering GmbH, ότι το προϊόν Capture Module 1000 High (TE-1176), πληροί την οδηγία 2014/30/ΕΕ. Το πλήρες κείμενο της δήλωσης συμμόρ φωσης ΕΕ διατίθεται στην ακόλουθη ιστοσελίδα στο διαδίκτυο:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Français

Le soussigné, Technica Engineering GmbH, déclare que le produit Capture Module 1000 High (TE-1176), est conforme la directive 2014/30/UE. Le texte complet de la déclaration UE de conformité est disponible l'adresse internet suivante:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Hrvatski

Technica Engineering GmbH ovime izjavljuje da je proizvod Capture Module 1000 High (TE-1176) u skladu s Direktivom 2014/30/EU. Cjeloviti tekst EU izjave o sukladnosti dostupan je na sljedećoj internetskoj adresi:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Italiano

Il fabbricante, Technica Engineering GmbH, dichiara che il prodotto Capture Module 1000 High (TE-1176), conforme alla direttiva 2014/30/UE. Il

testo completo della dichiarazione di conformità UE disponibile al seguente indirizzo Internet:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Latviešu

Ar šo Technica Engineering GmbH deklarē, ka produkts Capture Module 1000 High (TE-1176), atbilst Direktīvai 2014/30/ES. Pilns ES atbilstības deklarācijas teksts ir pieejams šādā interneta v ietnē:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Lietuvių

Aš, Technica Engineering GmbH,
patvirtinu, kad produktas sugavimo modulis Capture Module 1000 High (TE-1176), atitinka Direktyvą 2014/30/ES. Visas ES atitikties deklaracijos tekstas prieinamas šiuo internet adresu:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Magyar

Technica Engineering GmbH igazolja, hogy a termék Capture Module 1000 High (TE-1176) a 2014/30/EU irányelvnek. Az EUmegfelelőségi nyilatkozat teljes szövege elérhető a következő internetes címen:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Malti

B'dan, Technica Engineering GmbH,
niddikjara li l-prodott Capture Module 1000 High (TE-1176), huwa konformi madDirettiva 2014/30/UE. It-test kollu tad-dikjarazzjoni ta' konformit tal-UE huwa disponibbli f'dan l-indirizz tallnternet li ġej:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Nederlands

Hierbij verklaar ik, Technica Engineering GmbH, dat het Capture Module 1000 High (TE-1176) product voldoet aan richtlijn 2014/30/EU. De volledige tekst van de EUconformiteitsverklaring kan worden geraadpleegd op het volgende internetadres:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Polski

Technica Engineering GmbH niniejszym oświadcza, że produkt Capture Module 1000 High (TE-1176), jest zgodny z dyrektywą 2014/30/UE. Pełny tekst deklaracji zgodnośc I UE jest dostępny pod następującym adre sem internetowym:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Português

O(a) abaixo assinado(a) Technica Engineering GmbH declara que o produto Capture Module 1000 High (TE-1176), está em conformidade com a Diretiva 2014/30/UE. O texto integral da declarao de conformidade está disponível no seguinte endereo de Internet:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Română

Prin prezenta Technica Engineering GmbH declară ca produsul Capture Module 1000 High (TE-1176), este în conformitate cu Directiva 2014/30/UE. Textul integral al declarației UE de conformitate este disponibil la următoarea adresă internet:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Slovensko

Technica Engineering GmbH potrjuje, da je izdelek Capture Module 1000 High (TE-1176), skladen z irektivo 2014/30/EU. Celotno besedilo izjave EU o skladnosti je na voljo na naslednjem spletnem naslovu:
https://technica-engineering.de/en/produkt/capture-module-1000-high/

Slovensky

Technica Engineering GmbH týmto vyhlasuje, že produkt Capture Module 1000 High (TE-1176), je v slade so smernicou 2014/30/EÚ. Úplné EÚ vyhlásenie o zhode je k dispozícii na tejto internetovej adrese:
https://technica-engineering.de/en/produkt/capture-module-1000-high/